EXPERIMENTAL AND THEORETICAL INVESTIGATION OF THE POLARIZATION DEPENDENT SYSTEM PERFORMANCE IN MULTIMODE OPTICAL FIBER COMMUNICATION SYSTEMS

by

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Abstract

Multimode fiber communications systems are generally found in short-reach enterprise or campus networks serving as high-speed network backbones. Data rates through these systems can range from 100 Mb/s to 10 Gb/s with the newest 10 Gb/s multimode fiber Ethernet (10GbE) standards ratified in 2006. Throughout the development of the latest 10GbE standard, IEEE 802.3aq 10GBASE-LRM, there was much discussion on the effect of launch polarization on system performance. However, while many qualitative observations were made and conclusions drawn from them no clear cause of the polarization dependence multimode fiber system performance was realized. This thesis discusses experiments that were conducted using a new testing methodology to further investigate this polarization dependence and a novel theoretical model created to describe this effect for the first time and provide further insights into its causes.

A new methodology to investigate polarization dependent response in multimode fiber systems is proposed and used to further quantify the true effects of the polarization dependence but possible causes as well. Previous studies measured output eye diagrams and observed how rotating the linear input polarization changed the output. This method, however, does not provide enough information to determine the true polarization effect and its causes. The new method sends isolated 100 ps pulses down the fiber and measures how the power in each fiber mode group changes with input polarization. By measuring the power fluctuation in each multimode fiber mode group as the input polarization state is rotated, the experiments performed quantitatively determined how power fluctuates between mode groups in a multimode fiber system. For each mode group propagating in the fiber, this power fluctuation shows a sinusoidal dependence on the linear polarization state of the input pulse with a period of 180 degrees. The final conclusions drawn from the experiment confirm some of the conclusions made during the previous polarization studies and disprove others including the possible causes of the effect and how this polarization dependence actually affects performance in multimode fiber systems. Based on these experimental results, a theoretical model was constructed that explains the polarization effects from the direct excitation of multimode fiber by a singlemode fiber input and provides further insights into the polarization dependence in a multimode fiber that is preceded by a patchcord with mismatched fiber parameters or fusion splice. The effects modeled include an explanation of the lack of polarization response due to direct singlemode to multimode fiber excitation as well as possible explanations for the mode group power fluctuation. Future work based on these experiments and model will be a more in depth look at the power fluctuations of specific modes within mode groups and how these individual modes affect the polarization dependent fluctuations overall.

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Chapter One
Introduction

Multimode fiber communications systems are generally found in short-reach enterprise or campus networks serving as high-speed network backbones. Data rates through these systems can range from 100 Mb/s to 10 Gb/s with the newest 10 Gb/s multimode fiber Ethernet (10GbE) standards ratified in 2006. Throughout the development of the latest 10GbE standard, IEEE 802.3aq 10GBASE-LRM, there was much discussion on the effect of launch polarization on system performance. This thesis will discuss this polarization dependence and ultimately attempt to model its effect on system performance. To begin, this chapter first discusses the basics of multimode fiber communications systems, system impairments, and current high speed standards.

1.1 What is Multimode Optical Fiber?

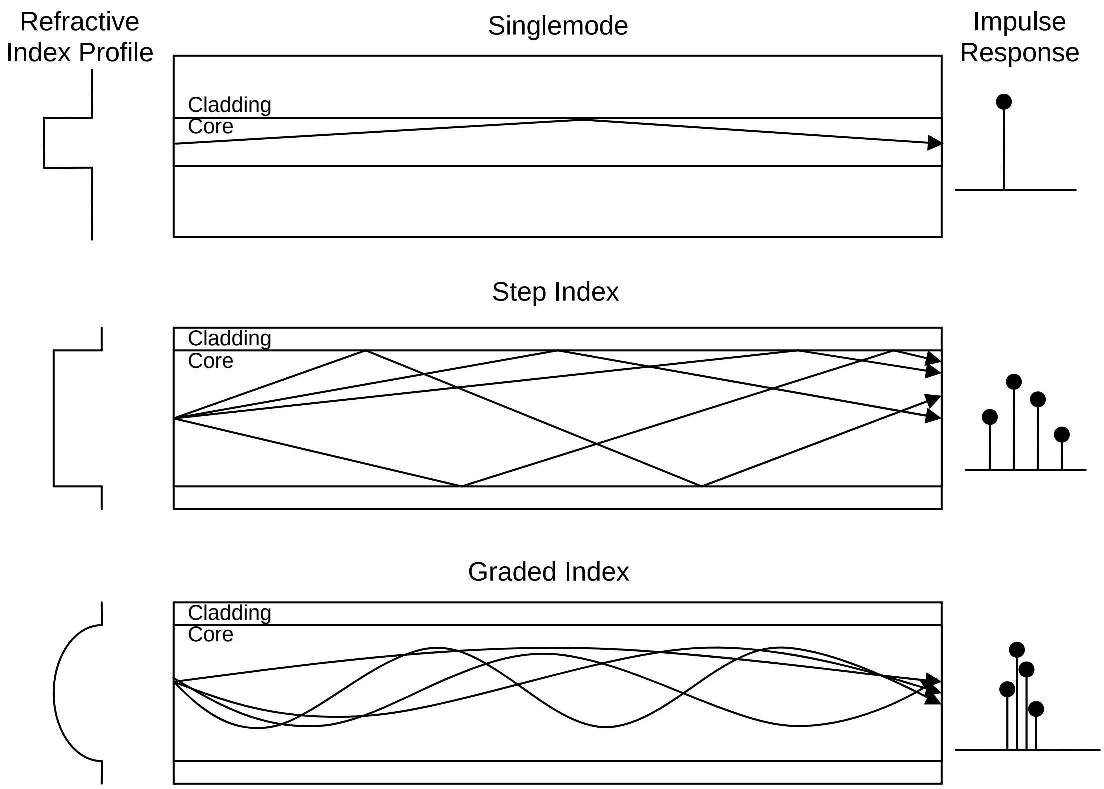

Fig. 1.1: Singlemode and Multimode fiber types: refractive index profiles, ray paths and impulse response based on intermodal dispersion.

Optical fiber is a cylindrical dielectric waveguide consisting of a high refractive index inner core surrounded by a lower refractive index cladding layer. Light coupled into the core of the waveguide will travel along the fiber and if launched at the correct angle will be bounded in the waveguide because of total internal reflection at the core/cladding interface. Singlemode fiber (SMF) consists of fiber with a core that has a small diameter (~ 9 μm) that allows for the propagation of light along only one path (or, in a single mode) at the appropriate wavelength (e.g. ~1550 nm for SMF communications). Multimode fiber (MMF) on the other hand has a relatively large core diameter (50 – 62.5 μm) that allows for light to propagate along many different paths (or modes). Aside from singlemode fiber, there are two basic types of multimode fiber: step index and graded index fiber, as shown in Fig. 1.1.

Step index fiber consists of a core region that has a constant refractive index and is surrounded by a cladding region of lower refractive index. Step index fibers support twice as many modes as a graded index fiber at a given wavelength and the number of modes in a step index fiber is approximately

$$N \cong \frac{V^2}{2}, \quad V = \frac{2\pi}{\lambda} a n_1 \sqrt{2\Delta}, \quad \Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \tag{1.1}$$

where N is the total number of modes supported by the fiber, V is the normalized frequency, λ is the free space wavelength, a is the fiber core radius, n_1 is the maximum refractive index of the core, n_2 is the refractive index of the cladding and Δ is the fiber height parameter. For singlemode fibers $\Delta < 0.01$ and for multimode fibers $0.01 < \Delta < 0.03$ [1, 2]. The uniform refractive index of the core means that each mode travels at the same speed but since they take drastically different paths through the fiber, each mode arrives at the end of the fiber at a different time, causing pulses launched into the fiber to spread out dramatically.

Graded index fiber consists of a core region that has a maximum refractive index at the center of the fiber and then decreases with increasing radius. The refractive index profile of such a fiber is usually given by a power law profile such that

$$n(r) = \begin{cases} n_1 \sqrt{1 - 2\Delta (r/a)^q}; & 0 < r \le a, \\ n_1 \sqrt{1 - 2\Delta}; & r \ge a \end{cases} \tag{1.2}$$

where r is the radial coordinate and q is the profile shape parameter. When $q = \infty$ the fiber is a step index parameter, when $q = 2$ it is a parabolic profile and when $q = 2 - 2\Delta$ the fiber has an optimum profile. Such graded index fibers only support half as many modes as their step index equivalent and are capable of reducing intermodal dispersion significantly. Table 1.1 lists the maximum pulse dispersion [2] for various profile shapes and Fig. 1.2 shows the group delays of each mode group present in the fiber for step and graded index fibers. Notice the significant reduction in maximum pulse dispersion (almost 80 times) even in a non-optimum graded index profile!

Table 1.1: Pulse dispersion for various power law profiles
with $n_1 = 1.444$, $\Delta = 0.017$

Shape Parameter (q)	Maximum Pulse Dispersion ($\Delta\tau$)	Maximum Pulse Dispersion (ns/km)
∞ (step)	$\frac{n_1\Delta}{c}L$	82.0
2 (parabolic)	$\frac{n_1\Delta^2}{2c}L$	0.70
2 - 2Δ (optimum)	$\frac{n_1\Delta^2}{8c}L$	0.17
2.06 (typical)	$\frac{n_1 L}{c}\frac{q-2}{q+2}\Delta$	1.21

The reason for this incredible decrease in pulse dispersion is best explained in terms of the optical ray model. Each excited mode in the fiber takes a different path, leading to intermodal dispersion as seen in Fig. 1.1. However, since the refractive index is now graded, instead of traveling all the way to the core/cladding interface (as in step index fiber) the rays (modes) of the graded index fiber are slowly curved away from the edge of the core as they propagate. This means that the higher order modes that generally have steeper ray trajectories (i.e. longer paths) spend much more of their time traveling through a medium of lower refractive index (thus traveling faster) than the lower order modes that spend most of their time traveling at lower speeds (in the shorter paths). If designed right, the steepest (longest) and shallowest (shortest) ray paths actually take the same amount of time to propagate a given length of fiber. This happens when the profile shape parameter is at its optimum value of 2 - 2Δ. As seen in Table 1.1 and Fig. 1.2 however, any kind of non-optimum graded index fiber is far superior to the equivalent step index fiber in terms of intermodal dispersion.

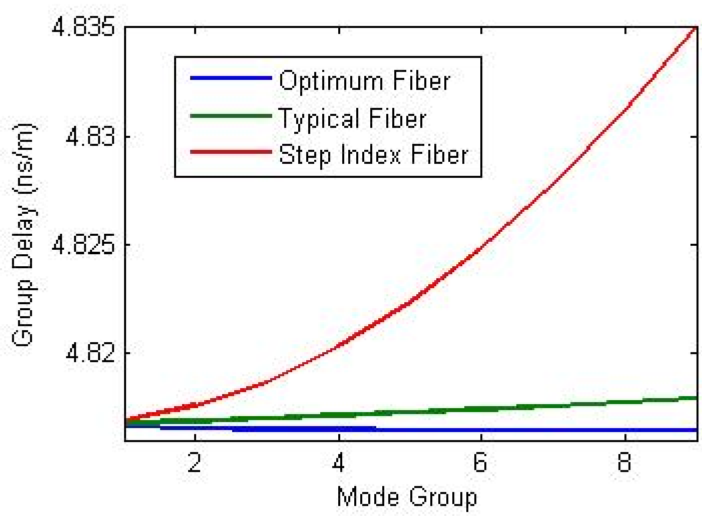

Fig. 1.2: Multimode fiber group delays for graded index optimum, typical and step index fibers.

Singlemode fiber of course does not suffer from such intermodal dispersion as only one mode (a single ray path) is excited. Long-haul singlemode fiber communications suffer from other types of impairments (intermodal/chromatic dispersion, polarization mode dispersion, etc.) but due to the severity of the intermodal dispersion in multimode fiber, the maximum distance of a multimode fiber based communication link is severely limited.

1.2 Why Choose Multimode Fiber?

As the previous section showed, compared to singlemode fiber, multimode fiber suffers from a severe performance impairment: intermodal dispersion. Even for multimode fibers with optimum profiles this greatly inhibits the transmission distance in communication links based on multimode fiber. Also, because of the advanced techniques necessary to create graded index fiber (especially newer, laser optimized fibers) and the lower market volume, the per-unit cost of multimode fiber is higher than single mode fiber. Given this information, why would any network engineer/IT department choose to work with multimode fiber? There are two very good reasons: 1) despite its higher per-unit cost, multimode fiber allows for the use of cheaper transceivers/electronics and its larger core diameter makes it easier to align and splice, subsequently lowering labour costs; 2) it is already widely installed in enterprise/campus/premise networks in North America.

1) Multimode Fiber System Cost:

The larger core of the multimode fiber, while allowing for the poor intermodal dispersion performance, makes it much easier to couple light into the fiber. Singlemode fiber communications require expensive distributed feedback (DFB) lasers for the efficient and stable coupling of light into the small diameter of the singlemode fiber. There are less expensive options however, such as light emitting diodes (LED) and vertical cavity surface emitting lasers (VCSEL). While significantly less expensive than DFB lasers, LEDs (with low-data rates at nominal operating conditions for LED) and VCSELs (low output power) have larger active areas and relatively low output powers. This means that to efficiently couple any of the light into an optical fiber, the core diameter of the fiber has to be large. To give an idea of the cost savings using a VCSEL over a DFB laser, VCSELs cost approximately $20 while high-end DFB lasers can cost more than $2,000 and since they consume less power, VCSELs and LEDs are cheaper in the long term as well.

2) Installed Multimode Fiber Base:

Because of the large intermodal dispersion inherent to multimode fiber, only short lengths of it at relatively low bitrates can be used without requiring some dispersion compensation scheme. However, because of the cost savings achievable using less expensive electronics, multimode fiber has been widely used in enterprise/campus networks (that require only short transmission links) for thirty years and as such already has a large installed base. To upgrade the enterprise/campus network to higher bitrates then would either require figuring out how to use the current installed fiber or to install new fiber, a very expensive task. Fig. 1.3, from [3], shows the current installed fiber base and a future projection in enterprise networks.

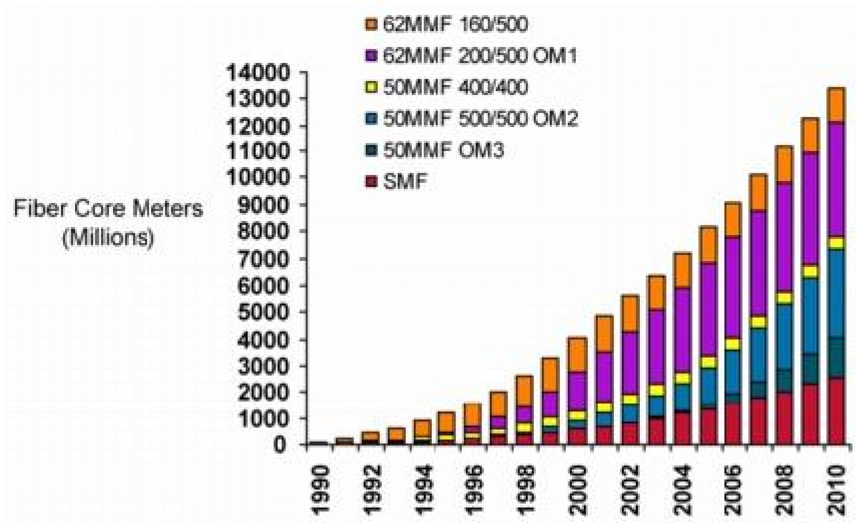

Fig. 1.3: Installed base of MMF and projected installation [3].

1.3 Graded Index Multimode Fiber Types

Table 1.2: Multimode fiber types and bandwidths

Fiber Type	Core Diameter (μm)	Bandwidth (MHz.km, 850/1300 nm)
FDDI	62.5	160/500
OM-1	62.5	200/500
OM-2	50	500/500
OM-2+	50	700/500
OM-3	50	1500/500
OM-3+	50	3500/500

There have been many improvements in the optical fiber manufacturing process since the first multimode fibers were placed in campus and enterprise backbones. As a result, each system can have a different quality of installed fiber or a combination of fibers of varying quality. Fiber performance is measured in a bandwidth distance product in units of MHz.km and ranges from 160 MHz.km for the oldest grade fiber up to 3500 MHz.km for some of today's laser optimized fibers; these fibers are labeled OM1, OM2 or OM3. Table 1.2 gives a list of the types of fiber available in current networks, their core diameters and their bandwidths for both 850 nm and 1300 nm sources. The bandwidth is measured assuming an overfilled launch (OFL), meaning that the all the modes of the fiber are excited and the bandwidth then measured as would happen using an LED as a source. Fiber distributed data interface (FDDI) and OM1 grade fiber was made for use with LEDs and has a larger core diameter and lower bandwidth as a result.

The new laser optimized fiber (OM3/3+) was designed specifically for 850 nm VCSELs and as a result its bandwidth are measured in terms of effective modal bandwidth (EMBW) since the laser sources only excite a subset of all possible fiber modes.

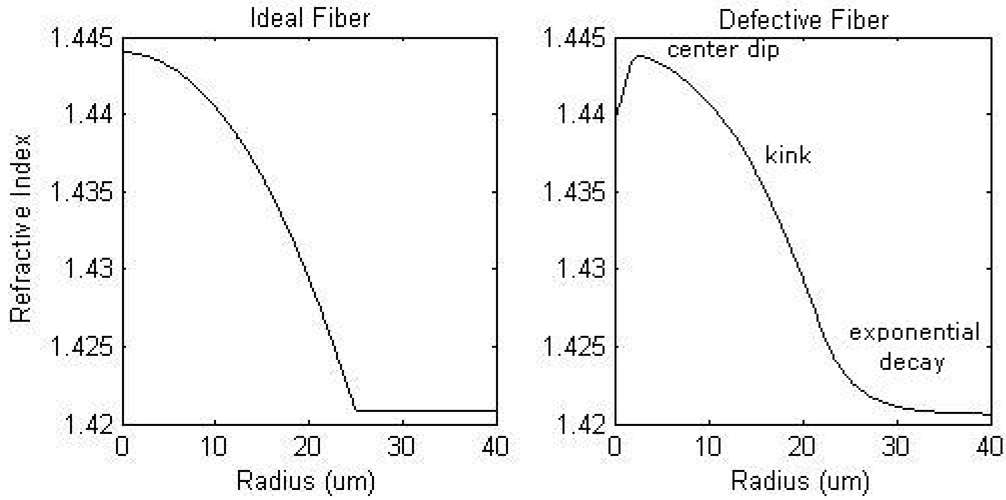

Fig. 1.4: Ideal vs. Defective refractive index profile.

Older fibers (FDDI – OM2+) can have one of many defects in the refractive index profile such as a center dip/peak, a kink or mismatched shape parameter or an exponential decay. Fig. 1.4 compares an ideal refractive index profile with that of one possessing some of these defects. OM3/3+ fibers are manufactured using a new crystal vapour deposition (CVD) method that eliminates these defects and their profiles are optimized for use with 850 nm lasers, greatly increasing bandwidth. To increase the bandwidth above the OFL bandwidth in older OM1/2 fibers, VCSELs can be used by physically offsetting the laser source away from possible defects in the fiber to avoid exciting modes that would be affected by them.

1.4 10 Gigabit Ethernet Standards

There have been three next generation 10 Gigabit Ethernet (10 GbE) standards ratified by the IEEE in the past five years, 802.3ae 10GBASE-SR, 802.3ae 10GBASE-LX4 and 802.3aq 10GBASE-LRM [4 – 6]. To achieve 10 GbE, LEDs can no longer be used as their direct modulation rate maximum is only 622.5 Mb/s. VCSELs on the other hand are used in the current 1/10 GbE standards as their modulation bandwidth exceeds 10 Gb/s.

10GBASE-SR is a short range standard that supports 10 GbE using 850 nm VCSELs over all fiber types. For FDDI and OM1 fiber, it specifies a range of 25 – 33 m, for OM2/2+ fibers it specifies a range of 66 – 82 m and for OM3 fiber it supports distances up to 300 m.

10GBASE-LX4 is a long range standard supporting transmission over distances of 300 m using all fiber types. To do this it uses wavelength division multiplexing (WDM) to transmit four different wavelengths around 1300 nm at an individual channel rate of 3.125 Gb/s for a total rate of 12.5 Gb/s where the extra bandwith is used for forward error correction coding overhead. Since this standard supports all new and legacy fiber types, it requires the use of a mode conditioning patch cord for the legacy fiber. This patch cord acts as a transverse offset, exciting higher order modes in the fiber that do not propagate near fiber defects and increasing overall performance. This standard also supports transmission over 10 km of SMF.

10GBASE-LRM is another long range standard that transmits at a rate of 10.3 Gb/s over a single wavelength (850 nm VCSEL) and supports all fiber types up to lengths of 220 m. While it does not support the same long range as the LX4 standard, it does not require the mode conditioning patch cords for the older fiber types. It does, however, specify the use of electronic dispersion pre- or post-compensation (EDC) for the worst-case fibers.

All of these standards only specify lengths up to a maximum of 300 m, while older, slower standards (e.g. 1 GbE) specify transmission distances up to 2000 m over MMF. This is because a survey by [7] showed that all link lengths in premise local area networks were less than 500 m, 88% of which were shorter than 300 m. Because of this, the 10 GbE standards only require link lengths up to this maximum.

1.5 Other Important Multimode Fiber Impairments

Aside from the intermodal dispersion already discussed, multimode fiber systems suffer from other important impairments as well. These include mode selective loss, modal noise, and polarization dependence.

Mode selective loss (MSL) arises from multimode fiber links that have connections or splices along them. If there is an offset between the two adjacent multimode fibers at a connector or splice, then it is possible that some power can be lost. This is because if power is traveling in a higher order mode that propagates near the core/cladding boundary, power from that mode can be coupled to the cladding of the fiber after the offset connection/splice. Once in the cladding, the power in the mode is quickly attenuated, increasing the total link loss.

Modal noise is caused by a dynamic speckle pattern present at the receiver where there is an element of mode selective loss present somewhere along the transmission link. The highly coherent laser sources used in the new 10 GbE standards have coherent lengths longer than the links they are used in. This allows both the longitudinal and transverse modes of the fiber to interfere, creating a speckle pattern during each pulse at the receiver. Fig. 1.5 shows an example of the speckle pattern at the end of a multimode fiber. If there is a dynamic exchange of power between modes (e.g. mode coupling due to fiber vibrations), the speckle pattern will change with time. If there is then an element of mode selective loss that preferentially filters out specific modes, then there will be a power fluctuation at the receiver: modal noise.

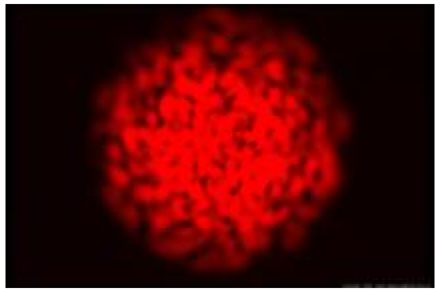

Fig 1.5: Multimode fiber speckle pattern [8].

The polarization dependence of multimode fiber performance has only recently been studied [1, 9 – 15] and there have been no extensive quantitative studies to date. Only qualitative experiments have been conducted and were presented during discussion of the IEEE 802.3aq standards. Previous work has shown that as the polarization of the input field to the multimode fiber is rotated, the performance of the link can be altered drastically. This has been displayed qualitatively as eye diagrams being completely open at one polarization and then completely closed at an orthogonal polarization. Fig. 1.6 gives an example of this effect. It shows the eye diagrams of a 10 Gb/s pseudo-random bit sequence over 500 m of OM3 fiber first at a launch polarization of zero degrees and then at ninety degrees. At ninety degrees the eye diagram is significantly more closed than at zero. The quantification of this phenomenon is the ultimate goal of this study on the polarization dependence of multimode fiber.

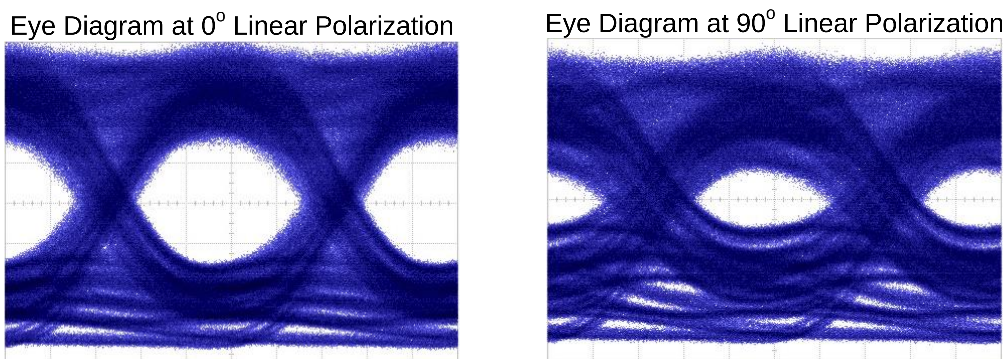

Fig. 1.6: Eye diagrams for 500 m OM3 fiber excited by 10 Gb/s SMF at an offset of 0 μm at two different linear polarizations, 0^o and 90^o.

1.6 Contribution and Organization

The goal of this thesis is to investigate the phenomenon of polarization dependence in multimode fiber systems and present an accurate model of this effect. The thesis is organized as follows:

- Chapter two presents relevant background information on the study of multimode fiber as well as a discussion of previous studies on the effects of polarization in multimode fiber systems;
- Chapter three discusses the motivation behind the methods of investigation and the ultimate experimental setup used to investigate the polarization dependence;
- Chapter four presents the experimental results and discusses their implications regarding the theoretical model;

- Chapter five presents the theoretical model;
- Chapter six presents numerical results generated using the newly developed model, and compares them with experimental data;
- Chapter seven summarizes the work presented and discusses relevant future work.

Other studies [9 – 15] have presented qualitative experimental results investigating the polarization effects seen in multimode fiber, or have presented simulations that attempt to explain it when the fiber is perturbed under special conditions [16]. To the best of the author's knowledge, this work is the first quantitative experimental and theoretical study of the polarization dependence in multimode fiber based on a full vectorial model of electromagnetic optical fields that shows the polarization effect is due to polarization dependent mode group power coupling at a single interface and not an accumulative effect present throughout the fiber. It also shows that the effect can be caused by fiber parameter mismatches, fusion splices and offset fiber connectors and that birefringence plays no significant role. The theory is in good agreement with the experimental observation, and serves as a good approach to model the effect of polarization dependence on system performance in short-reach MMF based fiber networks.

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Chapter Two
Background Theory and Previous Work

This chapter discusses important background theory needed for the discussion of system performance in multimode fiber communications systems. Starting with a discussion of the power law profiles that describes the refractive index of the multimode fiber core and determines the guided modes inherent to such profiles, we will look at previous work investigating the system performance dependence on input polarization in multimode fiber systems.

2.1 Multimode Fiber Guided Modes

As seen in Chapter One, there are many different kinds of multimode fibers, from those with step refractive index profiles to the many different flavours of graded-index multimode fibers. Due to their lower intermodal dispersion, the only fibers used in communications systems are graded index fibers, and as such these are the only fibers discussed in this chapter.

2.1.1 Power Law Refractive Index Profiles

The power law refractive index profile, under certain simple assumptions, provides very useful analytical solutions that allow for the complete characterization of the guided modes of multimode fiber. The power law profile is described by

$$
\begin{gathered}
n(r) = \begin{cases} n_1\sqrt{1-2\Delta(r/a)^q}\,; & 0 \le r \le a, \\ n_1\sqrt{1-2\Delta}; & r > a \end{cases} \\
\Delta = \frac{n_1^2 - n_2^2}{2n_2^2}
\end{gathered}
\tag{2.1}
$$

where n_1 is the maximum refractive index of the fiber, n_2 is the refractive index of the cladding, Δ is the fiber height parameter, a is the radius of the fiber core and q is the fiber shape parameter. When $q = 2$, it becomes a parabolic profile, when $q = \infty$ it is a step-profile and when $q = 2 - 2\Delta$ it is an optimal profile with the least amount of intermodal dispersion.

2.1.2 Wave Equation Solutions

In radial coordinates, the scalar modes of an optical fiber have the solution [2, 3]

$$
E(r,\phi,z,t) = E_r(r)\begin{Bmatrix} \cos l\phi \\ \sin l\phi \end{Bmatrix}\exp\{j(\omega t - \beta z)\}
\tag{2.2}
$$

where l is the azimuthal mode number and $E_r(r)$ are the solutions to the scalar wave equation in radial coordinates

$$\frac{d^2E_r(r)}{dr}+\frac{1}{r}\frac{dE_r(r)}{dr}+\left((k_0n(r))^2-\beta^2-\frac{l^2}{r^2}\right)E_r(r)=0\,. \tag{2.3}$$

where k_0 is the free space wave number and β is the propagation constant of each mode. From [2, 3], when $n(r)$ is an infinite power law profile with q = 2, the solutions to the radial equation are

$$E_r^{lm}(r)=N_{lm}r^l\exp\left\{-\frac{1}{2}\frac{V}{a^2}r^2\right\}L_{m-1}^l\left(\frac{V}{a^2}r^2\right) \tag{2.4}$$

where N_{lm} is a normalization constant and L_{m-1}^l are the associated Laguerre polynomials [4], with l = 0, 1, 2, … as the azimuthal mode number and m = 1, 2, 3, … as the radial mode number. Modes with the same $M = 2m + l - 1$ have the same propagation constant (and group delay) and are considered degenerate and are grouped together in the same mode group. The propagation constants for each mode can then be found using [2, 3]

$$\beta_{lm}=k_0n_1\left(1-\frac{2\sqrt{2\Delta}(2m+l-1)}{ak_0n_1}\right)^{1/2}. \tag{2.5}$$

Using WKB analysis, the propagation constants of the modes of a fiber with an arbitrary power law profile are [2, 3]

$$\beta_v\approx k_0n_1\left[1-2\Delta\left(\frac{2(q+2)}{qk_0^2a^2n_1^2\Delta}v\right)^{\frac{q}{q+2}}\right]^{1/2}. \tag{2.6}$$

From inspection with the propagation constants of the parabolic fiber when q = 2, it can easily be seen that $v = (2m + l - 1)^2$ and that the approximate propagation constants for any modes of a fiber with a power law refractive index profile in circular coordinates is

$$\beta_{lm}\approx k_0n_1\left[1-2\Delta\left(\frac{2(q+2)}{qk_0^2a^2n_1^2\Delta}(2m+l-1)^2\right)^{\frac{q}{q+2}}\right]^{1/2}. \tag{2.7}$$

The group delay of each mode are then $\tau_{lm} = d\beta_{lm}/d\omega$. For values of q near 2, the solutions to the radial wave equation are still very close to the Laguerre-Gaussian functions given in equation (2.4) [5, 6], even for fibers with severe defects.

2.1.3 Constructing Vector Modes

The scalar modes found in the previous section do not contain any polarization information and as they are solutions to the scalar wave equation are used as the linearly polarized (LP) solutions of the transverse electric field in the fiber. Also, each of the LP modes with same radial and azimuthal mode number (*m* and *l* respectively) have the same propagation constant and group delay and are degenerate. Since these modes are degenerate, it is possible to choose any modal field to be x- or y-polarized. To then find the vector modes from these degenerate scalar modes, the fields have to be constructed such that the tangential components at $r = a$ are continuous and give the correct ϕ dependence (either $\cos l\phi$ or $\sin l\phi$). To do this we follow the method of [3]; starting with

$$\begin{aligned} \mathbf{E}_1 &= E_r^{lm}(r)\cos l\phi\,\hat{\mathbf{x}}, \quad \mathbf{E}_3 = E_r^{lm}(r)\cos l\phi\,\hat{\mathbf{y}}, \\ \mathbf{E}_2 &= E_r^{lm}(r)\sin l\phi\,\hat{\mathbf{x}}, \quad \mathbf{E}_4 = E_r^{lm}(r)\sin l\phi\,\hat{\mathbf{y}}. \end{aligned} \tag{2.8}$$

as the degenerate vector modes, we can then find the even/odd HE/EH vector modes by choosing appropriate linear combinations of the above modes such that the tangential components of the field shows the correct dependence on ϕ. The z-component of each of the fields is not considered since the fibers under consideration are weakly guiding and the modal fields are nearly transverse. These modes can then easily be combined to form vector LP modes, and when brought back to rectangular coordinates the final vector LP modes are found as

$$\begin{aligned} &\text{even LP: } \mathbf{E}_{lm} = E_r^{lm}(r)\cos l\phi\,\hat{\mathbf{x}} \\ &\text{odd LP: } \mathbf{E}_{lm} = E_r^{lm}(r)\cos\big(l(\phi-\pi/2)\big)\,\hat{\mathbf{y}}\,. \end{aligned} \tag{2.9}$$

2.2 Constructing Fiber Impulse Response

A simple model for constructing the impulse response of a multimode fiber link is given in [7]. This model is used to easily characterize link performance and can later be extended to include polarization effects. The impulse response consists of an impulse representing each fiber mode, traveling with a different group delay and propagation constant as described above. The amplitude of each impulse is the power coupled into each mode by the input field whether it is a singlemode fiber input or the field directly from a laser such as a VCSEL. The power in each fiber mode by an input field φ is given by an overlap integral

$$a_{lm} = \left| \int_A E_{lm}(r,\phi)\varphi(r,\phi)\, dA \right|^2 \tag{2.10}$$

where the integral is done over the area A and E_{lm} is the field of the lm^{th} mode of the fiber. Ignoring the attenuation over the fiber link (since they are generally fairly short) the total fiber impulse response of the fiber is then found by summing the impulse response of each mode,

$$h(t) = \sum_{l,m} a_{lm} \delta(t - \tau_{lm} L) \tag{2.11}$$

where L is the length of the fiber link in meters, and τ_{lm} is the group delay of the fiber mode in ns/m. The transfer function of the fiber is then

$$H(f) = \sum_{l,m} a_{lm} \exp(-j2\pi\tau_{lm} Lf)\,. \tag{2.12}$$

When there are offset connectors involved, the fiber impulse response is found for each stretch of fiber and the power in each mode at each connector along the link is determined using a connector transfer matrix as further detailed in [7].

2.3 Previous Polarization Studies

There have not been many previous studies on the polarization dependence in multimode fiber. Discussions during the ratification of the IEEE 802.3aq standard first addressed it as not having a significant effect on performance for data rates less than 10 Gb/s [8 – 14] over the desired fiber range. Some of these results were then explored further in [15] which discussed the current understanding of the problem and gave further direction in its study. There is then finally the work presented in [5] that discusses the polarization dependence as a result of random polarization mode coupling throughout a given length of multimode fiber, similar to polarization mode dispersion (PMD) in singlemode fiber. This section briefly discusses some of the key observations in these studies.

2.3.1 IEEE 802.3aq Polarization Observations [8 – 14]

This section discusses the results presented in a series of task force meetings during the course of the development of the IEEE 802.3aq 10GbE standard [8 – 14]. All of these studies use a very similar setup when testing the polarization dependence in multimode fiber which is also similar to the setup presented in Chapter Three. It consists of a linearly polarized SMF launch that can be transversely offset and coupled to a MMF to excite different mode groups.

The first study in [8] shows very basic results showing that there is only strong polarization dependence in a few cases. That is, there will be strong polarization dependence if the field from the SMF is directly launched into the MMF and the MMF is of low quality (e.g. FDDI) or if there is a MMF patchcord placed between the SMF launch and the final MMF. If there is a direct SMF launch into a high quality OM3 fiber then there is very little observed polarization dependence.

The presentations of [9 – 11] reiterate the results presented in [8] and also discuss how fiber defects present in lower quality fiber induce strong polarization dependence in system performance and modes excited near regions of defect experience a much stronger polarization response. Also, when there is a direct SMF launch into the multimode fiber, there is no significant polarization response for any fiber when there is no transverse offset between the two fibers.

In [13], most of the same results are seen again, but there is a suggestion that the cause for the polarization response seems to be taking place only at the fiber-fiber interface regardless of whether there is a direct SMF – MMF launch or if there is a patchcord after the SMF launch.

2.3.2 Polarization Effects in MMF [15]

The section on the polarization effects in multimode fiber in [15] first discusses the underlying theoretical concepts necessary to the analysis. It then says how the mode power distribution excited in a multimode fiber (through any launch condition) is independent of launch polarization and the polarization effects are only due to defects (birefringence, local defects etc.) in the fiber breaking the degeneracy of the fiber modes.

2.3.3 Principal Modes in MMF with Polarization Mode Coupling [5]

The study of [5] expands the theory of polarization mode dispersion to multimode fibers. It breaks the multimode fiber link into many small sections each with its own separate birefringent axis randomly rotated relative to one another. This then allows for the random coupling between the even and odd polarization modes of the fiber. Since the fiber allows the propagation of multiple modes, at the junction between each fiber section there is also spatial mode coupling between the modes of the fiber. This creates a polarization dependent impulse response in the fiber. However, there is only significant polarization dependent response if the fiber is in the high coupling regime (i.e. significantly perturbed along the entire length of the fiber) and if there is a parameter mismatch between each short fiber section.

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Chapter Three
Experimental Setup and Methodology

This chapter describes a set of experiments, with the ultimate purpose to quantitatively evaluate the polarization dependent mode power coupling seen in previous studies and simulations [1 – 8]. To do this, the power in each mode group present and excited in a multimode fiber system will be measured as it fluctuates due to a change in the initial linear polarization. Overall, the experimental approach taken is very similar to experiments already conducted [1 – 7], the major difference being that systems under consideration are chosen such that individual mode groups can be resolved and the power in each mode group directly measured. This allows us to measure how power is coupled between each mode group under various mode excitation conditions.

3.1 Fiber Link Length

To quantify the polarization dependence in 10 GbE multimode fiber systems, a setup was needed such that each individual mode group could be individually measured to see how power fluctuated across the mode groups. With the current standards, using laser excitation the lengths of the fiber is too short to resolve individual mode groups using an input signal consisting of a single 10 Gb/s pulse. It is possible to see the effects of the polarization dependence but difficult to quantify as the accumulated intramodal dispersion is not great enough to allow for the proper separation of mode groups at the receiver using 10 Gb/s pulses. For this reason the fibers under test (FUT) used were longer than 750 m, usually 1 km. This allowed each of the individual mode groups to be resolved and the power fluctuation in each one to be easily measured.

3.2 Mode Group Excitation

To excite the mode groups, a fiber coupled 1550 nm DFB laser (> 2 km coherence length) was used to excite the MMF modes using a six-axis nano-positioner to align the two fibers. This wavelength was used not only because it was available, but because a maximum of only nine mode groups are able to propagate through the OM3/3+ fibers used, making it much easier to measure the polarization dependence. At 1300 nm, twelve mode groups and at 850 nm, 16 mode groups could have propagated through the MMF, requiring longer fiber lengths to resolve the individual mode groups. Unlike the LEDs used in older MMF systems, the launching of power from the SMF to the MMF only excited certain mode groups. Therefore to measure the polarization dependence through all the possible mode groups, a nano-positioning stage was used to create a transverse offset between the centers of the SMF and MMF. As the offset is increased, different mode groups are excited. The nano-positioner used was a Thorlabs six-axis nano-positioning stage with 37.5 nm resolution and 30 nm xyz axis repeatability. To align the nano-positioning stage with the center of the MMF, the output from a 1 km length of OM3 fiber was monitored. Since a MMF will capture 100% of the energy from a SMF up until an offset of approximately 15 μm, the alignment of the SMF/MMF fibers was determined by the power present in the fundamental mode group. When this power was maximized (at about 70% of total power), the two fibers were aligned along their center axis.

3.3 Test Fiber Types

Based on the previous work discussed in Chatper Two, it was found that there was no polarization dependent performance when using the newer laser optimized OM3/3+ fibers using a direct SMF – MMF coupling. However, when a mode-conditioning patchcord was used, there was a clear dependence on input polarization. Also, for older fiber types (FDDI – OM2) there was a definite polarization dependence regardless of the use of a patchcord between the SMF – MMF launch and the fiber under test. Because of this, five different FUT were used. First was a straight SMF – MMF launch where the MMF is an 750 m long spool of OM3 fiber. Second was the same as the first but using an older 1 km long FDDI fiber instead of the OM3 fiber. The third FUT consisted of a 1 km length of OM3/3+ fiber where a 1 m patchcord was inserted between the SMF – MMF launch and the final length of MMF. For the third and fourth FUT, two different fibers were used, one a 1 km length of OM3+ fiber and the second a 1 km length of OM3 fiber. The fifth FUT consisted of the same 750 m fiber, but with the first meter cut off then fusion spliced back onto the fiber, aligned along the fiber axis. Other lengths and configurations of fiber were also tested but for the sake of brevity, their results (being similar to results presented in chapter five) are not discussed and only presented in Appendix A. Table 3.1 shows the five different FUT configurations, their offset launch condition and whether or not a patchcord was used in the experiment.

Table 3.1: Experimental Setup FUT Configurations

Cases	Launch Condition	Patchcord Used	FUT Configuration
1	0, 7, 14, 21 μm	No	750 m OM3
2	0, 7, 14, 21 μm	No	1 km FDDI
3	0, 7, 14, 21 μm	Yes	1 m – 1 km OM3+
4	0, 7, 14, 21 μm	Yes	1 m – 1 km OM3
5	0, 7, 14, 21 μm	No	1 m – 749 m OM3

3.4 Polarization Control

A JDSU PR2000 polarization controller (PC) was used to linearly polarize and then rotate the polarization state of the signal being launched into the MMF. This controller has an insertion loss of 1.5 dB and operates from 1530 to 1560 nm. A polarization analyzer was used to determine that it is able to change the polarization state of a signal such that it can reach every point on the Poincaré sphere. This controller also has a 0.1 dB polarization dependent loss (PDL) and Fig. 3.1 shows the power fluctuation in the controller as the linear polarization is rotated.

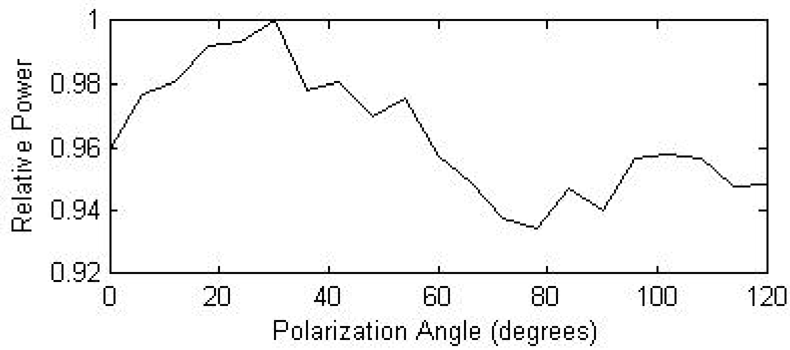


Fig. 3.1: Polarization dependent loss through the polarization controller.

3.5 Final Experimental Setup

Fig. 3.2 shows the final experimental setup. The Mach-Zehnder modulator (MZM) intensity modulated the output of the laser diode (LD). The pulse pattern generator (PPG) was set to generate a 100 ps FWHM pulse every 32 bit periods. This limited the length of fibers that could be tested to less than 2 km long before there was severe intersymbol interference between received pulses. The erbium doped fiber amplifier (EDFA) was used to compensate for link loss to allow for meaningful measurements. Loss mechanisms include insertion loss through each component, loss due to the SMF/MMF offset launch and possible loss through each MMF patchcord connector. The optical bandpass filter had a 0.55 nm bandwidth and was used to filter out noise generated by the EDFA. The variable optical attenuator (VOA) was used to finely tune the launch power to ensure that the total power received after each FUT was the same, especially when compensating for loss due to the offset fiber launch. The signal was then coupled into MMF through the nano-positioning stage and then through the FUT. The optical receiver has a MMF pigtailed front-end with a bandwidth of 12 GHz.

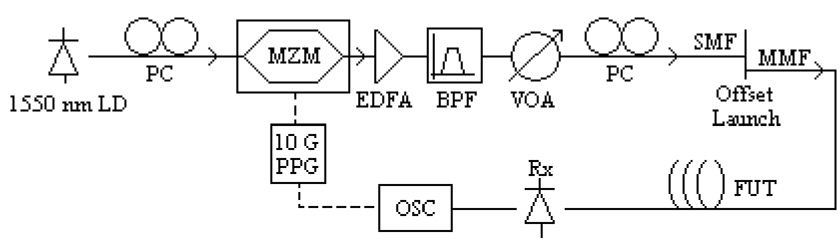


Fig. 3.2: Final experimental setup.

3.6 Experimental Methodology

To test the polarization dependence in multimode fiber, two different experiments were carried out on each FUT. The first was a series of differential mode delay (DMD) measurements for each fiber using an x-, y- and circular polarized input signal. The second set of measurements consisted of rotating the polarization of the pulsed input field at four different transverse offset locations: 0, 7, 14 and 21 μm.

DMD measurements consist of launching a single pulse at a fixed polarization from the SMF to the MMF and recording how the fiber response changes as the transverse offset between the SMF and MMF is increased from 0 – 24 μm. This measurement was done three times for each fiber to see the difference in fiber response for an input pulse that was x-, y- and circularly polarized. Fig. 3.3 gives an example of a DMD measurement. The total DMD of the fiber is then measured as the time between the first pulse at a 0 μm transverse offset and the last pulse at the final offset. Great care was taken to secure the fiber to the experimental setup to prevent any sources of mode or polarization coupling either due to periodic vibration or fiber stress. Securing of the fiber then allowed the results of the experiments to be repeatable to a high degree when using the same configuration, including using identical pathcords matched with each final length of fiber.

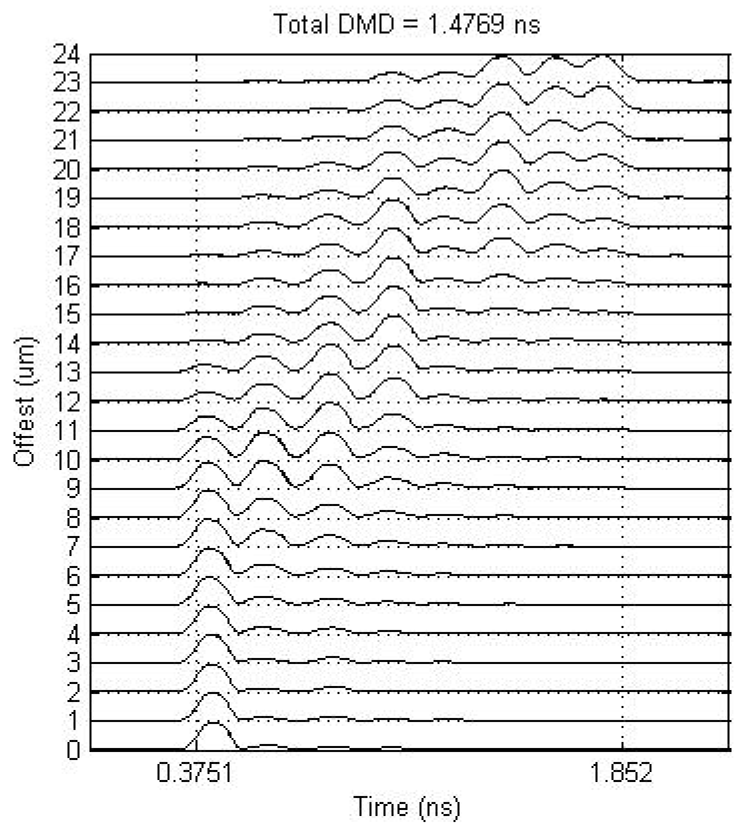


Fig. 3.3: Sample DMD measurement for an OM3, 1 km long fiber.

The polarization measurements then consist of launching a single pulse into the MMF at a given SMF transverse offset (0, 7, 14 or 21 μm) and recording the pulse response as the input polarization is rotated between zero and 240 degrees, allowing for the capture of at least one full period in the polarization response. Fig. 3.4 gives a sample polarization measurement. For the sake of clarity, while there may be many mode groups being excited at a given transverse offset (see DMD sample above), the measurements are displayed as consisting of only two mode groups, the dominant mode group that carries the most power compared to all others and then all the other mode groups combined. The total power fluctuation with respect to polarization is also given.

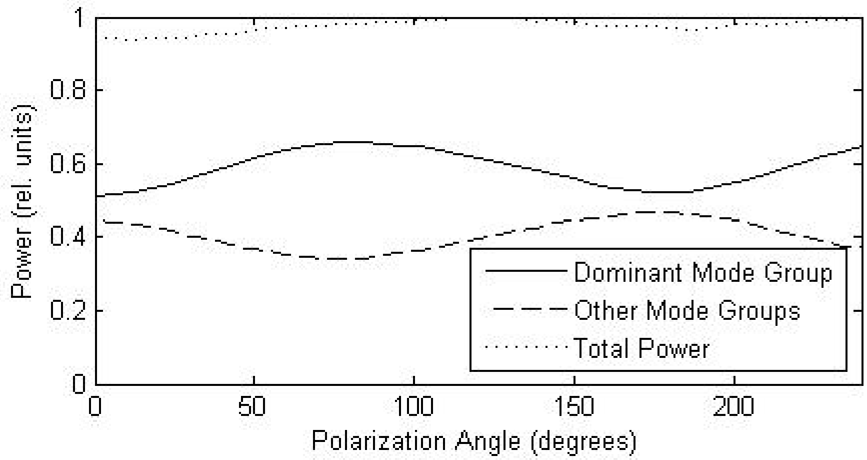
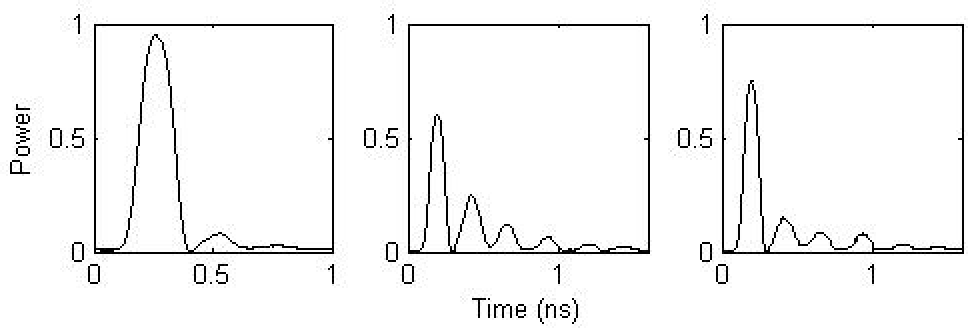


Fig. 3.4: Sample polarization measurement.

Fig. 3.5 shows a sample pulse response measurement. Fig. 3.5a is the input pulse coming directly from the SMF before the offset launch stage, and Fig. 3.5b shows the MMF pulse response after a 1 km length of OM3+ fiber. Fig. 3.5c shows the same pulse response but for the orthogonal input polarization. To then determine how the power in each mode group fluctuates as the polarization is rotated, a pulse measurement (e.g. Fig. 3.5b/c) is taken and the peak power of each pulse in the output is measured as the polarization is rotated to get curves similar to those of Fig. 3.4.



3.5a: Sample input *3.5b: 0^o output* *3.5c: 90^o output*

Fig. 3.5: Sample input and output at $0^o/90^o$ polarizations for a 1 km OM3+ MMF.

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Chapter Four
Experimental Results and Discussion

This chapter outlines the experimental results obtained from the methods outlined in Chapter 3. It will first present the differential mode delay (DMD) measurements made for each fiber. It will then look at the polarization measurements and discuss the reasoning behind the determined location of the apparent polarization dependent mode coupling and briefly discuss aspects of the theoretical model.

4.1 DMD Results

Each DMD measurement is taken at three different input polarizations: x-polarized, y-polarized and circularly polarized. The total DMD is then measured as the time difference between the rising edge of the first pulse at zero micron offset and the falling edge of the last pulse at the highest offset and is measured in ns. For the following measured fibers, all DMD measurements display a similar mode group order, the leftmost mode group is excited in the lower offset range and corresponds to the fundamental mode group and the rightmost groups are excited only in the high offset range and correspond to the higher order mode groups. This implies that all fibers measured have a fiber shape parameter, q, greater than 2, and show a roughly linear group delay relationship for the lower order modes.

4.1.1 DMD – Direct Excitation of 750 m OM3 Fiber

Fig. 4.1 shows the DMD measurements for the bare fiber spool of OM3 fiber under direct SMF excitation. For the bare fiber spool of OM3 fiber, there is no significant difference in the modal distribution due to offset at the various polarizations for most of the offsets. The only noticeable difference is at the higher offset range where for the y- and circularly polarized case, an extra mode group is able to be excited causing an increase in the DMD of 0.06 – 0.08 ns. This might be due to some local birefringence at the edge of the refractive index profile caused by a fiber defect such as an exponential decay of refractive index instead of a sharp profile transition to the cladding index. This birefringence then allows an additional mode to propagate only for y-polarizations (and thus circular) causing the increase in DMD. A relatively polarization independent mode power distribution is characteristic of all the measurements conducted using a direct excitation of OM3 fiber as seen in previous work [1 – 7] and in the additional results presented in Appendix A.1.

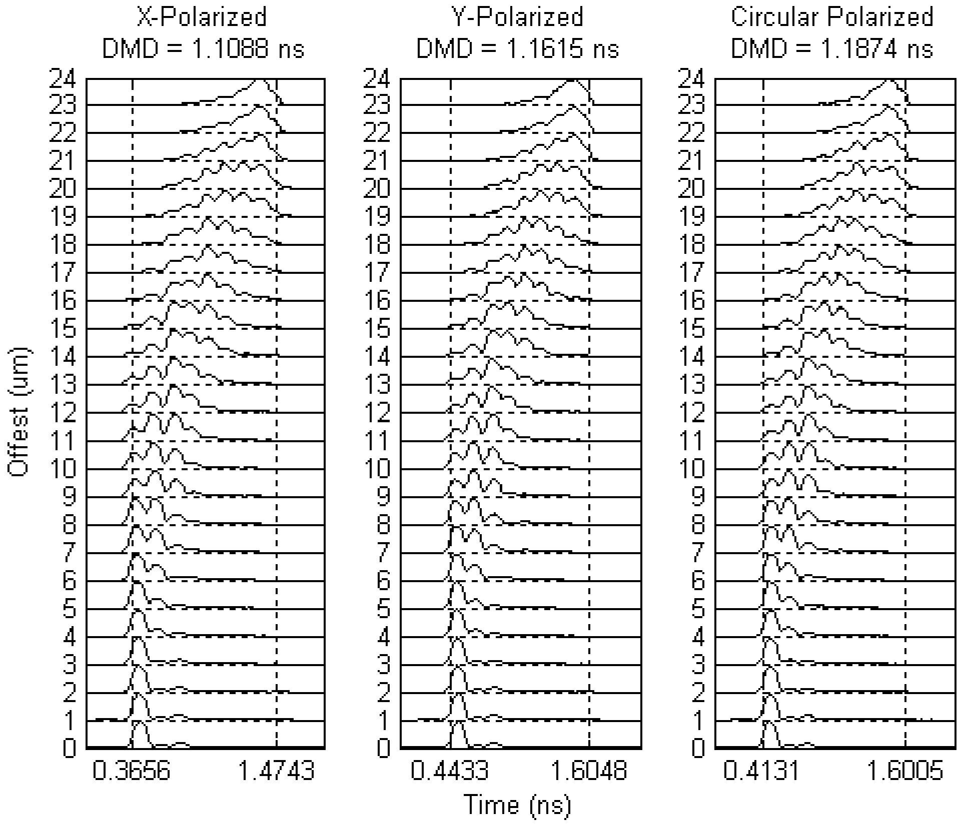

Fig. 4.1: DMD results, 750 m GRIN OM3 fiber.

4.1.2 DMD – Direct Excitation of 1 km FDDI Fiber

The FDDI grade fiber spool is a 1 km long 62.5/125 fiber with a central refractive index dip and refractive index kink. Fig. 4.2 shows the DMD measurements for this fiber. As is seen in Chapter 1.1, an ideal fiber with a power-law profile with a shape parameter, q, greater than 2, the group delays of the mode groups show a roughly linear relationship for the first lower order modes. Therefore, the center dip is apparent because the group delay of the first (fundamental) mode group in the DMD measurement strays vastly in time from a linear relationship with the rest of the mode groups. It is also possible to observe the kink in the refractive index profile because there is another noticeable deviation in the ideal near linear (e.g. see Chapter 1.1) group delay relationship between the second and third mode groups seen in Fig. 4.2. Unlike the OM3 fiber above, this fiber suffers from a relatively strong polarization dependent mode power distribution as can be seen in the middle offset range from 11 – 17 μm. Also, there was only a small 0.02 ns difference in measured DMD between polarization states although there was a noticeable difference in mode power distributions. Additional FDDI fiber measurements can be found in Appendix A.2.

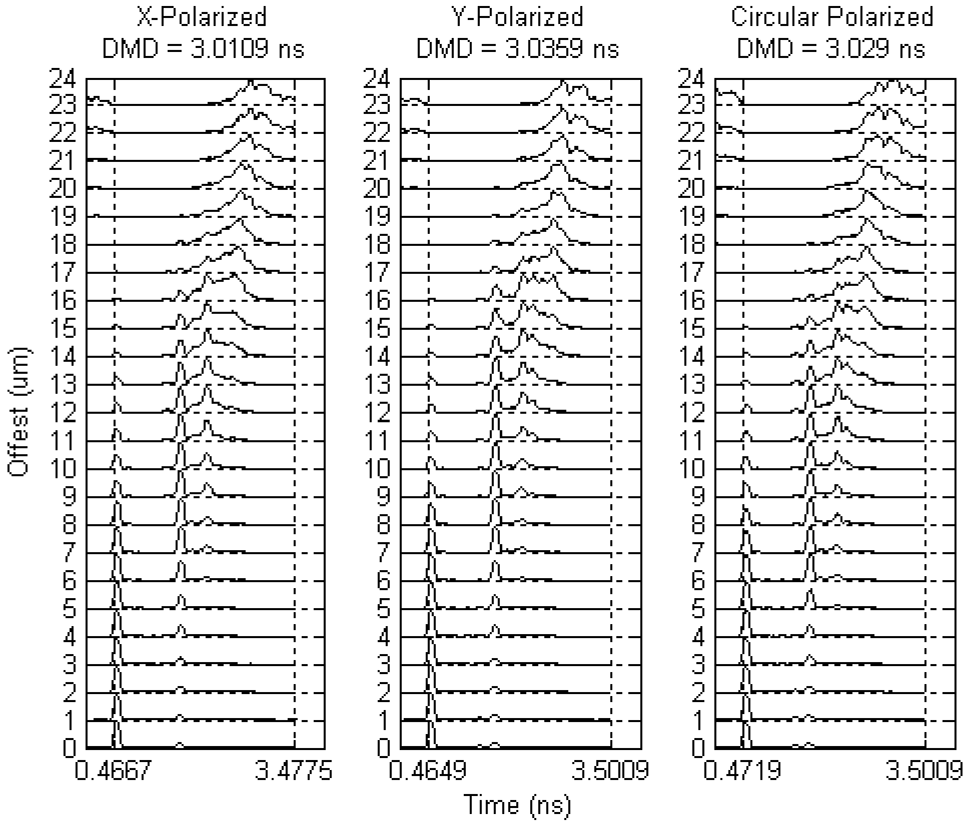

Fig. 4.2: DMD results, 1 km GRIN FDDI fiber.

4.1.3 DMD – 1 m Patchcord – 1 km OM3+ Fiber (Blue)

Fig. 4.3 shows the DMD results for the 1 km OM3+ fiber with SMF excitation first being coupled through a 1 m MMF patchcord. This fiber is close to ideal with a refractive index greater than 2 and has no significant defects. There is again only a 0.02 ns maximum difference in DMD between the three measured polarization states. There is, however, a significant polarization dependence on the modal power distribution as can be easily seen at almost any offset. Even though the powers of the DMD measurements are normalized, the distribution of power among the mode groups changes drastically between different polarizations. For example, at an offset of 10 μm, the x-polarized input has three almost equally excited mode groups while for the y-polarized input has three excited mode groups but with a single one having most of the power. Additional OM3+ DMD measurements can be found in Appendix A.3.

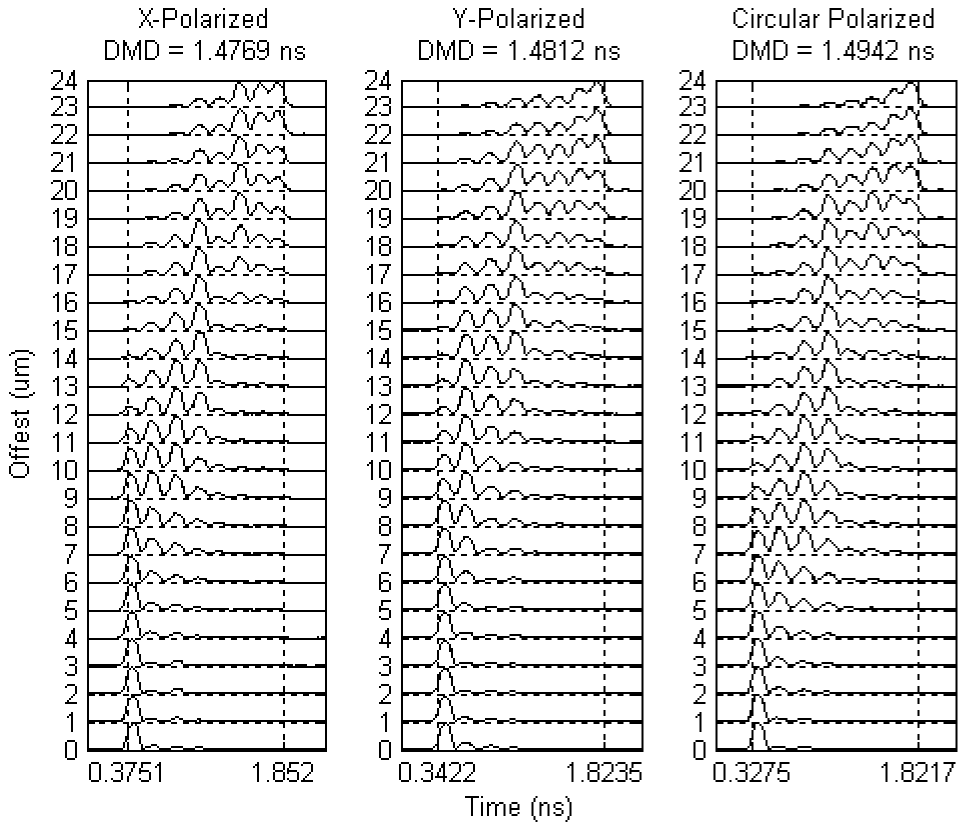

Fig. 4.3: DMD results, 1 km GRIN OM3+ fiber

4.1.4 DMD – 1 m Patchcord – 1 km OM3 Fiber (Orange)

Fig. 4.3 shows the DMD results for the 1 km OM3 fiber with SMF excitation first being coupled through a 1 m MMF patchcord. This fiber is close to ideal with a refractive index greater than 2 and has no noticeable defects. With this fiber as well there is only a maximum 0.03 ns difference in DMD. However, it is easily observed that there is again a significant polarization dependence on the modal power distribution similar to that of the previous OM3+ fiber. One important difference between this fiber measurement and the previous OM3+ fiber is that the circular polarized pulse response exhibits a lower total DMD than that of the y-polarization. The reason for this might be that there is an element of polarization dependent loss somewhere in the final MMF link, selectively attenuating higher order modes in only the x-polarization. This would also account for the lower DMD in the x-polarized pulses as well, since they contain no y-polarized fields. Additional measurements for this fiber can be found in Appendix A.4.

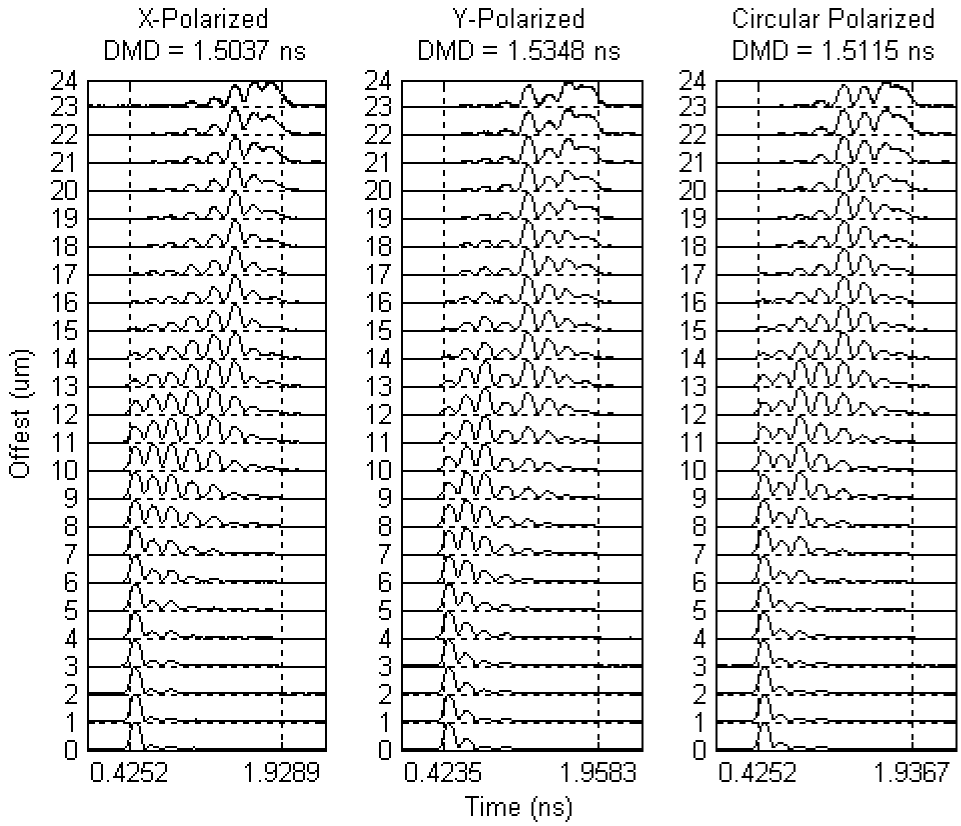

Fig. 4.4: DMD Results, 1 km GRIN OM3 fiber.

4.2 Polarization Results

Each polarization measurement is taken at four different offsets, 0, 7, 14, 21 μm and measures the power in each mode group as the input polarization is rotated from 0 - 240°. Because each system can have up to eight or nine different mode groups present, the results of the polarization dependent coupling is simplified for the sake of clarity to only show three curves: the "Dominant Mode Group" showing how power fluctuates in the mode group that carries the most power; the "Other Mode Groups" which is the summation of the power of the remaining mode groups; and "Total Power" which is the sum of the previous two curves and shows the total polarization dependent loss in the system. The extraction of this information from each of the mode groups in the pulsed output of the multimode FUT is discussed in Chapter 3.6. The polarization controller itself exhibits < 0.1 dB polarization dependent loss. Additional polarization results can be found in Appendices A.5 – A.8.

4.2.1 Polarization – Direct Excitation of 750 m GRIN OM3 Fiber

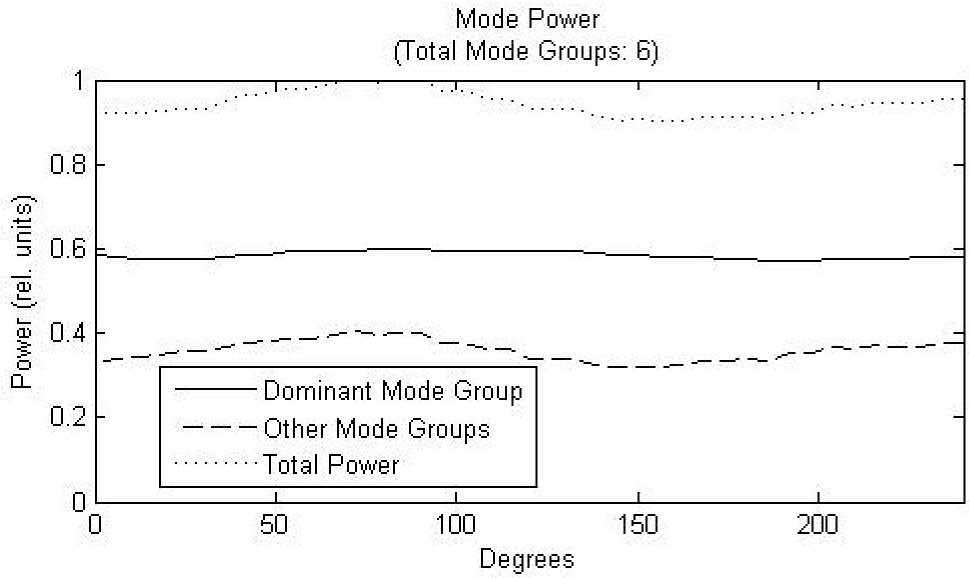

Fig. 4.5: Modal power polarization dependence at a 0 um offset.

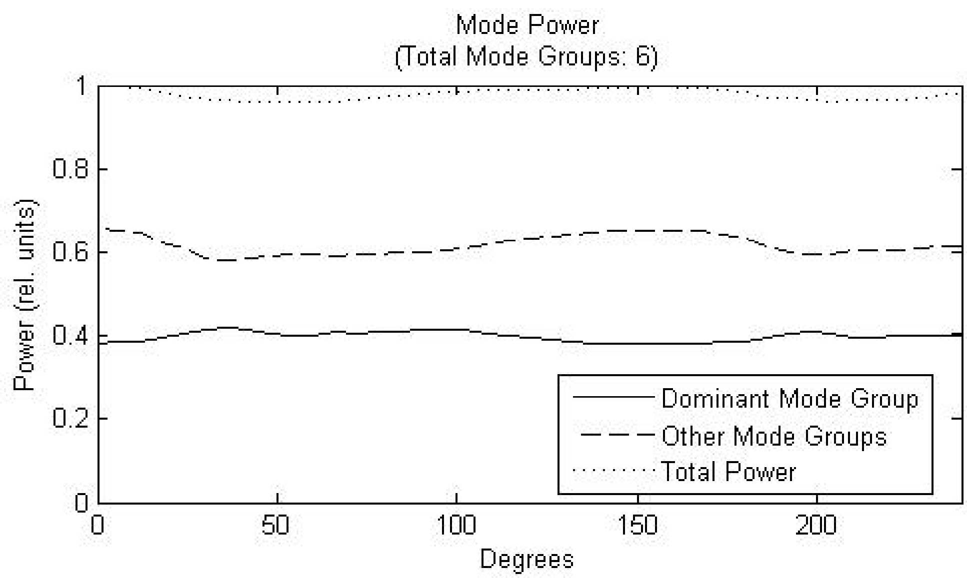

Fig. 4.6: Modal power polarization dependence at a 7 um offset.

Figs 4.5 – 4.8 show the input polarization dependence on the modal power distribution of the OM3 fiber under direct SMF excitation. For each offset there is only a slight variation in modal power distribution with no discernable relation.

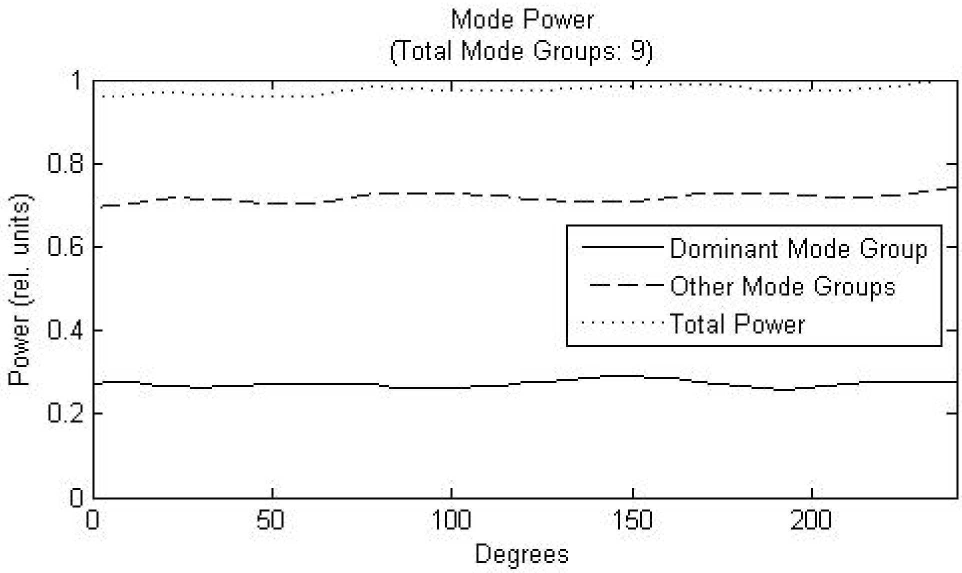

Fig. 4.7: Modal power polarization dependence at a 14 um offset.

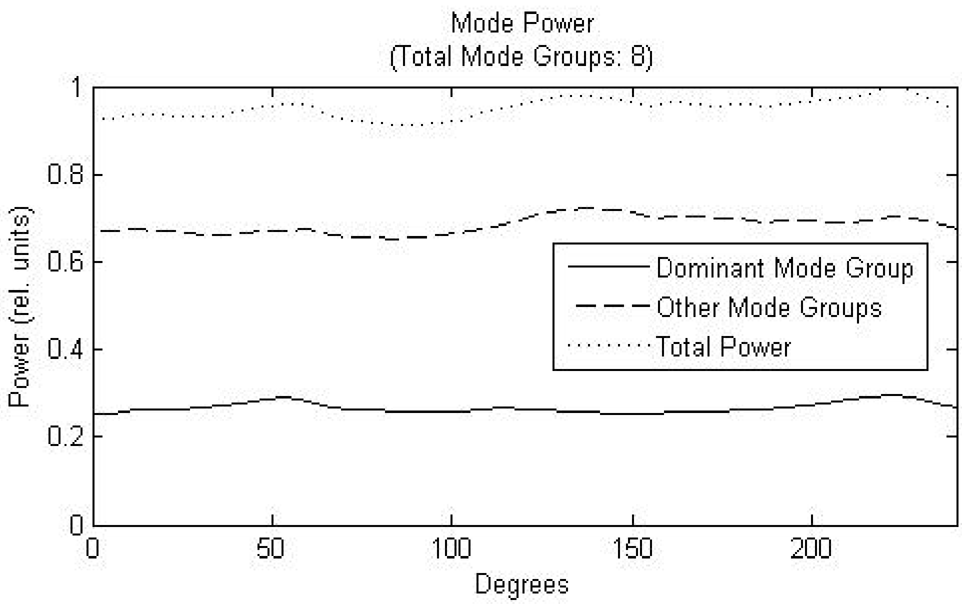

Fig. 4.8: Modal power polarization dependence at a 21 um offset.

4.2.2 Polarization – Direct Excitation of 1 km FDDI Fiber

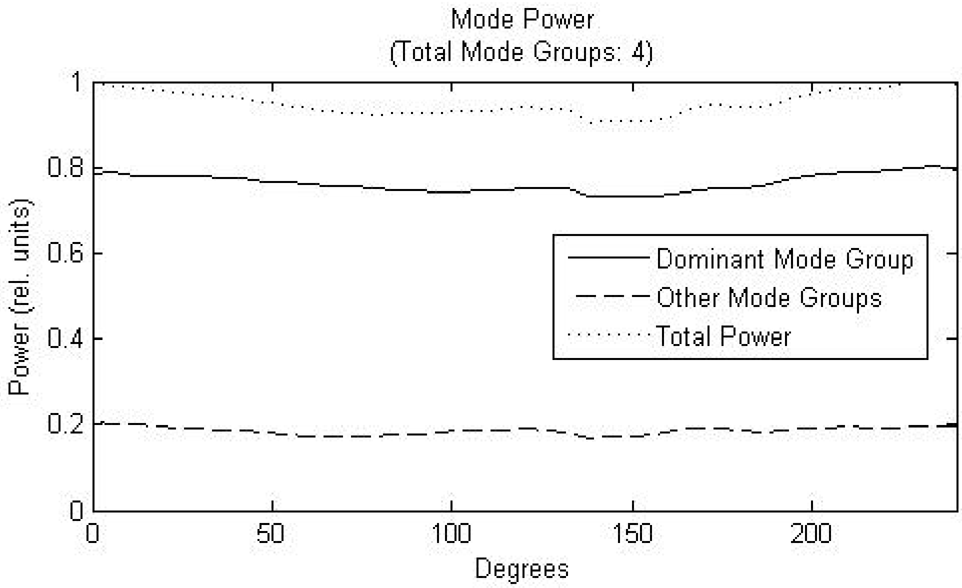

Fig. 4.9: Modal power polarization dependence at a 0 um offset.

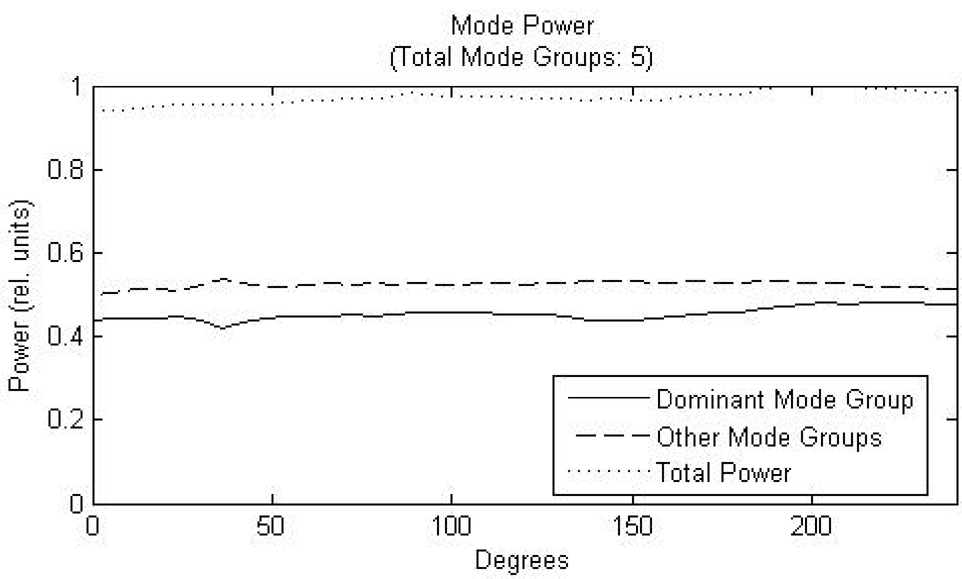

Fig. 4.10: Modal power polarization dependence at a 7 um offset.

Figs. 4.9 – 4.12 show the input polarization dependence of the FDDI fiber. Similar to the OM3 fiber, there is not much polarization dependence and it cannot be easily characterized. However, there can be significant polarization dependence as seen in the DMD measurements above. This dependence then it seems is not due to power fluctuation in each mode group, but due to power being transferred between modes with similar but not identical group delays, causing the received pulses to drift as the polarization is rotated. This is demonstrated in Fig. 4.13 where the pulse shown is the fundamental mode excited by a SMF input at an offset of 7 µm.

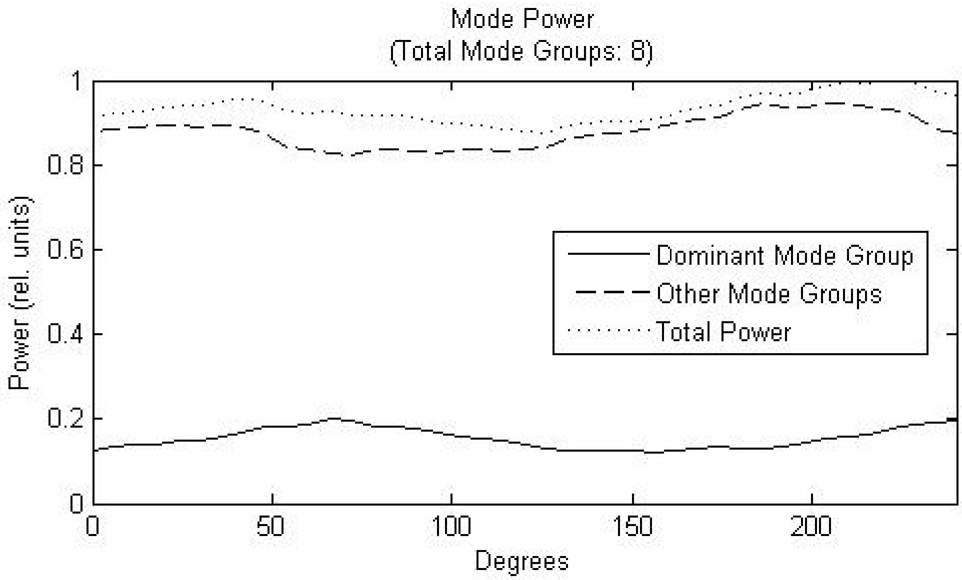

Fig. 4.11: Modal power polarization dependence at a 14 um offset.

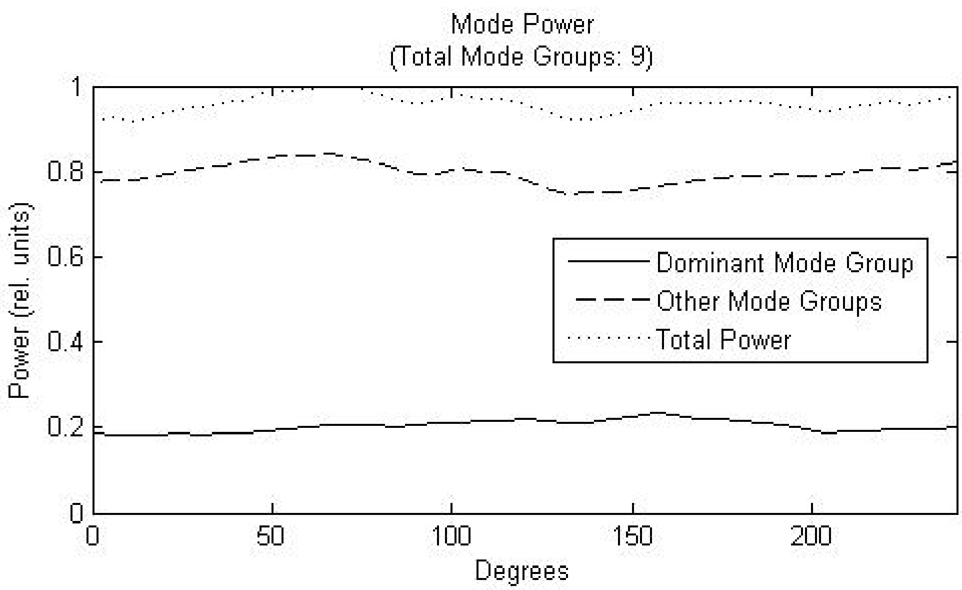

Fig. 4.12: Modal power polarization dependence at a 21 um offset.

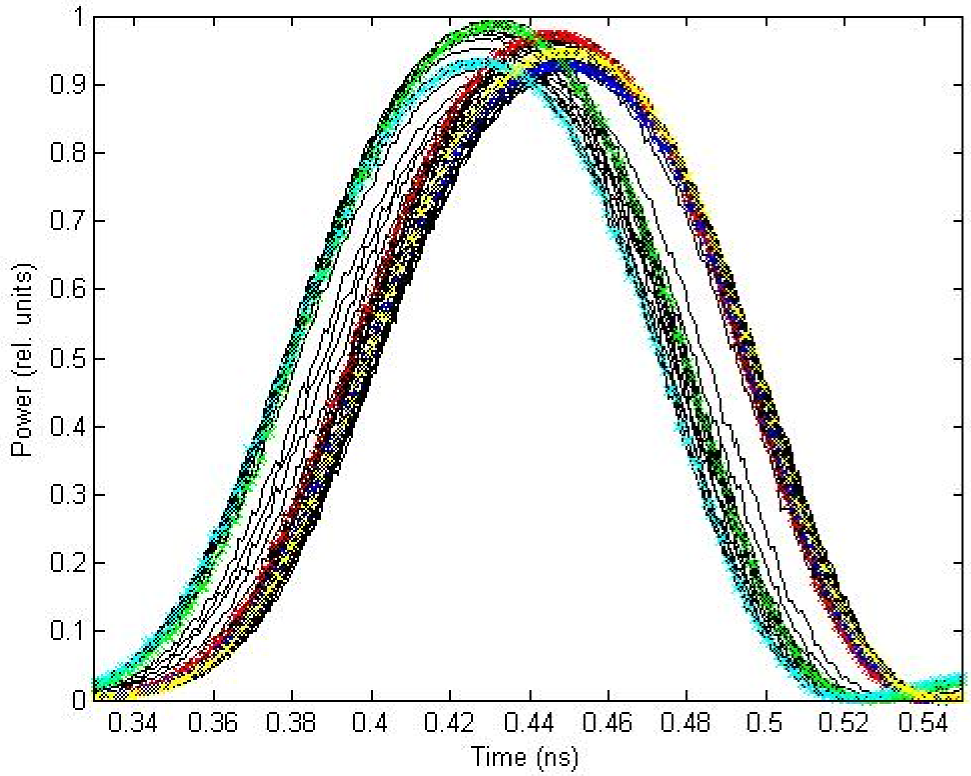

Fig. 4.13: Fundamental mode group of FDDI fiber with polarization rotation. Red = 0^o, yellow = 60^o, blue = 120^o, cyan = 120^o, green = 240^o.

Fig. 4.13 shows the fundamental mode excited in the FDDI fiber by a SMF launch offset by 7 μm. Over the course of the input polarization rotation, it is clear that the relative position of the pulse drifts in time. This polarization dependent temporal drift is caused by the excitation of the non-degenerate x- and y-polarized modes in the fiber. Since these non-degenerate polarization modes have different group delays, power in one polarization will arrive at a different time as another, and as the input polarization is rotated, different powers will be coupled into each mode, causing the pulse at the end of the fiber to drift.

4.2.3 Polarization – 1 m Patchcord – 1 km OM3+ Fiber (Blue)

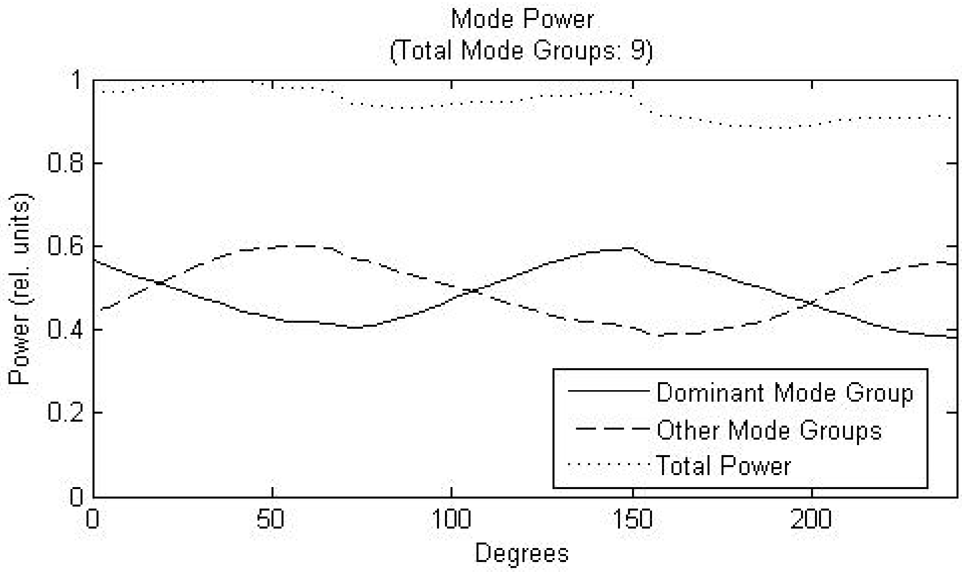

Fig. 4.14: Modal power polarization dependence at a 0 um offset.

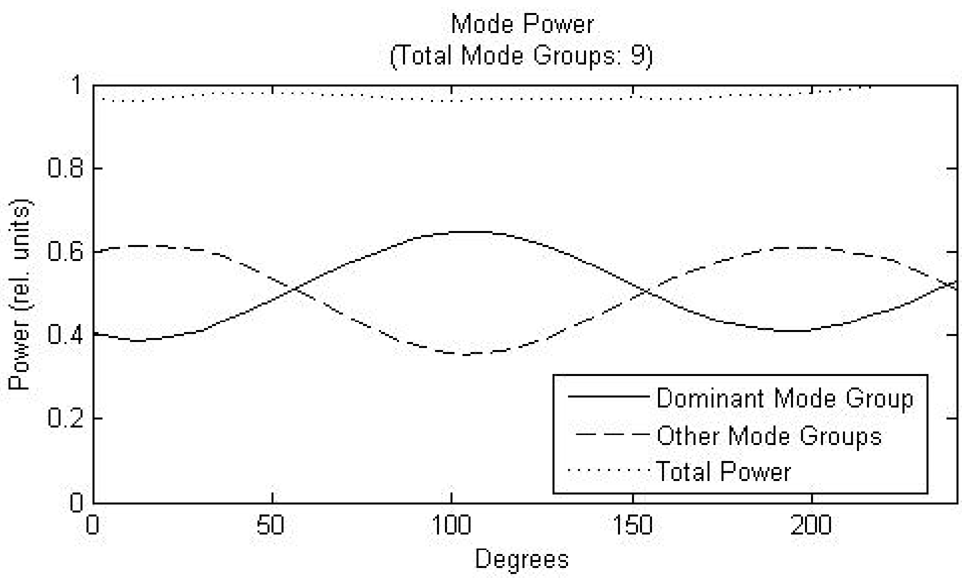

Fig. 4.15: Modal power polarization dependence at a 7 um offset.

Fig. 4.14 – 4.17 shows the input polarization dependence of modal power of a 1 km length of OM3+ fiber excited by a 1 m MMF patchcord that is first excited by a SMF offset launch. Unlike the direct excitation of the fiber, these figures show that modal power has a very clear dependence on input polarization. In each of these figures there are nine mode groups present but again for clarity the dominant mode group is shown with the rest being grouped together. Both the dominant and other mode group curves in each figure display a sin(2α) mode power dependence on input polarization (where α is the input polarization angle).

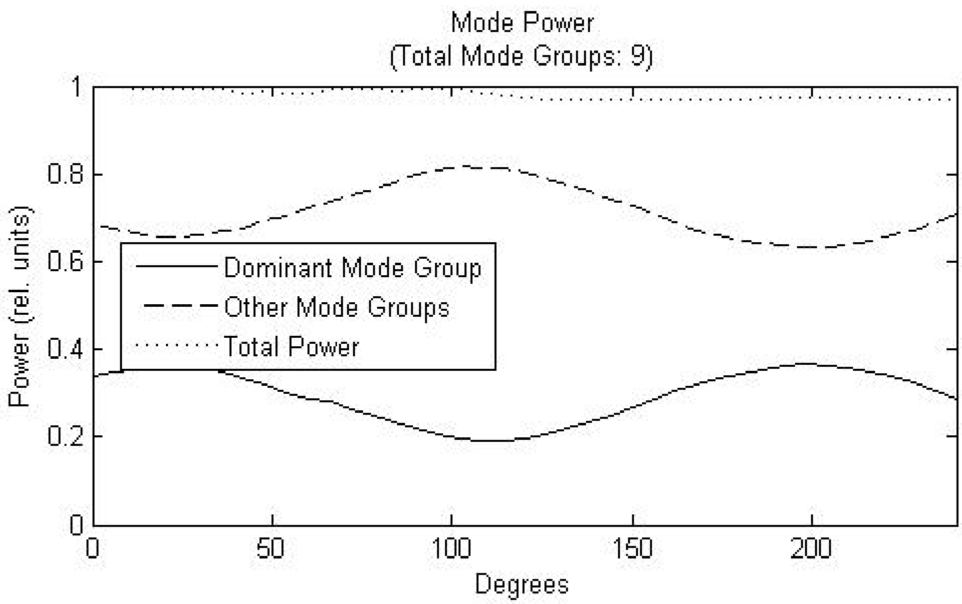

Fig. 4.16: Modal power polarization dependence at a 14 um offset.

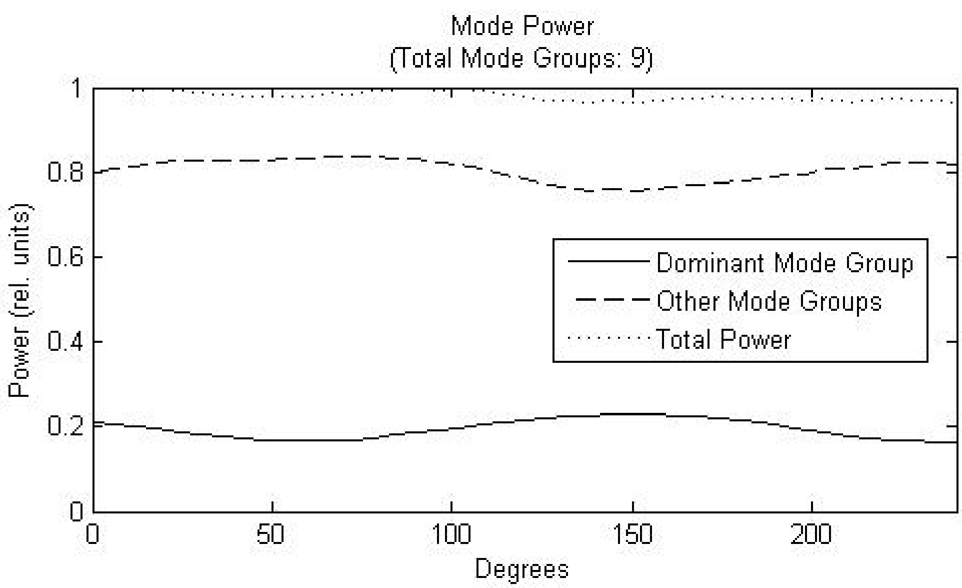

Fig. 4.17: Modal power polarization dependence at a 21 um offset.

4.2.4 Polarization – 1 m Patchcord – 1 km OM3 Fiber (Orange)

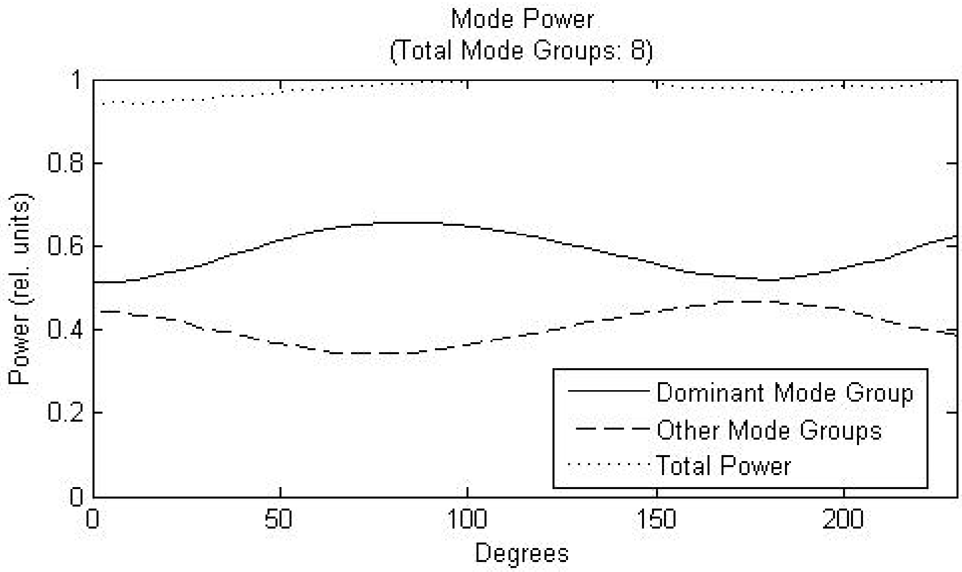

Fig. 4.18: Modal power polarization dependence at a 0 um offset.

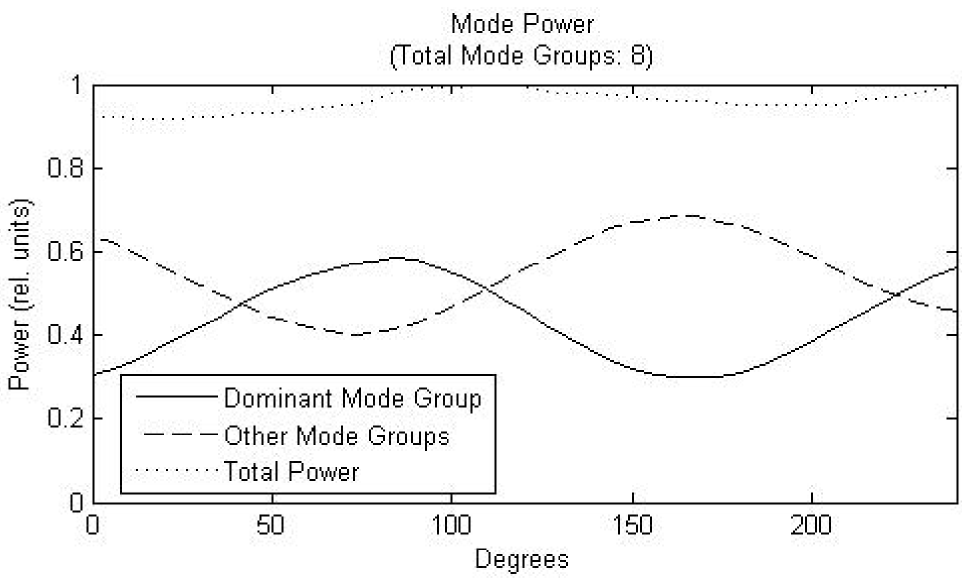

Fig. 4.19: Modal power polarization dependence at a 7 um offset.

Figs. 4.18 – 4.21 show the modal power dependence on input polarization for the 1 km OM3 fiber excited by a 1 m MMF patchcord that is first excited by a SMF offset launch. Again, the power in each mode group displays a sin(2α) dependence (where α is the input polarization angle). For this setup, as power in the dominant mode group decreases with rotating input polarization, more power is coupled into the other mode groups and vice versa. To get a difference perspective on this, Fig. 4.22 shows the pulses corresponding to the first two mode groups for a zero offset SMF launch into the 1 m patchcord and then into the 1 km OM3 fiber.

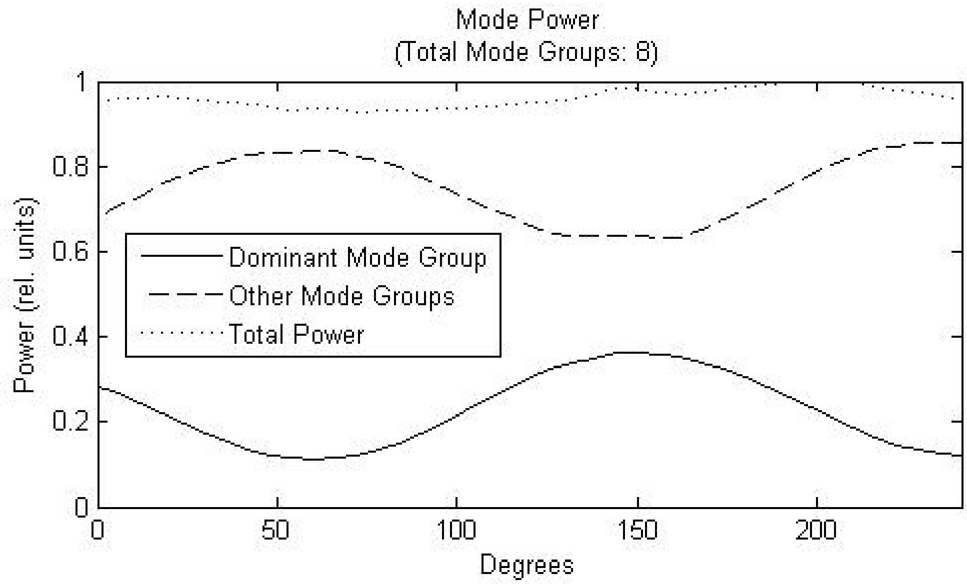

Fig. 4.20: Modal power polarization dependence at a 14 um offset.

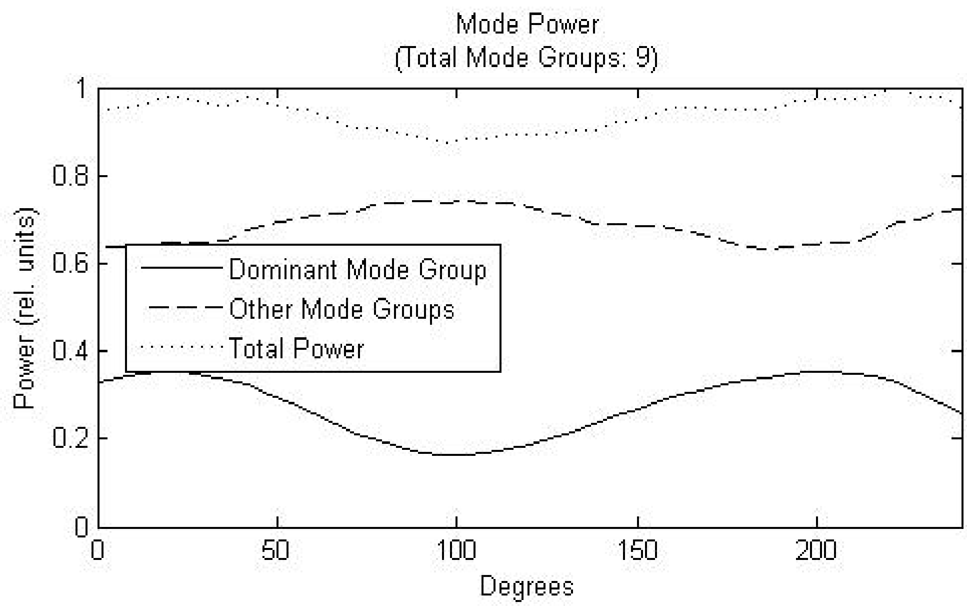

Fig. 4.21: Modal power polarization dependence at a 21 um offset.

Fig. 4.22 shows the first two pulses received after the 1 km OM3 fiber excited by an offset SMF launch through a 1 m MMF patchcord as the input polarization is rotated. Unlike the case of the FDDI fiber, it is apparent that this fiber is not at all birefringent as the pulses do not drift with polarization; the only effect is that as the input polarization rotates power is clearly being coupled from the first pulse into the second.

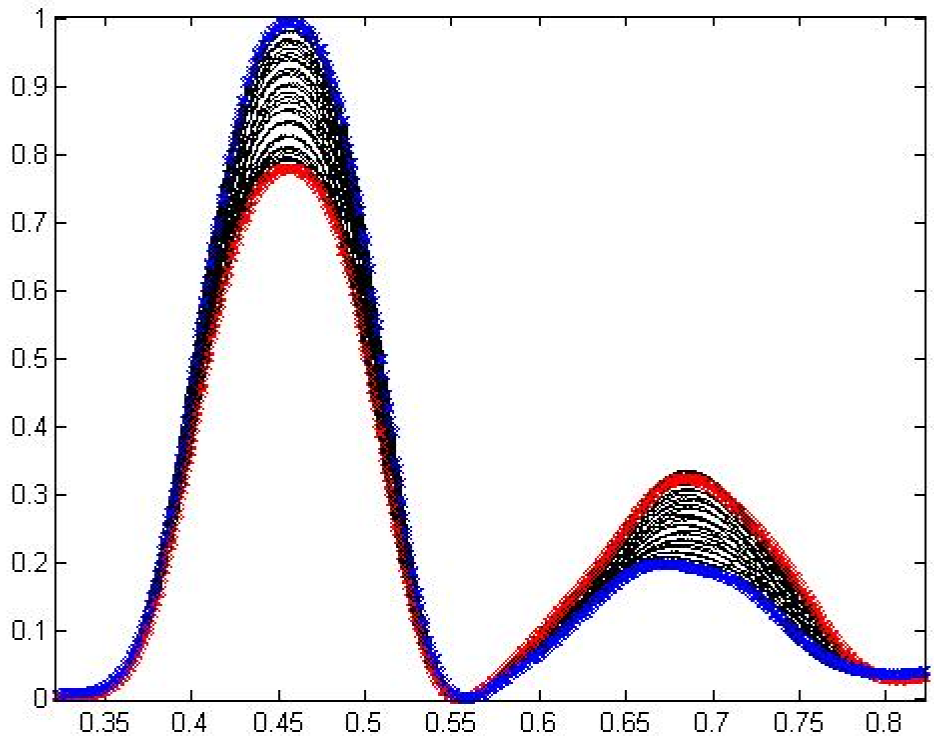

Fig 4.22: First two pulses in the 1 km OM3 fiber setup as polarization is rotated. Red = 0^o, blue = 90^o.

4.2.5 Polarization – 1 m Fusion Spliced Fiber – 749 m OM3 Fiber

Figs. 4.23 – 4.26 show the modal power dependence on input polarization for the 749 m OM3 fiber excited by a 1 m MMF section that was cut from the original 750 m spool and then fusion spliced back on to the final length of fiber. This fiber is first excited by a SMF offset launch. Similar to the previous fiber experiments, the presence of a fusion splice in an otherwise near ideal fiber creates the same kind of polarization dependent mode coupling that we see above when there is a patchcord present. This suggests that any kind of perturbation that causes mode coupling to be present at a specific point will result in this polarization dependence.

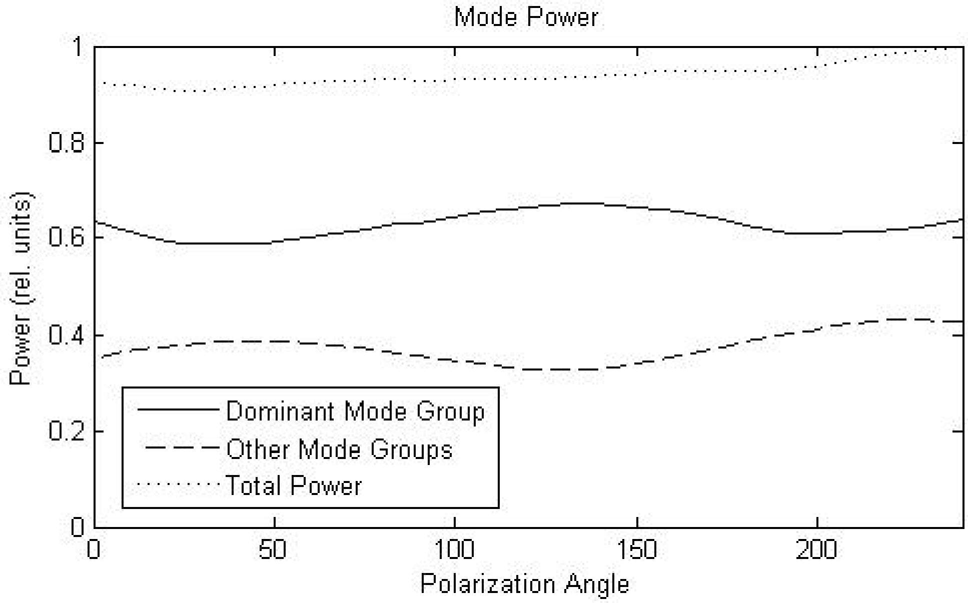

Fig. 4.23: Modal power polarization dependence at a 0 um offset.

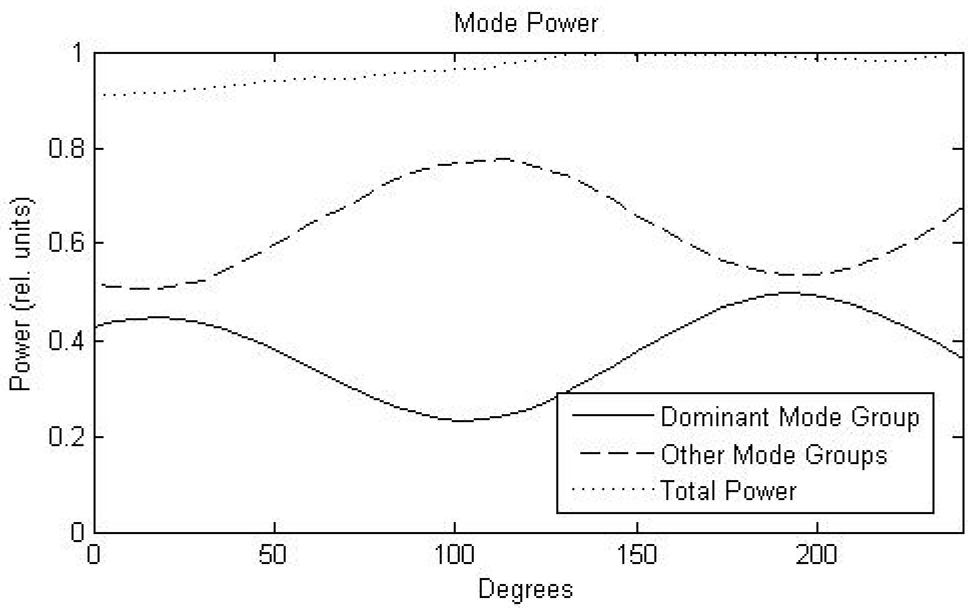

Fig. 4.24: Modal power polarization dependence at a 7 um offset.

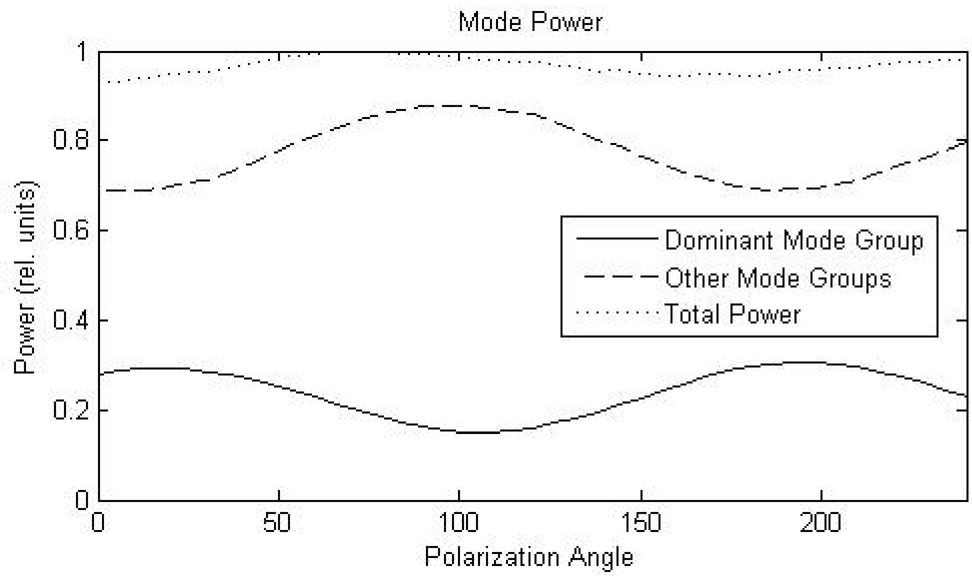

Fig. 4.25: Modal power polarization dependence at a 14 um offset.

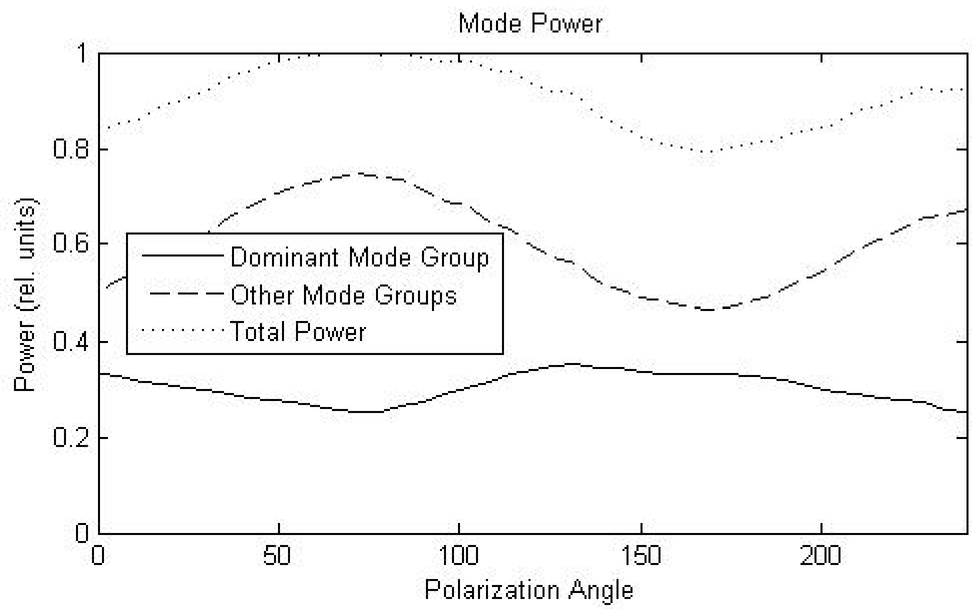

Fig. 4.26: Modal power polarization dependence at a 21 um offset.

4.3 Discussion of Results

The DMD measurements in section 4.1 show that there is a clear polarization dependence on pulse response for all the fibers except the OM3 fiber that is directly excited by the offset SMF launch. The direct excitation of the FDDI fiber yielded polarization dependence due solely to the birefringence inherent in the fiber and there was no significant power difference between mode groups. However, for the OM3/3+ fibers that were excited there was a significant amount of polarization based mode power dependence. Looking again at the DMD measurements it is easy to say that there was no significant mode coupling happening throughout the OM3/3+ fibers as each mode group arrived at the receiver as a distinct pulse separated almost linearly according to its group delay. From this we conclude that the polarization dependence on mode power must be occurring at the beginning of the fiber system. Also, since there was no mode group power dependence for the fiber spools that were not excited by the MMF patchcord or that had a fusion splice in them, we conclude that the polarization dependence is due to the MMF patchcord or presence of a fusion splice. To ensure that this was not unique to a single patchcord both the OM3/3+ fibers were tested with different patchcords (as were the extra fibers in the Appendices) and identical results were obtained. Because this polarization dependence was not due to a specific patchcord and is also seen when there is a fusion splice in an otherwise near ideal fiber, it is then concluded that the polarization dependence arises at the 1 m patchcord – 1 km MMF fiber connector or fusion splice and is due to polarization dependent mode excitation based on the broken symmetry because of the perturbation introduced in the fiber by the fusion splice or the mismatched parameters of the fiber patchcord.. The theoretical work in the next chapter is based upon this observation. Table 4.1 contrasts the current experimental results with that of previous studies regarding polarization dependence in MMF.

Table 4.1: Comparison of previous observations and current experimental results on polarization dependence.

Previous Polarization Observations	Current Experiments	Reference
No dependence with direct excitation of high quality (OM3/3+) fiber.	Confirmed in current experiment.	[1 – 3]
Large dependence with direct excitation of low quality (FDDI) fiber.	Confirmed in current experiment.	[1 – 7, 8]
Large dependence when patchcord used over any quality fiber.	Confirmed in current experiment.	[1 – 3]
Large dependence when fusion splice present in high quality fiber.	Confirmed in current experiment.	[1 – 3]
Due to birefringence in fiber breaking mode degeneracy resulting in polarization based mode coupling.	Confirmed, but power coupling between modes only when patchcord or fusion splice used.	[8]
No power is coupled between mode groups.	Current experiment shows strong power coupling between groups.	[8]
Aggregate effect throughout the length of the fiber.	No pulse spreading shows that the effect occurs at beginning of fiber.	[8, 9]
Due to fiber parameter mismatch and fiber in high coupling regime.	Fibers used were not perturbed; in low coupling regime.	[9]

These experimental results also confirm some of the observations made in [1 – 7], namely that there is very minimal polarization dependence in higher quality fiber under a direct SMF – MMF launch (at any transverse offset) and that there can be quite a strong response when using a lower quality FDDI grade fiber. Also, they confirm the observation that there is a significant polarization dependent response for all fibers when there is a MMF patchcord inserted between the SMF launch and the final fiber under test.

There is, however, a significant conflict between some of the results mentioned in [8] as discussed in Chapter Two. The study in [8] says that there can only be a polarization dependent response when there is significant inherent birefringence in the fiber, breaking the degeneracy of the modes within mode groups. The results presented above, however, show that while there is dependence in the presence of fiber defects, there is a greater polarization effect when all fibers are near ideal and there is a patchcord between the SMF launch and the final MMF (where there is no polarization dependence under direct SMF excitation). Furthermore, this dependence is not due to birefringent fiber, because as the polarization of the input field is rotated, there is no timing jitter in the received pulses through the fiber and the polarization dependence is in actuality a polarization dependent mode coupling happening at the beginning of the fiber link. The previous study [8] also says that no power is coupled between fiber mode groups as the polarization is rotated, but we have shown above that there is indeed significant power flow between mode groups. This power flow in one mode group can be measured as having a $a.\cos^2(\alpha) + b.\sin^2(\alpha)$ (with α the input polarization angle and a and b being different power coefficients) dependence with power going into the other mode groups present.

In addition, contrary to the theoretical findings of [9], it appears that the major polarization effect occurs not as a result of intrinsic accumulative PMD effects in the many multimode fiber segments, but at the beginning of the fiber link where the launch condition is crucial.

Chapter 4 References

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Chapter Five
Theoretical Setup

The two main experiments conducted consisted of testing spools of fiber directly coupled to a SMF input and testing spools where the SMF was coupled first to a MMF connector which was then coupled to a longer spool. These two cases show very different results with regards to any polarization dependence. When the SMF is directly coupled into a MMF spool, if the MMF is high quality OM3/3+, then there is no polarization dependence, but if it is lower grade OM2 or FDDI fiber, there is a strong response to input polarization. However, when there is a MMF connector between the SMF and the final MMF spool or a fusion splice present in the fiber, regardless of fiber quality, a strong polarization response is seen.

This chapter deals with the constructing a theoretical model that explains the above phenomena. It will describe three models that explain how a polarization dependence can exist when exciting multimode fiber from a linearly polarized singlemode input and also how it is possible for no dependence at all.

5.1 Problem Description and Coordinate Systems

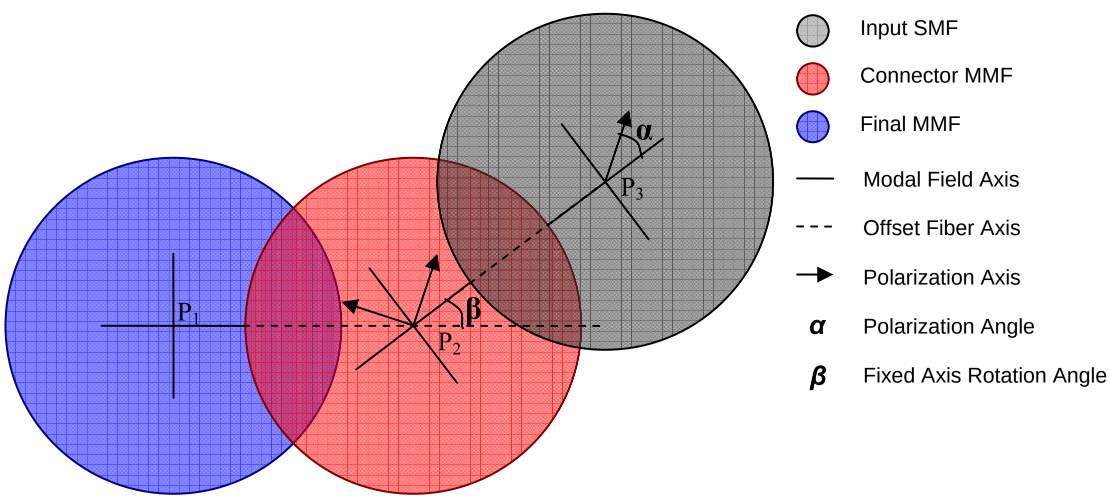


Fig 5.1: Theoretical setup of three offset fibers.

From the experimental results there are some conclusions that can be drawn. The first is that the polarization dependence results in mode coupling that can only be present at the beginning of a fiber system and the second is that this mode coupling takes place due to two offset connector fibers. From these we set up the simple theoretical problem of a two connector system consisting of a singlemode offset launch and a short Connector fiber coupled to the Final MMF.

Every problem consisting of three offset fibers can be reduced to the situation as shown in Fig. 5.1. Each of these fibers has its own axis (centered on points P_1 (x_1,y_1), P_2 (x_2,y_2), or P_3 (x_3,y_3)) and each has its own local coordinate system (r, θ), (r', θ'), (r'', θ''), respectively. The global coordinate system used is centered on P_1 with coordinates (x, y) and each local coordinate system is related to the global system by a simple rotation and translation. The local coordinate systems are used for the drawing of the modal fields present in each fiber given a certain local axis defined by its relative coordinates (e.g. (r', θ')). The rotation of any local system with respect to the global (at P_1) is dependent on the relative position of the fiber preceding the current system. That is, the local coordinate system of a fiber is created based on the offset of the fiber coupled to it. For example, the local coordinate system of the Connector MMF in Fig. 5.1 is centered on the fiber axis and is aligned along the offset between the Input SMF and the Connector MMF. Similarly, the local system of the Final MMF is centered on the fiber axis but aligned along the offset between the Connector MMF and the Final MMF. Given this, the three local coordinate systems are related to the global system by:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan} 2\left(\frac{y}{x}\right); \tag{5.1}$$

$$r' = \sqrt{(x - x_2)^2 + (y - y_2)^2}, \quad \theta' = \operatorname{atan} 2\left(\frac{y}{x - x_2}\right) - \beta; \tag{5.2}$$

$$r'' = \sqrt{(x - x_3)^2 + (y - y_3)^2}, \quad \theta'' = \operatorname{atan} 2\left(\frac{y - y_3}{x - x_3}\right) - \beta. \tag{5.3}$$

where β = atan2(y_3 / (x_3 - x_2)). When doing integration over these systems, the integration is always carried out with respect to the global coordinate system (x, y) using coordinates (r, θ).

A fundamental assumption made in the construction of this model is that the fiber is sufficiently long such that each mode group can be resolved individually. This is shown experimentally for the OM3/3+ results in the DMD experiment in section 4.1 where there is no overlap between different mode groups. This allows us to neglect effects that would inevitably arise due to the interference between the modal fields of different mode groups.

5.2 SM – MM Fiber Excitation

As was shown through the experiment, as well as through other work [1-7], when the multimode fiber being coupled into is of high quality, i.e. OM3/3+, no polarization dependence is exhibited. However, when the MMF used is a lower OM2 or FDDI grade, there is a strong polarization dependence. The major difference between the two grades of fiber is the amount of birefringence present in each fiber as well as local fiber defects. It is this birefringence that accounts for most of the polarization dependence in multimode fiber, however, near the local defects (e.g. a center dip or peak), there can be increased dependence as well.

5.2.1 Ideal Fiber Excitation

For OM3/3+ fibers, advances in manufacturing techniques have allowed for the defects present in the fiber to be drastically reduced, including any birefringence [1, 4, 6]. Because of the near ideal nature of these high grade fibers and their near perfect circular symmetry, no set fiber axis is present in the fiber. Due to this circular symmetry, any chosen fiber axis is completely arbitrary. In single mode fibers this is not an issue because there is no azimuthal dependence on the fiber electric field distribution and the field can be drawn along the same axis as it is polarized with no noticeable effect (over the distance and bit rates of multimode fibers at least). In multimode fibers however, at least one mode in every higher order mode group has a field distribution that is dependent along the azimuth coordinate. This means that the choice of a fiber axis along which all the modal field distributions are drawn is very important and as will be shown later is part of the reason for the polarization dependence in multimode fiber.

Now, when dealing with the excitation of an ideal multimode fiber by a single mode fiber, there should be no polarization dependence exhibited. This arises because of the choice of fiber axis. In an ideal fiber the choice of axis is generally arbitrary, but in the case of SMF – MMF coupling, the axis is set along the direction of the offset between the two fiber centers. This can be seen in Fig 5.2. An axis created in such a way is now the set axis along which the fiber modal electric field distributions are *drawn*.

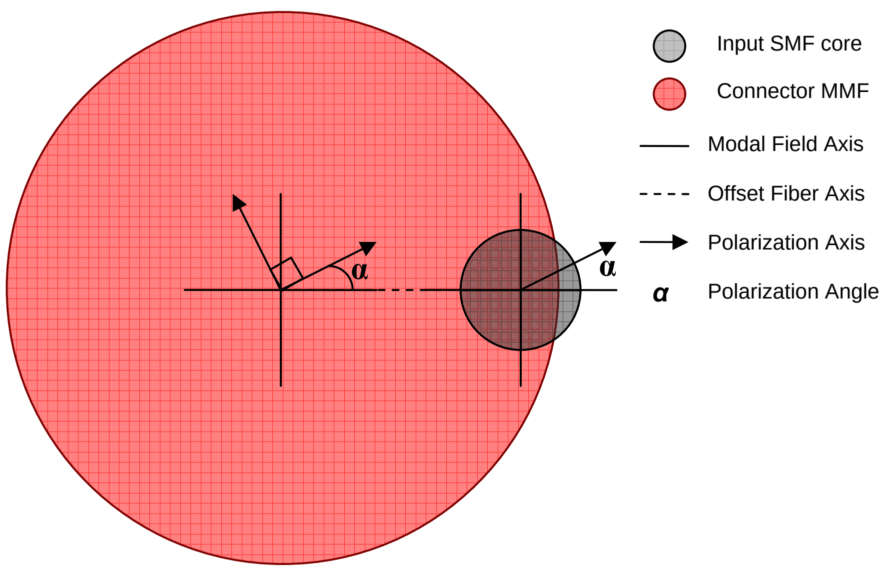


Fig 5.2: SMF Excitation of ideal MMF.

With the ideal MMF, we choose the degenerate modes (modes with the same propagation constant and group delay) and place them into mode groups. When solving the scalar wave equation in polar coordinates, this occurs when they have the same mode group number given by $N = 2m + l - 1$ [8-10]. Moreover, since the fiber is ideal, the x- and y-polarized modes of the same group number are also degenerate. We can then construct LP^{lm} vector modes by grouping these degenerate x- and y-polarized modes and using the methods in [8-10] we end up with:

$$\boldsymbol{\psi}^{lm}(r,\theta) = \begin{bmatrix} \cos(l\theta) \\ \cos(l(\theta - \pi/2)) \end{bmatrix} \psi^{lm}(r) \tag{5.4}$$

where $\psi^{lm}(r)$ are the solutions to the scalar wave equation in polar coordinates. These vector modes are then used as an orthonormal basis of the fiber where,

$$\iint_{r\,\theta} \boldsymbol{\psi}^{lm} \bullet \boldsymbol{\psi}^{jk}\, r\, dr\, d\theta = \delta_{lj}\delta_{mk}, \tag{5.5}$$

with δ_{jk} as the Kronecker delta function. The polarization of each one of these vector modes is not easily described except for the simplest case, that of the fundamental mode, where the polarization state is simply pointed along the vector **x** + **y** and shaped by the field distribution over all space. Fig 5.3 shows the x- and y-polarized field distributions and total polarization states for some of the lower order modes.

With these modes constructed, their polarization state (or polarization axis) can easily be rotated while the axis on which their field distributions (modal field axis) are drawn remains the same. If this fiber had no set axis, the polarization axis, as seen in Fig. 5.2, would rotate along with the modal field axis, causing the polarization state to rotate along with the modal field distributions. This is demonstrated for the LP^{11} mode in Fig. 5.4. This happens if the two fibers were perfectly aligned, as the polarization state of the input SMF rotated, both axes would be able to freely rotate because of the circular symmetry of the fiber. Once that symmetry is broken, e.g. by offsetting the fibers (creating a fiber offset axis), the modal field distribution axis can no longer freely rotate to cause the polarization state to rotate. However, the polarization state of the LP vector modes is still free to rotate.

To facilitate this polarization rotation along a fixed axis, we simply multiply every LP^{lm} vector mode by a rotation matrix

$$\mathbf{R}(\alpha) = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \tag{5.6}$$

where α is the angle of the input SMF polarization relative to the established modal field axis and denotes a counterclockwise rotation. To simplify the notation, *all rotations are done with respect to the local coordinate system of the vector being rotated.* Fig. 5.5 now shows the static field distributions after they have been rotated using $\mathbf{R}(\alpha)\boldsymbol{\psi}^{11}(r,\theta)$. Keep in mind however that these two figures are just an example of the different ways the polarization state can rotate (relative to mode field axis) and there is no connector offset considered in either of the figures.

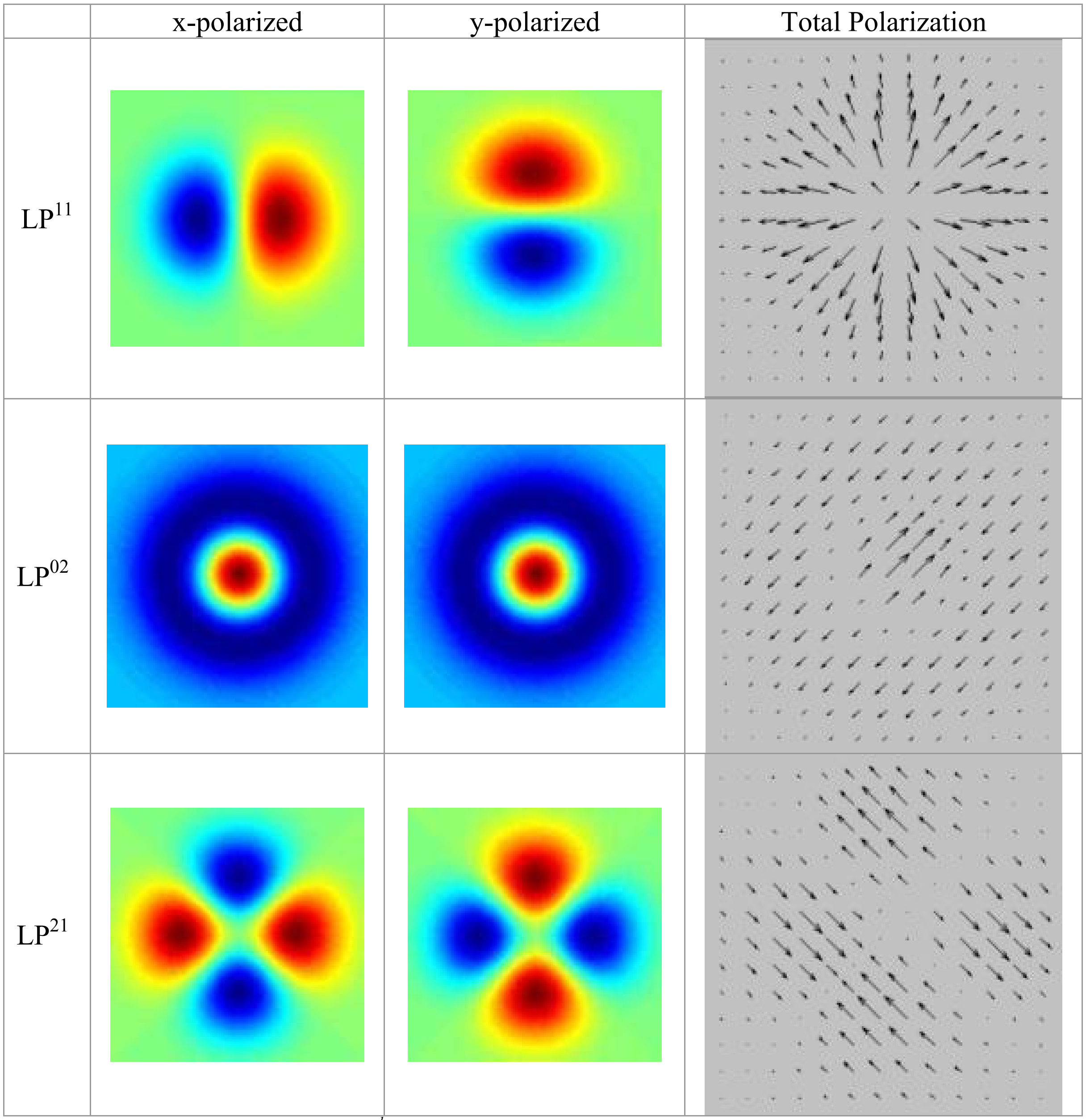
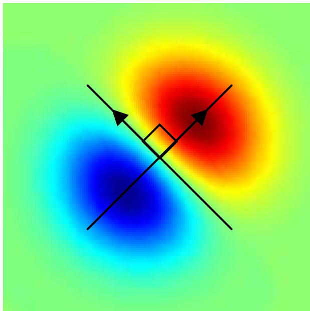
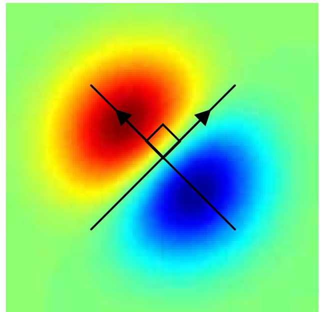
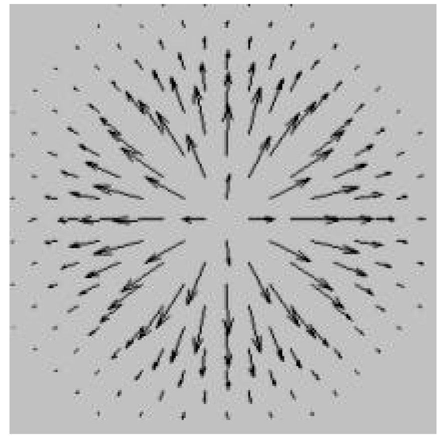
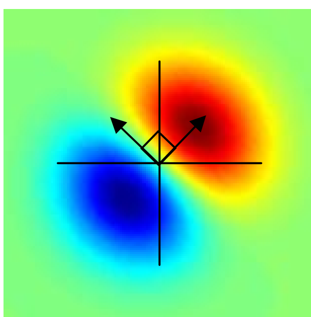
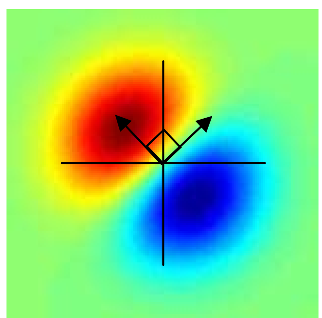
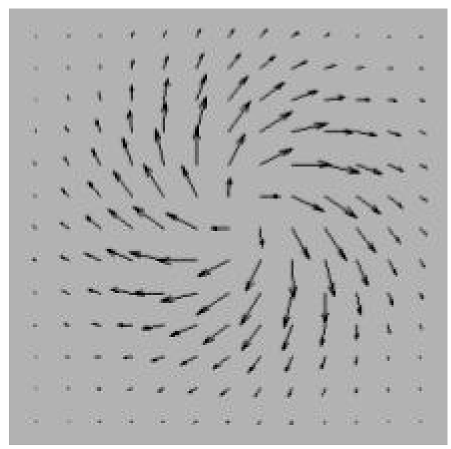


Fig. 5.3: Some LP^{lm} vector modes and their total polarization states.

	x-polarized	y-polarized	Total Polarization
LP11			

$$\boldsymbol{\psi}^{11} = \begin{bmatrix} \cos(l\theta - \alpha) \\ \cos(l(\theta - \pi/2) - \alpha) \end{bmatrix} \psi^{11}(r)$$

Fig 5.4: LP11 Field distributions and polarization state with modal field and polarization axis rotated by45°

	x-polarized	y-polarized	Total Polarization
LP11			

$$\boldsymbol{\psi}^{11} = \mathbf{R}(\alpha) \begin{bmatrix} \cos(l\theta - \alpha) \\ \cos(l(\theta - \pi/2) - \alpha) \end{bmatrix} \psi^{11}(r)$$

Fig 5.5: LP11 Field distributions and polarization state with only polarization axis rotated by 45°.

The field distributions in Figs. 5.4 and 5.5 are identical as a result of the rotation, but since the polarization has rotated around a constant field axis in Fig. 5.5 (indicated by the matrix rotation R), the polarization results between the two figures are different. With modes lacking an azimuthal dependence (such as the fundamental LP01 mode), rotating the axes along which the modes are drawn and rotating just the polarization state are equivalent. The significance of this result becomes apparent when determining the total power in each mode. When coupling power from one field to another, the modal field coefficient in each resultant mode is found through an overlap integral [11]:

$$k^{lm} = \iint_{r\theta} \boldsymbol{\psi}^{lm} \bullet \boldsymbol{\varphi}\, r\, dr\, d\theta. \tag{5.7}$$

where $\boldsymbol{\varphi}$ is the field of the input SMF and integration is always done over the global coordinate system used as in equation (5.1). The total power in each mode is then found by $|k^{lm}|^2$. When coupling from one fiber to another, the polarization of both the excited fields in the MMF and the input field of the SMF have to remain the same assuming the glass of the fiber is linear and isotropic [12]. When two fibers are coupled together where there is an offset between the two, the symmetry of this system is now broken creating an axis between the two fibers. Given this setup of a constant axis created by the offset between the two fibers (the offset axis), the axis upon which the full LP vector modes are polarized (the polarization axis) can be rotated in two ways. First, by rotating the axis upon which the modal fields are drawn (the field axis), essentially changing the direction the **x** and **y** unit vectors are pointing (e.g. Fig. 5.4); second, by rotating the polarization axis relative to the field (or offset) axis. To illustrate this point and to show how it is possible for there to be no polarization dependence, we consider three possible models for the excitation of modal fields in MMF, as shown in Fig. 5.6:

1. Both the polarization and modal field axes of the MMF are determined solely by the polarization state of the input SMF. That is, they are allowed to rotate freely but must follow the polarization axis of the exciting SMF as its polarization state rotates and must be aligned with one another.
2. Both the polarization and modal field axes of the MMF are fixed and aligned along the vector of the offset between the two fibers.
3. The modal field axes of the MMF and SMF are fixed along the vector of the offset between the two fibers but their respective polarization axes are allowed to freely rotate but must be pointing along the same vector, determined by the SMF polarization state.

For the remainder of this section, the coordinate systems used are slight modifications to those laid out in section 5.1 to account for the simplified two fiber examples:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}\left(\frac{y}{x}\right); \tag{5.8}$$

$$r' = \sqrt{(x - x_2)^2 + (y)^2}, \quad \theta' = \operatorname{atan2}\left(\frac{y}{x - x_2}\right);. \tag{5.9}$$

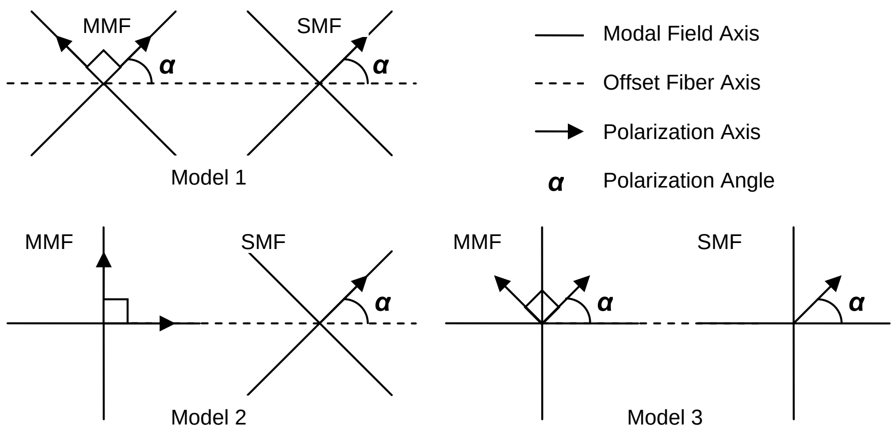


Fig. 5.6: Three SMF – MMF excitation models under consideration.

Model 1:

First we construct the necessary fields on the rotated modal field and polarization axes drawn relative to the fixed offset axis as:

$$\boldsymbol{\psi}^{lm} = \begin{bmatrix} \cos(l\theta - \alpha) \\ \cos(l(\theta - \pi/2) - \alpha) \end{bmatrix} \psi^{lm}(r), \quad \boldsymbol{\varphi} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \varphi(r') \tag{5.10}$$

where r' is the radial coordinate centered on the SMF center axis as seen in equation (5.9), and α is the polarization angle relative to the constant offset axis. We then rotate the polarization state of each vector field using the rotation matrix $\mathbf{R}$ and take the dot product:

$$\begin{aligned} \mathbf{R}(\alpha)\boldsymbol{\psi}^{lm} \bullet \mathbf{R}(\alpha)\boldsymbol{\varphi} &= \begin{bmatrix} \cos(\alpha)\cos(l\theta - \alpha) + \sin(\alpha)\cos(l(\theta - \pi/2) - \alpha) \\ -\sin(\alpha)\cos(l\theta - \alpha) + \cos(\alpha)\cos(l(\theta - \pi/2) - \alpha) \end{bmatrix} \bullet \begin{bmatrix} \cos(\alpha) \\ -\sin(\alpha) \end{bmatrix} \psi^{lm}(r)\varphi(r') \\ &= \left(\cos^2(\alpha)\cos(l\theta - \alpha) + \sin^2(\alpha)\cos(l\theta - \alpha)\right)\psi^{lm}(r)\varphi(r') \\ &= \cos(l\theta - \alpha)\psi^{lm}(r)\varphi(r') \end{aligned} \tag{5.11}$$

which clearly has a dependence on α meaning that the field coefficient and thus the power in the lm^{th} mode will have a dependence on α which is contrary to experiment.

Model 2:

For the second example, we construct the fields as though the MMF field and polarization axes are fixed along the set offset axis and only the SMF polarization axis is allowed to rotate:

$$\boldsymbol{\psi}^{lm}(r,\theta) = \begin{bmatrix} \cos(l\theta) \\ \cos(l(\theta - \pi/2)) \end{bmatrix} \psi^{lm}(r), \quad \boldsymbol{\varphi} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \varphi(r') \tag{5.12}$$

where r' is the radial coordinate centered on the SMF center axis as seen in equation (5.9), and α is the polarization angle relative to the constant offset axis. In the previous model, the basis vectors for both the SMF and MMF coordinate systems were pointing in the same direction; however, in this second model the basis vectors are different for the MMF and SMF where the SMF vectors are rotated by α relative to the MMF and the common offset axis. Now with the polarization axis of the MMF fixed such that it cannot follow the input polarization, only the polarization of the SMF field has to be rotated around the common offset axis and:

$$\begin{aligned}\boldsymbol{\psi}^{lm} \bullet \mathbf{R}(\alpha)\boldsymbol{\varphi} &= \begin{bmatrix} \cos(l\theta) \\ \cos(l(\theta-\pi/2)) \end{bmatrix} \bullet \begin{bmatrix} \cos(\alpha) \\ -\sin(\alpha) \end{bmatrix} \psi^{lm}(r)\varphi(r') \\ &= \left(\cos(\alpha)\cos(l\theta) - \sin(\alpha)\cos(l(\theta-\pi/2))\right)\psi^{lm}(r)\varphi(r') \end{aligned} \tag{5.13}$$

which also has an α dependence and is contrary to experiment.

Model 3:

In the final model, the MMF field axis is fixed along the common offset axis and the SMF axis is allowed to rotate. This rotation by the SMF field can also be modeled as though the SMF were drawn along the fixed offset axis and just the polarization state has been rotated. Because of the azimuthal symmetry of this fundamental fiber mode, both models are equivalent. But now as the polarization state of the SMF field rotates, the polarization state of the MMF modal fields are also allowed to rotate but the axis upon which all fields are drawn remains constant. We construct the fields similar to the previous model:

$$\boldsymbol{\psi}^{lm}(r,\theta) = \begin{bmatrix} \cos(l\theta) \\ \cos(l(\theta-\pi/2)) \end{bmatrix} \psi^{lm}(r), \quad \boldsymbol{\varphi} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \varphi(r') \tag{5.14}$$

where r' is the radial coordinate centered on the SMF center axis as seen in equation (5.9), and α is the polarization angle relative to the constant offset axis. The major difference in this model though, is that the basis vectors of both coordinate systems are now pointing in the same direction and aligned along the common fiber offset axis. We now find the dot product between the two vector fields:

$$\begin{aligned}\mathbf{R}(\alpha)\boldsymbol{\psi}^{lm} \bullet \mathbf{R}(\alpha)\boldsymbol{\varphi} &= \begin{bmatrix} \cos(\alpha)\cos(l\theta) + \sin(\alpha)\cos(l(\theta-\pi/2)) \\ -\sin(\alpha)\cos(l\theta) + \cos(\alpha)\cos(l(\theta-\pi/2)) \end{bmatrix} \bullet \begin{bmatrix} \cos(\alpha) \\ -\sin(\alpha) \end{bmatrix} \psi^{lm}(r)\varphi(r') \\ &= \left(\left(\cos^2(\alpha)+\sin^2(\alpha)\right)\cos(l\theta) + \left(\sin(\alpha)\cos(\alpha) - \sin(\alpha)\cos(\alpha)\right)\cos(l(\theta-\pi/2))\right)\psi^{lm}(r)\varphi(r') \\ &= \cos(l\theta)\psi^{lm}(r)\varphi(r') \end{aligned} \tag{5.15}$$

which of course has no α dependence, meaning that under this model there is no polarization dependence in multimode fiber response when coupling linearly polarized light directly from a singlemode fiber to an *ideal* multimode fiber, which is confirmed by experiment.

5.2.2 Non-Ideal Fiber Excitation

In non-ideal multimode fibers there can be any number of defects present in the refractive index profile. These range from center dips and peaks, to an exponential decay as the index approaches the cladding as well as kinks along the center of the refractive index profile [1, 3-4, 6, 13]. While it has been shown that these defects can cause a local polarization dependence [1, 3], we will show that it is the inherent birefringence in older, lower grade fibers that is the source of the overall polarization dependence observed.

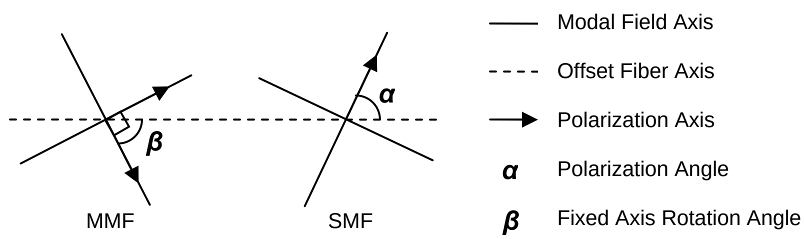


Fig. 5.7: SM – MM fiber excitation with birefringent MMF.

Fig. 5.7 shows the setup of a SM – MM fiber excitation scheme where there is significant birefringence in the multimode fiber. This setup is very similar to that of Model 2 in the previous section, except that this time it is possible for the set multimode fiber axis to be offset by a rotation of β from the common fiber offset axis. The use of Model 2 is justified because the birefringent fiber is no longer isotropic and the modal fields have to be drawn along the fixed birefringent axis. The birefringent fiber also causes the breaking of the degeneracy of the x- and y-polarized modes. This forces us to consider the excitation of each mode separately and we can no longer form the LP^{lm} vector modes as we did in the previous section. Since there are no LP^{lm} vector modes, the polarization of these modes cannot rotate to follow the input polarization of the singlemode fiber field since we are left with separate x- and y-polarized fields (whose polarization can rotate only with the rotation of the field/birefringent axis). It is because we have to consider both polarization components of the MMF separately that we are forced to deal with the situation in Fig. 5.7 with a constant and fixed polarization and modal field axes. This time we set up our modes a little bit differently:

$$\begin{aligned} \boldsymbol{\psi}_x^{lm} &= \begin{bmatrix} \psi_x^{lm}(r,\theta) \\ 0 \end{bmatrix} = \begin{bmatrix} \cos(l\theta-\beta) \\ 0 \end{bmatrix} \psi^{lm}(r), \\ \boldsymbol{\psi}_y^{lm} &= \begin{bmatrix} 0 \\ \psi_y^{lm}(r,\theta) \end{bmatrix} = \begin{bmatrix} 0 \\ \cos(l(\theta-\pi/2)-\beta) \end{bmatrix} \psi^{lm}(r) \end{aligned} \tag{5.16}$$

What we have done here is constructed the modes relative to the constant offset axis, with field distributions rotated by β around the local coordinate system (r, θ). Now, similar to the second model in section 5.2.1 we rotate the polarization state of the SMF input and project it onto the constant fields of the MMF:

$$k_x^{lm} = \iint_{r\,\theta} \boldsymbol{\psi}_x^{lm} \bullet \mathbf{R}(\alpha)\boldsymbol{\varphi}(r',\theta')\, r\, dr\, d\theta$$
$$= \iint_{r\,\theta} \cos(\alpha)\psi_x^{lm}\varphi\, r\, dr\, d\theta$$
$$= k_{xx}^{lm}\cos(\alpha) \tag{5.17}$$

where

$$k_{xx}^{lm} = \iint_{r\,\theta} \psi_x^{lm}\varphi\, r\, dr\, d\theta \tag{5.18}$$

The power of each mode is then found

$$a_x^{lm} = \left|k_x^{lm}\right|^2 = (k_{xx}^{lm})^2\cos^2(\alpha) \tag{5.19}$$

where a_x is the power in an x-polarized mode, and of course power in a y-polarized mode is found similarly:

$$a_y^{lm} = \left|k_y^{lm}\right|^2 = (k_{yy}^{lm})^2\sin^2(\alpha) \tag{5.20}$$

with k_{yx} and k_{yy} defined as

$$k_{yy}^{lm} = \iint_{r\,\theta} \psi_y^{lm}\varphi\, r\, dr\, d\theta. \tag{5.21}$$

It is the sin(2α) term in each of these modal power equations that gives us the correct period response that is seen in experiment and the other two terms ensure that the power cannot of course be negative. Because each the x- and y-polarized modes in the birefringent fiber are non-degenerate and each have different group delays, as the polarization is rotated, a fraction of the total power will be coupled into either polarized mode and the final pulse will appear to drift as seen in Section 4.2.2. The model presented in this section has no inherent conflict with previous models where β was equal to zero. This is because of the inherent birefringence in the fiber which causes the breaking of the degeneracy of the modes and the polarization dependence is then mainly presented as pulse drift due to differing arrival times of the x- and y-polarized modes.

5.3 MM – MM Fiber Excitation

Traditional mode coupling devices operate due to phase matching conditions and work over a certain length, whether it's a few millimeters for long period gratings or up to centimeters as in evanescent field couplers. However, as was seen in experiment, the polarization effect appears to be taking place at a single point near the beginning of the fiber link, also implying that there is no phase matching condition necessary. This is known because there was no pulse spreading throughout the length of the MMF link (other than due to intermodal group delay) and each initial pulse maintained its initial shape. Now, to model the polarization mode coupling characteristics of multimode fiber given a sharp fiber perturbation such as a fusion splice or the direct coupling of two fibers (either with a parameter mismatch or relative offset between fibers), we adapt the current mode coupling theory [23 - 26] to include a term that accounts for a power transfer between modes dependent on their initial polarization state. Starting with the full expression for the electric field of a vector mode present in the fiber

$$\mathbf{E}^{lm}(\theta,r,z,t) = k^{lm}\begin{bmatrix}\cos(l\theta)\\ \cos(l\theta-\pi/2)\end{bmatrix}\psi^{lm}(r)\exp\{j\beta z\}\exp\left\{\frac{-at^2}{2}\right\} \tag{5.22}$$

where β is the propagation constant, a is a shaping coefficient to give the correct pulsewidth of the mode, and the implicit time dependence $\exp\{j\omega t\}$ is omitted. We then simplify this by integrating the vector field over the area of the fiber to eliminate the θ and r variables. We are now left with

$$\mathrm{E}^{lm}(z,t) = k^{lm}\exp\{j\beta z\}\exp\left\{\frac{-at^2}{2}\right\} \tag{5.23}$$

Now, using the conventional coupled mode theory equations governing the amplitudes of the mode groups (including an additional term to account for mode coupling due to polarization rotation), the coupled mode equations are

$$\left(\frac{\partial}{\partial z}+\frac{\partial}{\partial t}+\frac{\partial}{\partial \alpha}\right)\mathbf{K} = j\beta\mathbf{K} - at\mathbf{K} + j\mathbf{H}\mathbf{K} \tag{5.24}$$

where α is the polarization angle of the field input to the multimode fiber system; z is the distance along the fiber; t is the time; a is a Gaussian shaping coefficient (to determine proper pulsewidth, 100 ps); **K** is a vector of mode field excitation (mode amplitude) coefficients whose initial values are determined by the overlap integral between the SMF input field and the MMF fields i.e. equation (5.7), and **H** is the mode coupling matrix of the system whose coefficients determine the coupling strengths between modes,

$$\mathbf{K} = \begin{bmatrix}k_1\\ \vdots\\ k_N\end{bmatrix};\quad \mathbf{H} = \begin{bmatrix}h_{11} & \cdots & h_{1N}\\ \vdots & \ddots & \vdots\\ h_{N1} & \cdots & h_{NN}\end{bmatrix} \tag{5.25}$$

where N is the total number of mode groups in the system. Since we model each mode as a vector field with x- and y-polarized components and each of these modes is normalized accordingly (i.e. equation (5.5)), the modal amplitude coefficients (i.e. the elements of **K**) are representative of the complete vector modes as described in equation (5.4) and it is not necessary to include modal amplitude terms that separately describe the x- and y-polarized fields. If the vector components of the modal fields were considered individually (each with their own k value, k_x or k_y, with 2N total elements in **K**), then the total modal power coefficient (in this case given by $|k_x + k_y|^2$, instead of $|k|^2$) would itself have an initial polarization angle dependence, contradicting *Model 3* established in the previous section. In this model, attenuation is negligible and the associated terms are ignored; the implicit time response is also ignored. To further simplify this model we look at the results of the experimental study. We know from the experiment that because there is no modal pulse spreading in the final pulse response, that the polarization effect takes place in a discrete spot at the beginning of the fiber and is not an accumulative effect spread throughout the fiber link. Because of this, we can neglect the z-dependant terms in (5.24). Also, since there is no observable time dependence seen in the experiment, we neglect the time terms as well. Instead of dealing with the complete set of orthogonal modes, we now reduce the system to its mode groups, where each full vector mode group (defined by (5.4)) is represented by its initial mode field excitation (mode amplitude) coefficient, k. This value is determined for each mode group by finding the average power shared by each mode in the group similar to the method presented in [13]. This then simplifies the set of coupled equations to:

$$\frac{d}{d\alpha}\mathbf{K} = j\mathbf{HK} \tag{5.26}$$

Now, to determine the values of the coupling coefficients in **H**, a few things have to be considered. We again make an assumption: that there is no significant source of mode selective loss that would cause severe polarization dependent loss in the system. This is seen in experiment for most cases of mode excitation where the SMF source does not excite higher order modes near the core/cladding interface in the MMF. If there is no loss mechanism then the model has to take into account that there will be constant total power as the initial polarization angle is rotated such that,

$$P_{tot} = \sum_{n=1}^{N} |k_n|^2 = constant. \tag{5.27}$$

where n is the mode group number and N is the total number of mode groups. For this to be possible in this current model, we know from coupled-mode theory [23 - 26] that the upper and lower triangles of the matrix **H** have to be complex conjugates of one another and that the diagonal terms have to be real. For example, in a two mode system,

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}, \quad h_{12} = h_{21}^{*}; \{h_{11}, h_{22} \in \Re\} \tag{5.28}$$

or more generally,

$$h_{mn} = h_{nm}^{*}; \{h_{nn} \in \Re\}. \tag{5.29}$$

In other words, **H** is Hermitian, but the diagonal terms are necessarily real and the off diagonal terms are necessarily complex and are needed to maintain a constant total power. To determine what possible values these coupling coefficients take, some insight can be obtained by looking at the experimental results. From the experiment, the power in the dominant mode can be described as

$$P_n(\alpha) = p_{n1}\cos^2(\alpha - \sigma_n) + p_{n2}\sin^2(\alpha - \sigma_n) \tag{5.30}$$

where n is the mode group number, p_{n1} is the initial mode power, p_{n2} is the mode power at $\alpha = \pi/2$ and σ_n is the initial phase of the curve. This relationship is true for every mode group excited in the system. One way to find a mode power curve such as this is if the mode excitation coefficient, k_n, takes the form

$$k_n(\alpha) = a_{n1}\cos(\alpha - \sigma_n) + ja_{n2}\sin(\alpha - \sigma_n) \tag{5.31}$$

where $P_n = |k_n|^2$, $p_{n1} = a_{n1}^2$ and $p_{n2} = a_{n2}^2$. Given the current system of (5.23), a general solution of the form of (5.31) can be found if all the elements of the upper and lower triangles of **H** are identical and real and all the diagonal elements are identical and real. **H** now takes the form

$$\mathbf{H} = \begin{bmatrix} h_d & h & h & \cdots & h \\ h & h_d & h & \cdots & h \\ h & h & h_d & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & h \\ h & h & \cdots & h & h_d \end{bmatrix} \tag{5.32}$$

and the analytical solution for the field coefficients (for N > 2) is

$$k_n = \exp\left\{ j\left(\frac{N-2}{2}h - h_d\right)\alpha \right\}\left(k_n(0)\cos\left(\frac{N}{2}h\alpha\right) + j\frac{1}{N}\left(2\sum_{m\neq n} k_m(0) - (N-2)k_n(0)\right)\sin\left(\frac{N}{2}h\alpha\right)\right) \tag{5.33}$$

where N is the total number of mode groups, n is the mode group index, h is the value of the matrix **H** that is not on the diagonal, h_d is the value of the diagonal elements of **H**, $k_n(0)$ is the initial field excitation coefficient of mode group n. From this it can be seen that to get the proper sinusoidal dependence of the modal power with period π, the value for the off diagonal terms of **H**, h, has to be equal to $2/N$. If any one of these h coefficients deviates even a small amount from the others, the mode coupling that occurs is no longer sinusoidal and an analytical solution cannot be found. Fig 5.8 shows the results of a nine mode case, where the elements in (5,1) and (1,5) are set to zero instead of 2/9. As you can see, since the matrix remains Hermitian, the total power remains constant, but the mode power fluctuation is no longer sinusoidal. Fig. 5.9 shows an example of this model given the initial conditions of a SMF – MMF – MMF launch with an initial SMF offset of 7 μm.

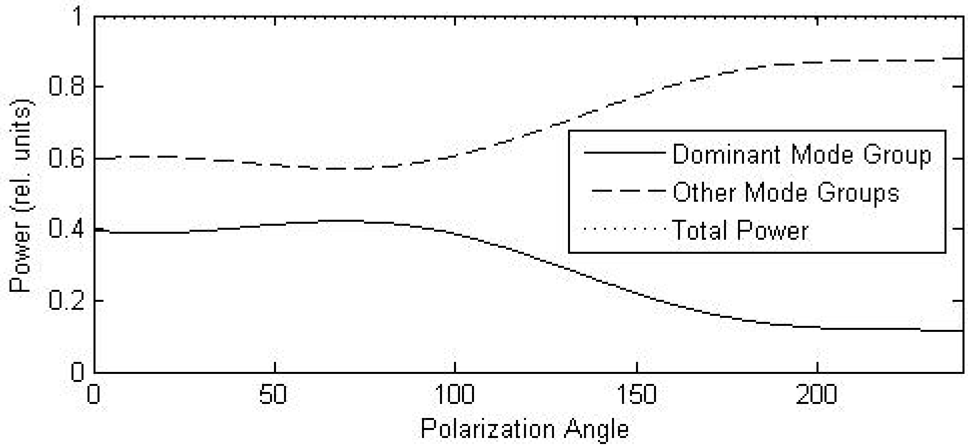


Fig. 5.8: Theoretical result with parameter deviation.

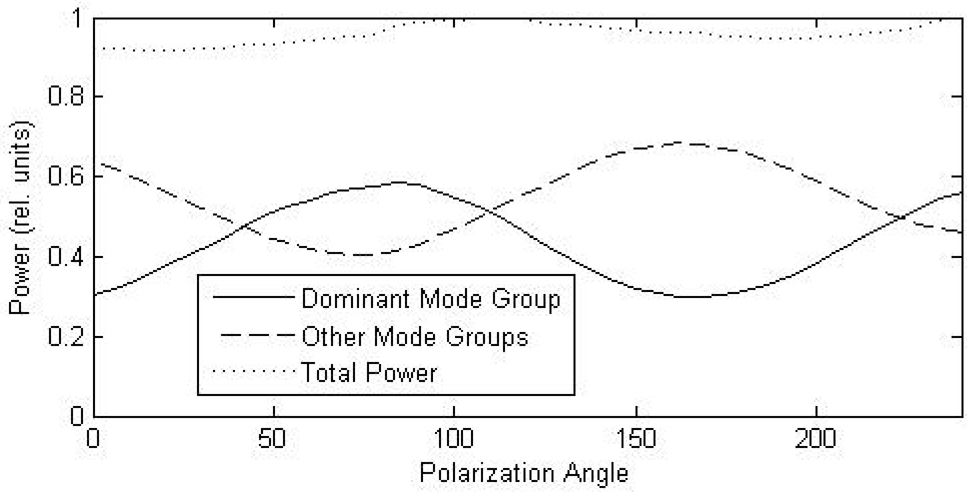


Fig. 5.9a: Experimental Results, 7 um SMF offset with SMF – MMF – MMF setup.

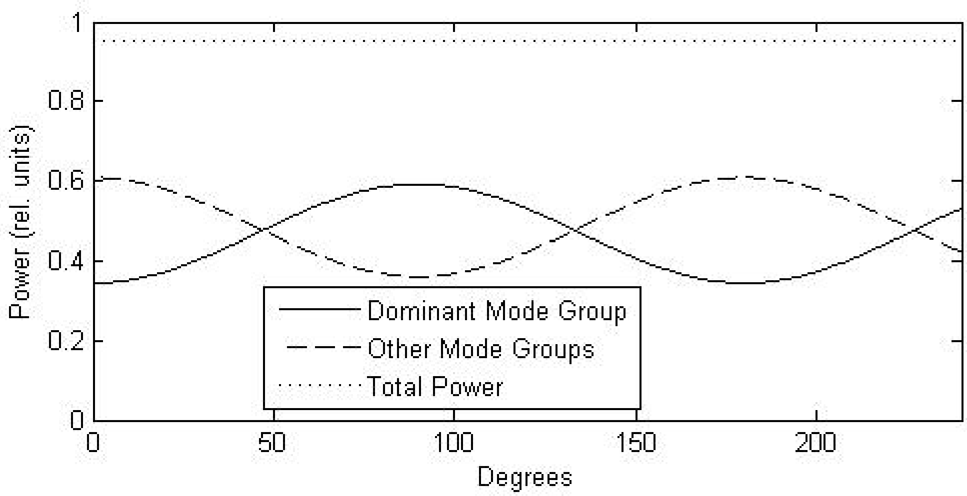


Fig. 5.9b: Results from model, 7 um offset, with initial conditions determined by experiment.

In the above figures, the initial conditions for the model, the vector **k(0)**, whose elements are the initial field excitation coefficients of each mode group, was determined by the initial mode power distribution of the experiment at a polarization angle of zero degrees. These initial values were taken as real numbers, but in practice could be complex. If this is the case, the relative phase of each mode group curve might be different (other than just in or out of phase), but because of the constraints on the mode coupling matrix **H**, constant power and correct sinusoidal response will be maintained.

5.3.1 Discussion of Model

From the general solution (for N > 2), equation (5.33)

$$k_n = \exp\left\{ j\left(\frac{N-2}{2}h - h_d \right)\alpha \right\}\left(k_n(0)\cos\left(\frac{N}{2}h\alpha \right) + j\frac{1}{N}\left(2\sum_{m \neq n} k_m(0) - (N-2)k_n(0) \right)\sin\left(\frac{N}{2}h\alpha \right) \right) \qquad (5.33)$$

we can see a few things. First, since the diagonal elements of the matrix **H** are all the same, the only effect these elements have on the final solution is on the phase as seen in the first exp{…} term. However, if h_d is set to (N-2)h/2, this phase term is set to unity and has no effect on the power in each mode. When finding the power of each mode group ($P_n = |k_n|^2$) this phase term has no effect either and since the phase is not measured in the experiment we cannot say for certain what these diagonal elements might be. Second, we know from experiment that the power fluctuates with a period of π which allows us to write the power in each mode group in the form of equation (5.30). This means that in order for equation (5.33) to eventually give the correct form of the power in each mode group, the value of h has to be equal to 2/N. Lastly, looking at the sine term in the above equation, the magnitude of this term ultimately determines the power left in this mode group when $\alpha = 90^o$. The interesting thing about this is that if we assume that this equation is indeed a description of the physical system, and given that we can represent each mode group completely by its excitation coefficient, it seems that the entire polarization mode coupling effect, instead of being due to the overlap of different modes (e.g. at an offset connector) it is due to the interference of all mode groups on some kind of interface, whether it's an offset connector, two fibers with a parameter mismatch or a fusion splice.

To be able to get the general mode coupling equation in its current form and with the correct sinusoidal response, the mode coupling coefficients for each mode have to be the same, meaning that all the modes couple into each other with the same probability. This is consistent with the idea that the polarization effect is due to the interference of the modes at a single interface. For this to be the case, all of these modes have to be coherent at the interface. Also, the amount of power that can couple between each individual mode is identical on this interface and scales with the total number of modes present.

Numerical Example

For the example with an initial SMF offset of 7 µm, we can find directly from the experiment the initial relative power in each mode group, given that there are eight total mode groups, as

	n	Power
	1	0.3443
	2	0.3108
	3	0.1932
$\|\mathbf{k(0)}\|^2$	4	0.0911
	5	0.0019
	6	0.0008
	7	0.0002
	8	0.0001

which then leads us to find the initial vector **k(0)** as

	n	Power
	1	0.5868
	2	0.5575
	3	0.4396
k(0)	4	0.3018
	5	0.0431
	6	0.0284
	7	0.0136
	8	0.0107

If we then take this vector and input it directly into the model (with h = 2/8 = 0.25 and h_d = 0.75 to eliminate the phase term) as the initial conditions, we then get a solution as shown in Fig. 5.10 below. In Fig. 5.11 we see how each of the eight individual mode groups fluctuate with input polarization angle. In Fig. 5.11, each curve represents a different mode group with the dominant mode group (defined as having the most initial power) in this case being the fundamental group.

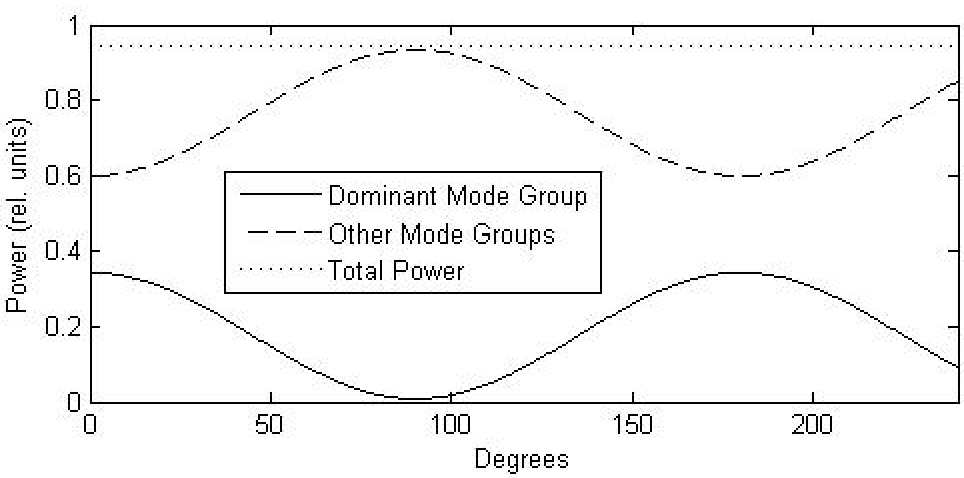


Fig. 5.10: Model results with all mode groups in phase.

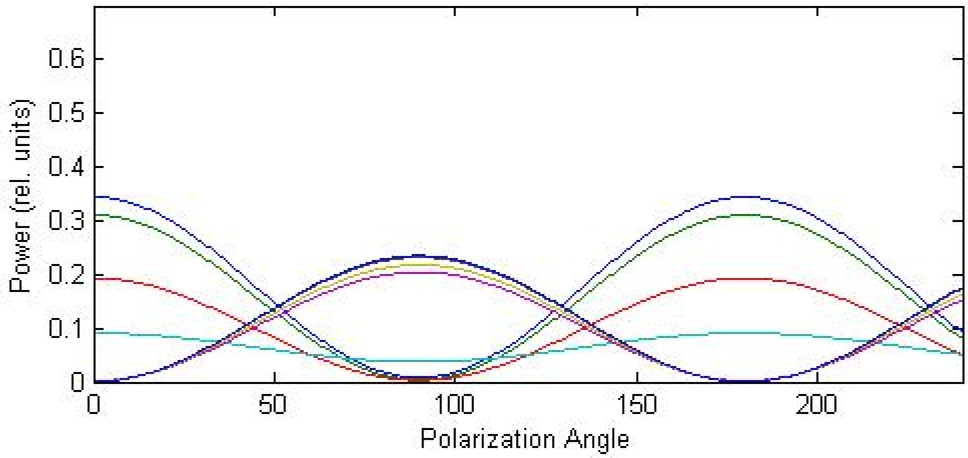


Fig. 5.11: Mode group power fluctuation with all mode groups in phase.

These results are not at all what we see in the experiment even though the initial mode power distribution is the same with the fundamental mode in this case being the dominant one. This relationship between the modes arises only when all of the mode groups are in phase with one another. However, if we adjust the vector of initial values **k(0)** such that some of the mode groups are completely out of phase with the others, we can see results that are in line with experiment. If we take in this case **k(0)** as

	n	Power
k(0)	1	0.5868
	2	-0.5575
	3	-0.4396
	4	-0.3018
	5	0.0431
	6	0.0284
	7	0.0136
	8	-0.0107

then the mode group fluctuation matches up incredibly well with experiment as shown in Fig. 5.12 below. From Fig. 5.12 we can see that the model in this case matches up with the experimental curves quite well, except for the fluctuation of (in this case) the higher order mode groups. In the experiment they do fluctuate, but only very slightly, compared to the relatively higher fluctuation as seen in the model. The problem with this approach however, is that other than looking at the experimental results and matching up which groups are out of phase from the rest, there is, as of this moment, no way to predict the mode groups that will be out of phase.

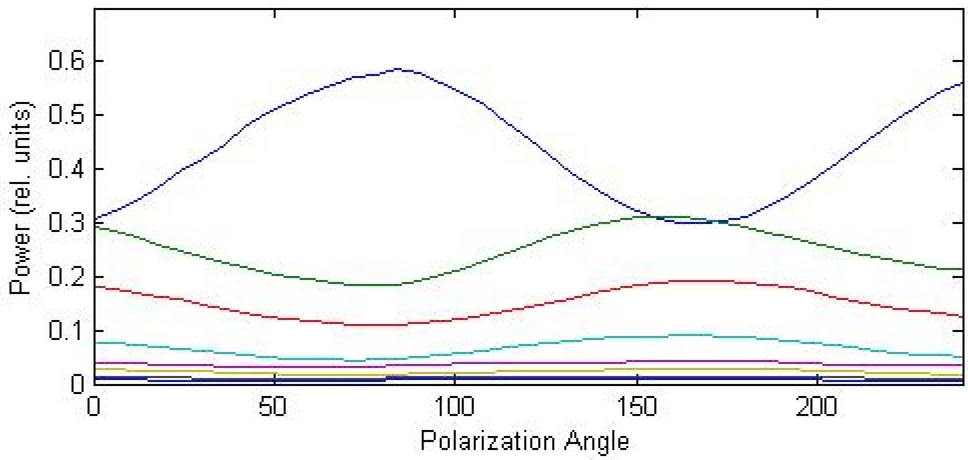


Fig. 5.12a: Experimental mode group fluctuation.

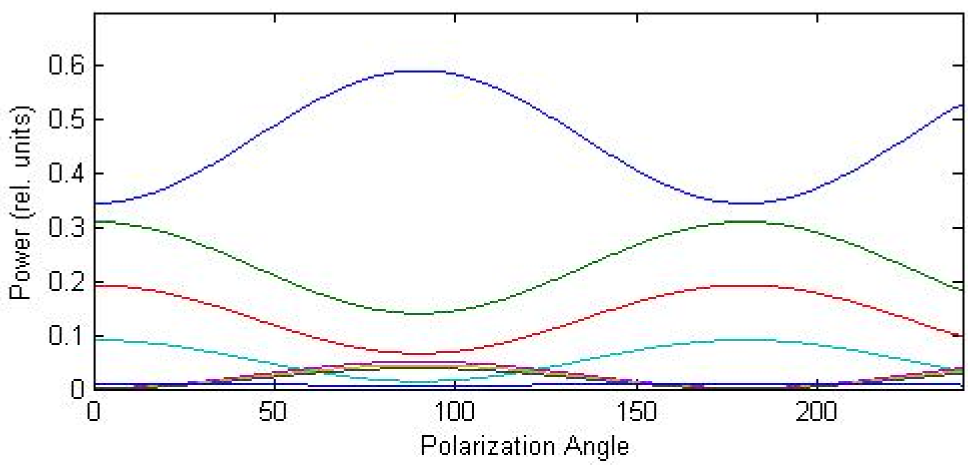


Fig. 5.12b: Mode group fluctuation given by model.

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Chapter Six
Theoretical Results

Following from Chapter 5, this chapter presents results from the theoretical approach used to model the polarization dependence in multimode fiber in the presence of a mode power coupling interface (Chapter 5.3) and compares them to the experimental results as presented in Chapter 4. Also, more detailed polarization experiment results are presented in this chapter and compared to the theory. For the sake of brevity only one fiber case is considered, that of the OM3 (orange) fiber, as the results and interpretations for the other fibers are similar and are presented in Appendix B. The results for the direct excitation of the OM3 and FDDI fibers are not presented as they exhibit no polarization dependent mode coupling as explained in Chapter 5.2.

The theoretical results presented in this section are displayed in two ways. First, they are presented similar to that of the original discussion of the experimental results, comparing how the overall polarization dependence matches with theoretical results with regard to the power transfer between the dominant mode group (greatest initial power) and all other lower and higher order mode groups. The second set of results for each offset launch condition then examines the power fluctuation in each individual mode group, comparing experiment with theory. The theoretical results for each individual mode group in this chapter are generated using equation (5.32) given an initial excitation vector (e.g. Table 6.1).

6.1 0 µm Transverse Offset – 1 m – 1 km OM3 Fiber

Fig. 6.1 shows the experimental (first seen in Fig. 4.18) and theoretical polarization results of a 1 km OM3 fiber excited by a SMF input at a transverse offset of 0 µm followed by a 1 m long patchcord. In this case, the LP01 mode group is the dominant mode group. Fig. 6.1a is the experiment and 6.1b the theory. Fig. 6.2 shows the complete power fluctuation for all excited mode groups, 6.2a/b is the experiment and 6.2c/d the theory. These theoretical results were generated using the initial excitation vector in Table 6.1.

Table 6.1: 0 µm Offset Excitation Vector

	n	k_n
k(0)	1	0.8104
	2	-0.3599
	3	-0.2713
	4	0.2854
	5	0.1868
	6	-0.1382
	7	0.1273
	8	0.1171

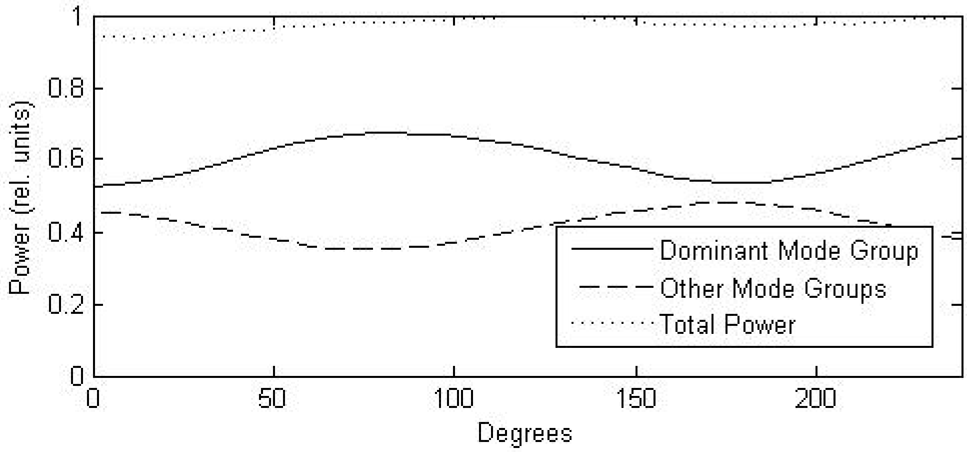


Fig. 6.1a: Experimental polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 0 um offset.

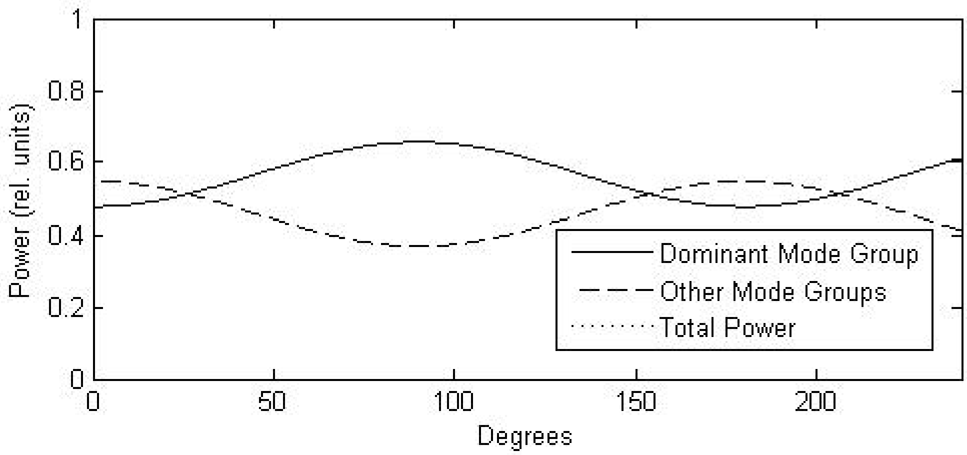


Fig. 6.1b: Theoretical polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 0 um offset.

The theoretical curves presented in Fig. 6.1 match well with the experimental results. The only discrepancy with these curves is that the average power in the dominant mode given by the theoretical approach is five percent lower than that of the experimental results. This can also be seen by looking at the power fluctuation in each individual mode group as in Fig. 6.2. Fig. 6.2a shows the first four mode groups, LP01 – LP31 seen in experiment and Fig. 6.2c shows the theoretically generated results. Both these figures show very good agreement, with the majority of power flowing from LP01 to LP11 and LP21 with a small amount of power flow due to LP31. The major discrepancy in the 0 µm results is the major power fluctuation in the theoretical LP51 (Fig. 6.2d) compared to experiment. Fig. 6.2b shows the higher order mode groups exhibit very little polarization dependence while 6.2d shows an almost ten percent power fluctuation in the LP51 mode group. It is the relatively large amount of power in this mode group that accounts for the discrepancy between the fundamental modes of theory and experiment.

a)

b)

c)

d)

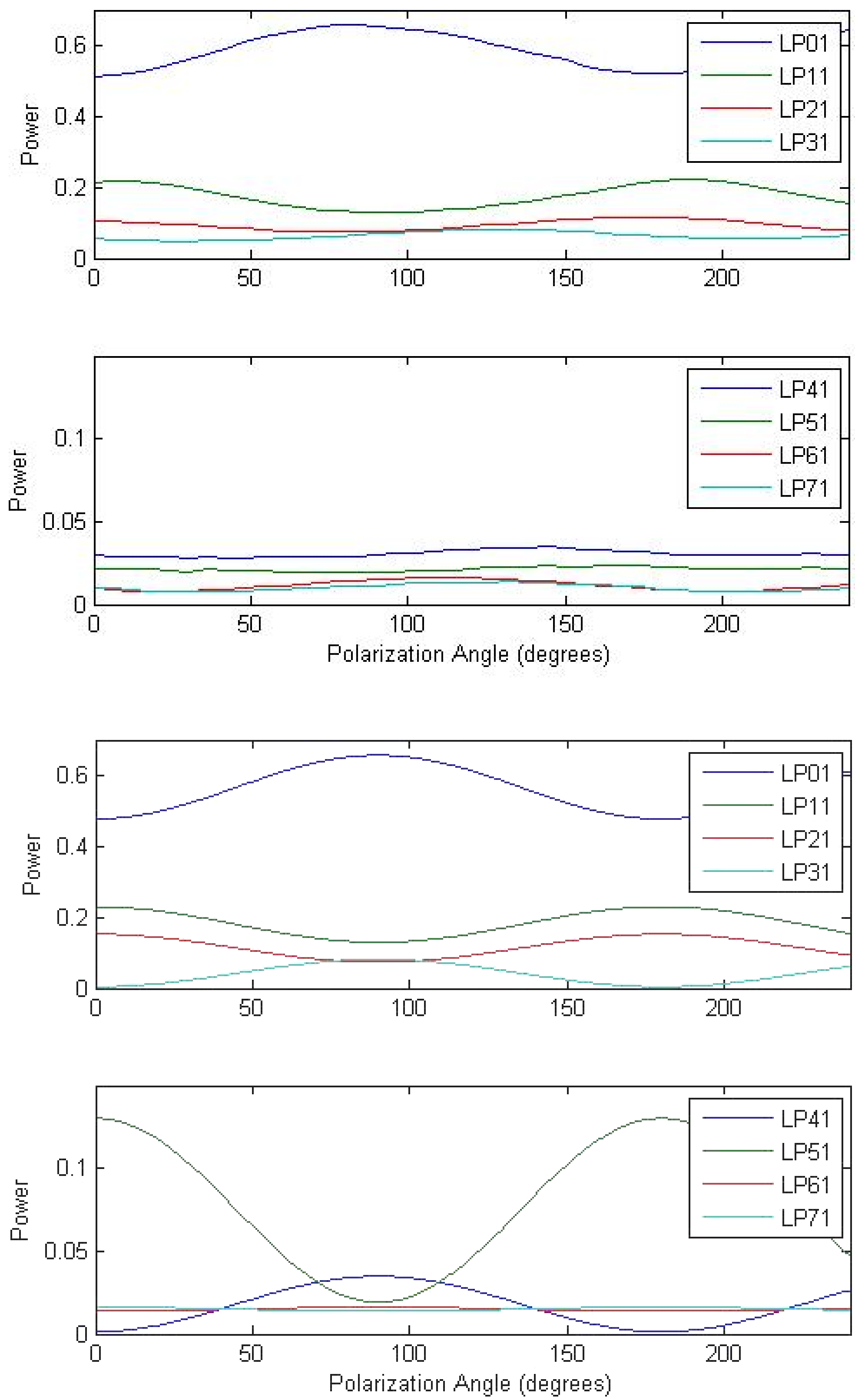


Fig. 6.2: Experimental and theoretical polarization dependent mode group power fluctuation, 0 um offset.

6.2 7 μm Transverse Offset – 1 m – 1 km OM3 Fiber

Fig. 6.3 shows the experimental and theoretical power fluctuation between the dominant mode group and all other mode groups combined when the MMF is excited by a 7 μm offset SMF into a 1 m MMF patchcord. The LP01 mode is again the dominant mode group. Fig. 6.4 shows the power fluctuation of each mode group for both experiment (a/b) and theory (c/d). Table 6.2 shows the excitation coefficients used to generate the results in Fig. 6.3 and 6.4.

Table 6.2: 7 um Offset Excitation Vector

	n	k_n
k(0)	1	0.5868
	2	-0.5575
	3	-0.4396
	4	-0.3018
	5	0.0431
	6	0.0284
	7	0.0136
	8	-0.0107

Similar to the results presented in the previous section, there is a very good match between the experimental and theoretical power flow between the dominant and all other higher or lower order mode groups as seen in Fig. 6.3. There is however a slight discrepancy; the experimental dominant mode shows five percent more power than the theoretical results. This discrepancy in the dominant mode group can be explained by looking at the mode power results for each individual mode group in Fig. 6.4. Fig. 6.4a shows the experimental results for the first four mode groups, LP01 – LP03 and they match very well with the theoretically produced results in Fig. 6.4c. In Fig. 6.4b, the experimental mode groups show very little power fluctuation compared to their theoretical counterparts in Fig. 6.4d. Even though there is very little total power in any of these modes, the relatively large fluctuation in the theoretical mode group curves accounts entirely for the discrepancy seen in Fig. 6.3. This large power fluctuation for the higher order mode groups is a consequence of simplifying the theoretical model of the polarization dependent power coupling between modes. The original assumption was that simplifying the model such that each mode group can be represented as a single excitation (or power) coefficient would be a fair representation of the system. However, this might not be the case for these higher order modes. For example, the eighth mode group, LP07, contains seven modes within it and simplifying these seven modes to be represented by one coefficient cannot express the proper mode coupling dynamics within the seven modes of the mode group. One possible explanation for the relatively lower power flow between the higher mode groups in experiments is indeed power flow between individual modes within each mode group. If the power fluctuation is dominated by this intramodal power coupling in the higher order mode groups, then there will be less power available to flow between adjacent mode groups.

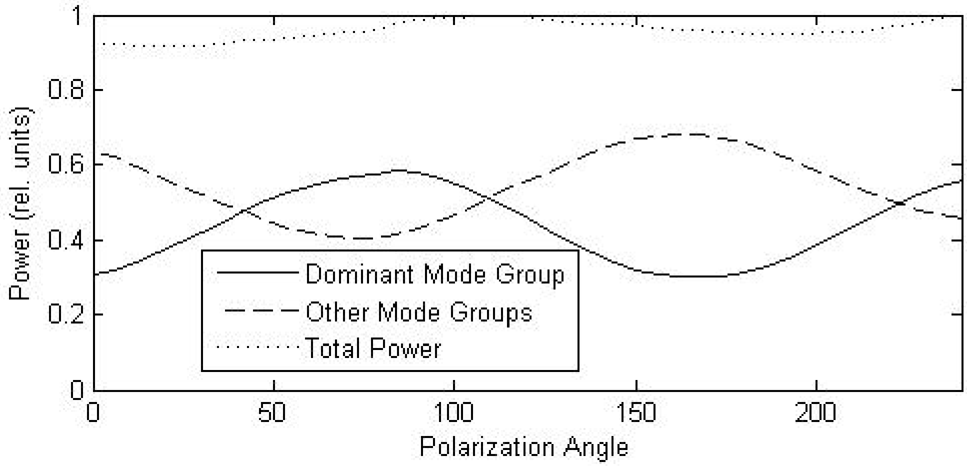


Fig. 6.3a: Experimental polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 7 um offset.

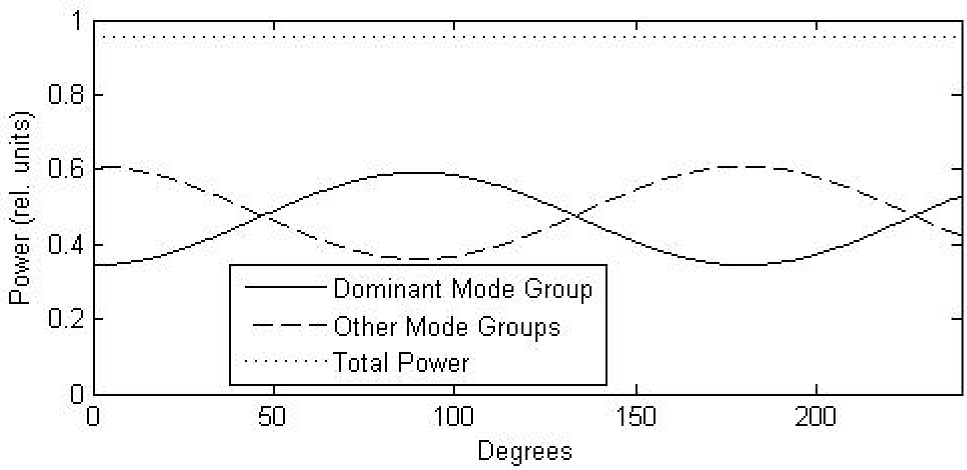


Fig. 6.3b: Theoretical polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 7 um offset

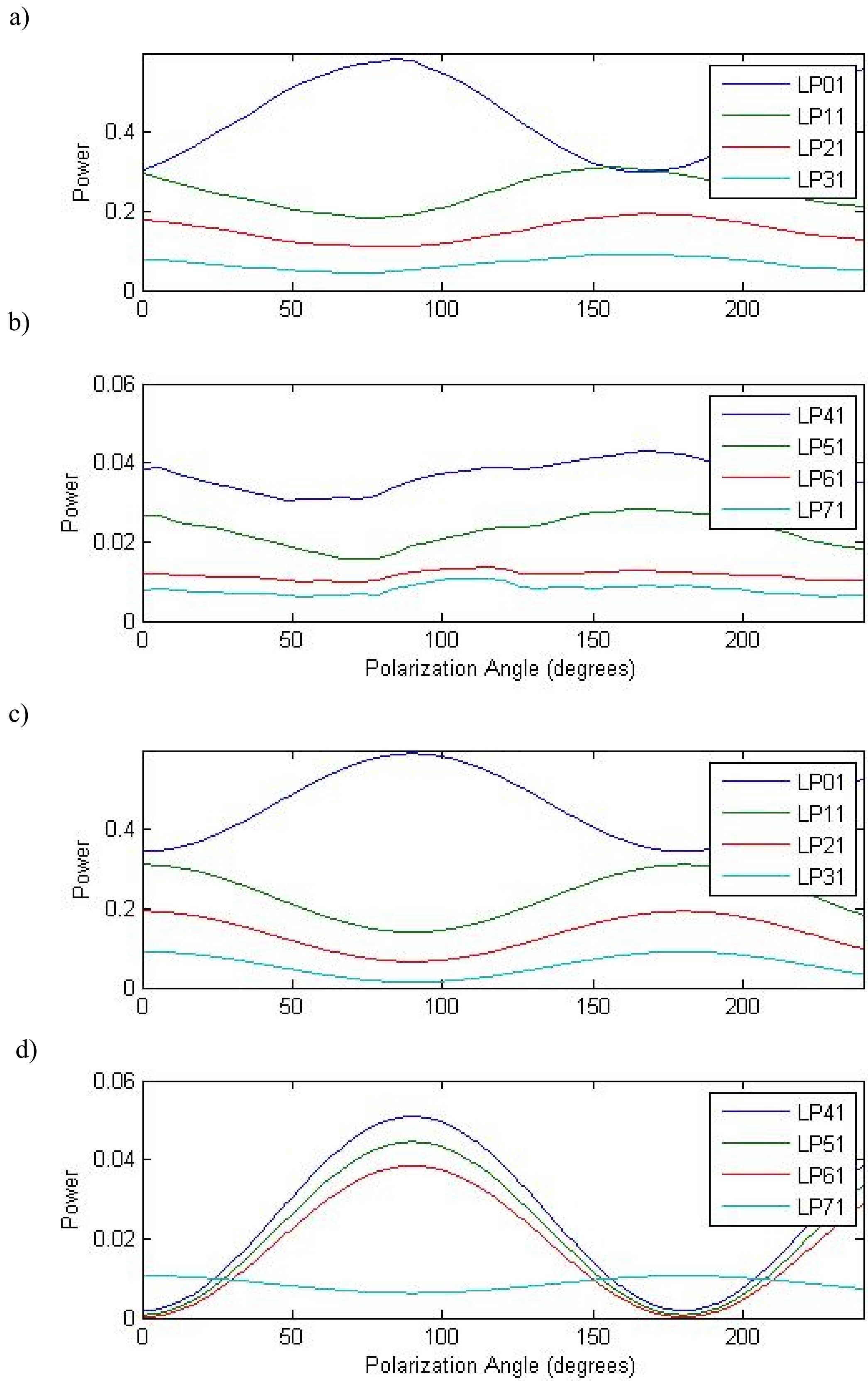


Fig. 6.4: Experimental and theoretical polarization dependent mode group power fluctuation, 7 um offset.

6.3 14 μm Transverse Offset – 1 m – 1 km OM3 Fiber

Fig. 6.5 shows the experimental and theoretical power fluctuation between the dominant mode group and all other mode groups combined when the MMF is excited by a 14 μm offset SMF into a 1 m MMF patchcord. The LP21 mode group is the dominant mode group in this case. Fig. 6.6 shows the power fluctuation of each mode group for both experiment (a/b) and theory (c/d). Table 6.3 gives the excitation coefficients used to generate the theoretical results presented in Fig. 6.5 and 6.6.

Table 6.3: 14 um Offset Excitation Vector

	n	k_n
k(0)	1	0.2993
	2	0.4995
	3	-0.3339
	4	$-0.3526e^{j\pi/4}$
	5	$0.4345e^{j\pi/4}$
	6	0.3202
	7	0.2528
	8	0.1566

Fig. 6.5 shows the experimental and theoretical results detailing the power flow between the dominant mode group and all other higher and lower mode groups. Like the results for the previous offset launches, the experiment and theory match up very well for this set of results. When looking at the mode power fluctuations for all mode groups in Fig. 6.6, however, it is apparent that the only single mode group that matches the experimental results is the dominant mode, LP21. This is because, as seen for the previous results as well, the power fluctuation in the higher order mode groups generally does not match up with the experimental results. A further complication in this particular instance is that some mode groups of the experiment appeared to be slightly out of phase with the dominant mode compared to previous results where mode power fluctuations were either in phase or completely out of phase with the dominant mode. To model this slight phase variation, the initial excitation coefficients are made to be complex as explained in Chapter 5.3. To create a result that matched the theoretical and experimental dominant modes, the excitation coefficients of the 4th and 5th modes (LP31, LP41) were given an additional phase angle of 45^{o}. Using this excitation vector resulted in theoretical power fluctuation curves that matched only for the first three mode groups, with the other mode groups deviating greatly from experiment.

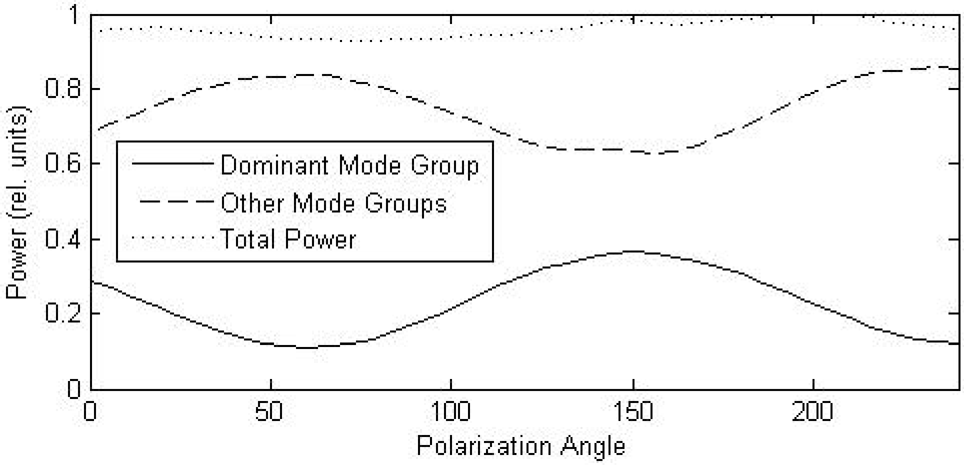


Fig. 6.5a: Experimental polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 14 um offset

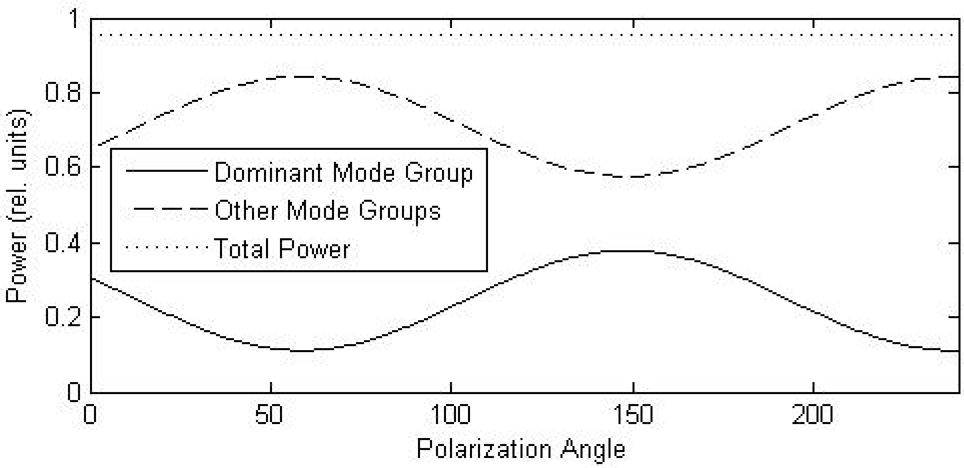
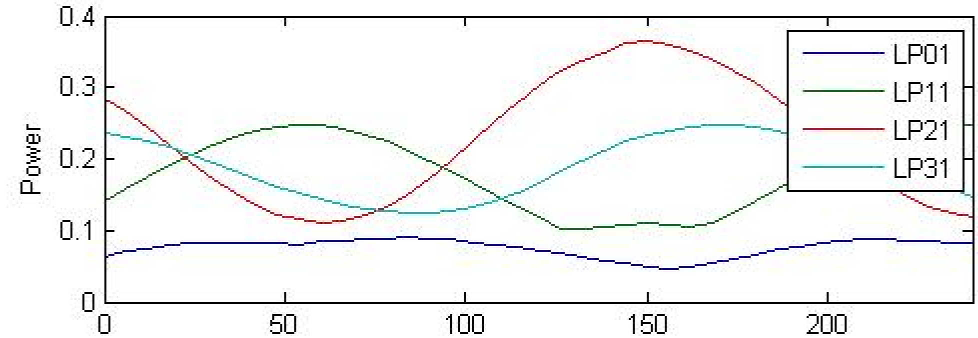
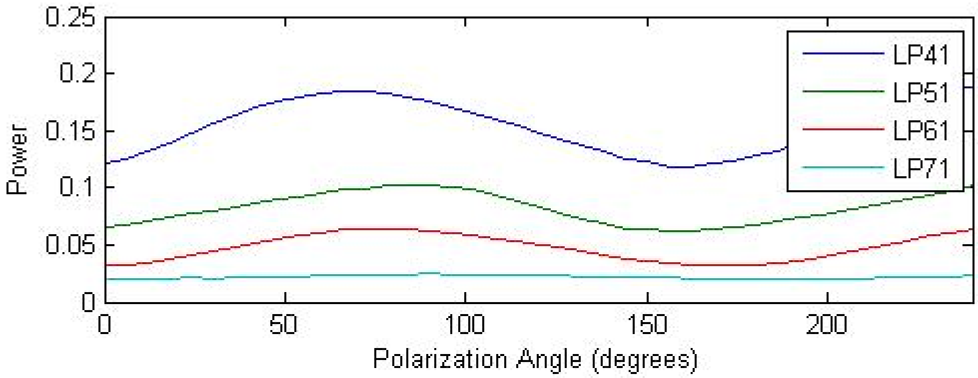


Fig. 6.5b: Theoretical polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 14 um offset

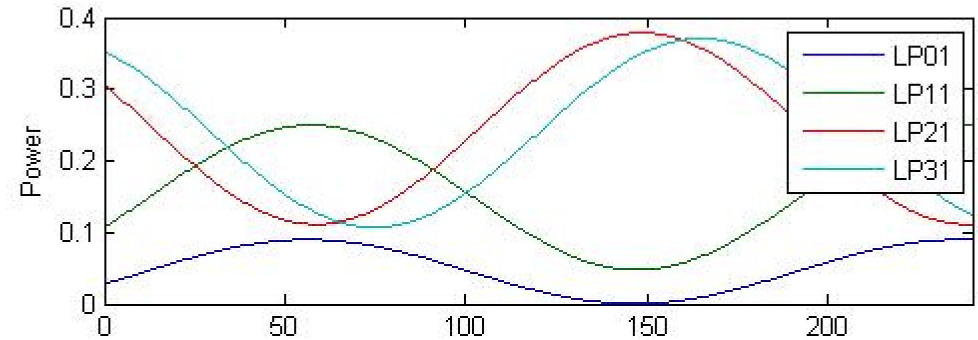
a)



b)



c)



d)

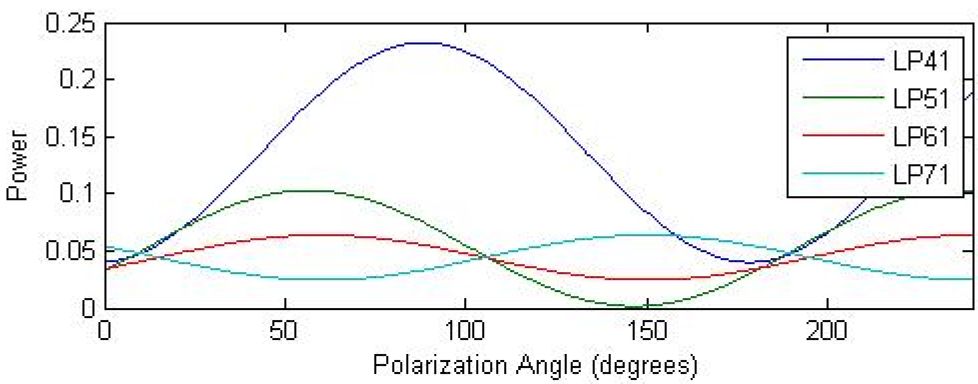


Fig. 6.6: Experimental and theoretical polarization dependent mode group power fluctuation, 14 um offset.

6.4 21 µm Transverse Offset – 1 m – 1 km OM3 Fiber

Fig. 6.7 shows the experimental and theoretical power fluctuation between the dominant mode group and all other mode groups combined when the MMF is excited by a 21 µm offset SMF into a 1 m MMF patchcord. The LP41 mode is the dominant mode group. Fig. 6.8 shows the power fluctuation of each mode group for both experiment (a/b) and theory (c/d). Table 6.4 shows the excitation coefficients used to generate the results in Fig. 6.7 and 6.8.

Table 6.4: 21 um Offset Excitation Vector

	n	k_n
k(0)	1	0.1410
	2	0.2370
	3	-0.2201
	4	-0.2822
	5	-0.4065
	6	0.5391
	7	0.3910
	8	0.2999

The results presented Fig. 6.7a and 6.7b show fairly good agreement, the only major discrepancy being the total power fluctuation seen in experiment due to polarization dependent mode selective loss because of the high offset launch condition. The only mode groups in Fig. 6.8 that show good agreement are the dominant LP41 and LP51 mode groups. All other mode groups deviate from the experiment, partly due to the simplifying assumptions of the theoretical model combined with the added complexity of the additional individual modes in the higher order mode groups as described in the previous sections.

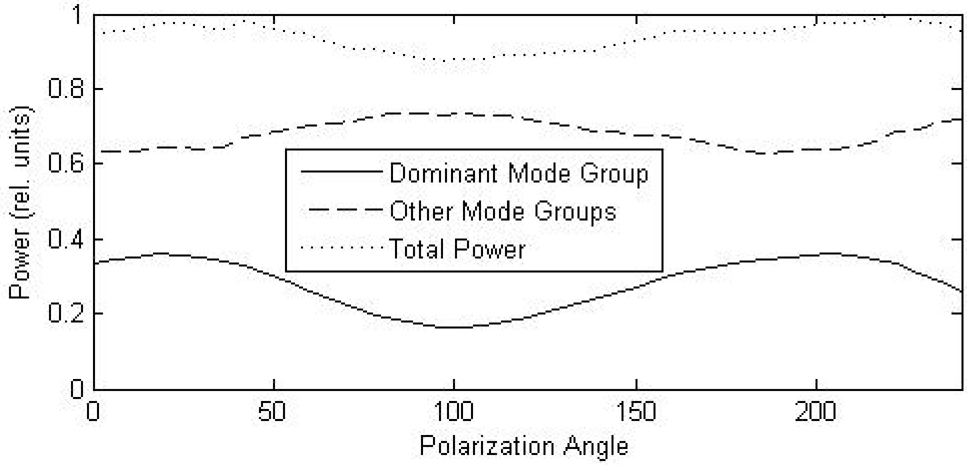


Fig. 6.7a: Experimental polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 21 um offset

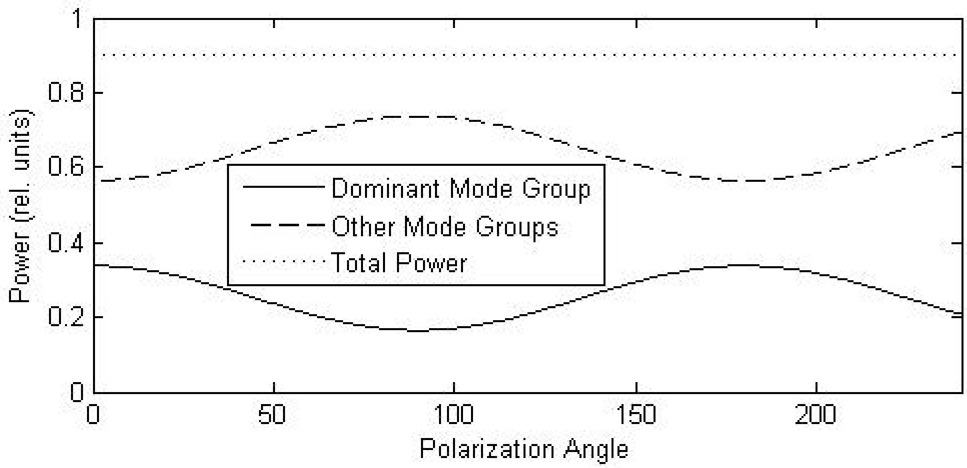


Fig. 6.7b: Theoretical polarization dependent power fluctuation between dominant mode group and all other excited lower and higher mode groups for 21 um offset

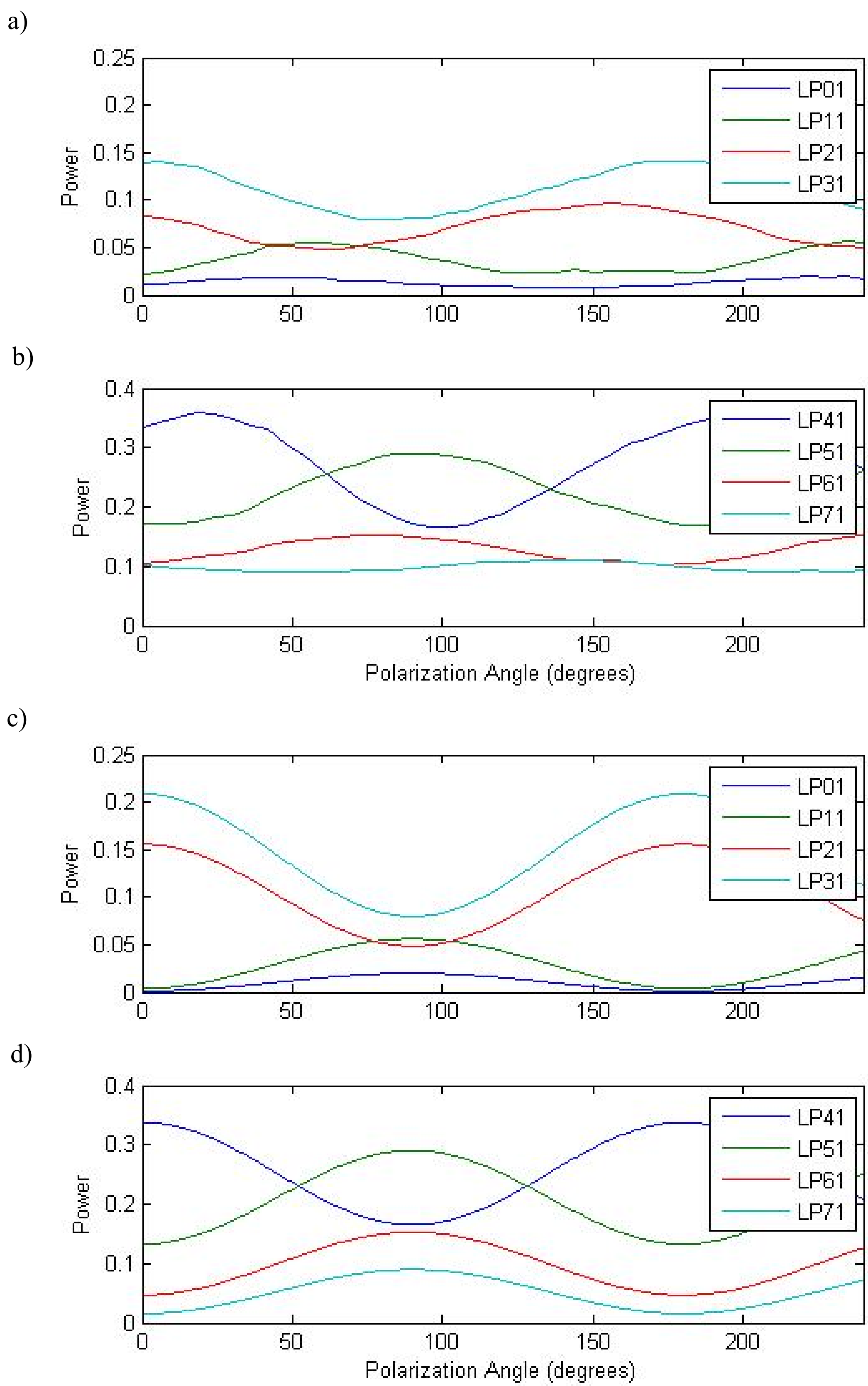


Fig. 6.8: Experimental and theoretical polarization dependent mode group power fluctuation, 21 um offset.

6.5 Discussion of Results

The results for the theoretical model presented above match very well for all the low excitation regimes such as 0 μm and 7 μm but begin to break down for the 14 μm and 21 μm cases. What we see in these higher launch offset regimes is that the dominant modes that hold most of the power can be made to match up quite well with experimental results, but the other mode groups can be off by quite a bit. It is also possible for the phases of each mode group to be off slightly from the other curves as seen in the 14 μm offset launch case. This can be modeled as having complex excitation coefficients but is hard to match with experiment since a change in the phase of an excitation coefficient does not translate directly to the final phase of the power curve for the mode group. The main reason for the large discrepancy between theoretical and experimental results is the increasing complexity of the higher order mode groups. For example, the first four mode groups only carry a total of six individual LP modes while the fifth, sixth, seventh and eighth mode groups carry fourteen total modes. This model, however, assumes that each mode group comprises of only one mode as represented by its initial excitation coefficient. As a result, it is impossible to resolve the polarization dependent interactions between the modes comprising any given mode group. To modify the model to include all the individual modes, each represented by its own excitation coefficient instead of combining them into mode groups does not produce accurate results either. This is because it is not possible to determine the phase or amount of power carried by each individual mode (only the mode groups) from the experimental data.

The model also cannot predict the specific phase relationships between mode groups which determines how power flows as the input polarization is changed. For example, in Fig. 6.4a, power flows from the fundamental LP01 mode group into all other mode groups. Given the current setup of the model, there is no way of determining how power will flow between mode groups. Since there are a number of ways to see this polarization dependent mode power coupling (e.g. fusion splice, fiber parameter mismatch) the mode coupling relationship of the fiber system will be highly dependent on the specifics of the perturbation that causes the mode coupling to begin with. The only insight the model provides is that the effect may be due to interference between the modes at the power coupling interface.

Despite the complexity the higher order mode groups introduces into the model, most real world multimode fiber systems rarely have offsets greater than 7 μm. The model as presented therefore can still represent the polarization dependent mode power coupling effects seen in multimode fiber systems in the low offset launch regime.

Chapter Seven
Conclusions and Future Work

As enterprise and campus backbone Ethernet networks progress past 10 Gb/s, a number of effects seen in multimode fiber networks can degrade overall link quality, making this progression to higher data rates increasingly difficult. One of these effects is the polarization dependence on system performance in multimode fiber. During the course of the development of the IEEE 802.3aq standard, many studies were conducted in an attempt to explain this effect, but none were able to directly determine the cause of this effect and what exactly the performance dependence on input polarization is. The purpose of this thesis was to further study the polarization dependence in multimode optical fiber, determine its root causes and present a theoretical model that explains the total polarization dependence. Table 7.1 compares the current contributions of this thesis with previously presented polarization observations.

Table 7.1: Comparison of Current Contributions and Previous Observations

Significant Current Contributions	**Previous Polarization Observations**
New methodology to quantify polarization effect: direct measurement of mode groups	No dependence with direct excitation of high quality (OM3/3+) fiber.
Polarization effect determined to be a sinusoidal power coupling between modes.	Large dependence with direct excitation of low quality (FDDI) fiber.
Possibly caused by fiber parameter mismatch, fusion splice, offset connectors.	Large dependence when patchcord used over any quality fiber.
Mathematical model created that shows good agreement with experiment in low offset regime.	Due to birefringence in fiber breaking mode degeneracy resulting in polarization based mode coupling.
Possibly due to an interference effect between modes at a single interface.	Large dependence when fusion splice present in high quality fiber.
All modes can couple power between one another at the coupling interface.	No power is coupled between mode groups.
Determined effect occurs at discrete point along the fiber link.	Aggregate effect throughout the length of the fiber.
Effect present even in short lengths of fiber.	Due to fiber parameter mismatch and fiber in high coupling regime.

To determine the true polarization effect present in multimode fiber, a series of experiments was performed on five different fiber configurations. These were the direct excitation of OM3 MMF, FDDI MMF followed by excitation of a MMF patchcord coupled to an OM3 or OM3+ MMF and the direct excitation of an OM3 MMF with a fusion splice after the first meter. First, DMD measurements were made for four different fiber setups and the change in DMD was recorded for two orthogonal and circular initial polarization states. The DMD results showed that the major cause of the change in DMD between orthogonal polarizations was the transfer of power between mode groups as the input polarization changed. Following these DMD measurements, a second set of experiments was conducted to observe how power was being transferred between the mode groups. The power in each mode group for each of the fiber configurations at four different offsets between the SMF input and the MMF system was then measured. This showed that for the direct excitation for both the OM3 and FDDI fibers there was no significant transfer of power between mode groups with input polarization rotation and that the performance change in FDDI fiber was due to the inherent birefringence of the lower quality fiber. For the last three configurations where there was a one meter length of fiber excited by an offset SMF input and then coupled or fusion spliced to a final length of OM3/3+ MMF, there was significant power transfer between mode groups. Because there was no pulse spreading through the fiber system other than what was due to the differential group delay of the fiber mode groups, it is known that the polarization effect takes place at the start of the fiber link and is not an aggregate birefringence effect inherent in multimode fiber. Also, since this effect was not seen under direct excitation conditions, the cause is known to be due to the presence of the multimode fiber patchcord or fusion splice. The exact cause is not known, but it could be due to the fiber parameter mismatch of the patchcord and final fiber sample, an inherent offset present in the patchcord – fiber coupling region, or induced birefringence in the fiber (especially for the case with the fusion splice). These findings agree with the basic qualitative observations made in the previous studies [1 – 7] in that there is no polarization dependence when a high quality OM3 fiber is directly excited by a SMF input; that there is an observable dependence when a low quality FDDI fiber is directly excited by a SMF input; that there is a large polarization dependence when a patchcord is placed before a length of any quality fiber; that the dependence can be caused by a fusion splice in the fiber and that it might be caused by birefringence breaking the mode degeneracy.

Based on these findings, a model was constructed to attempt to explain the effects. The model uses the full vectorial LP mode representation of each individual mode that propagates through the fiber. Using overlap integrals to model the direct excitation of these full vector multimode fiber modes, it was found that under polarization rotation (modeled using a rotation matrix) the mode power coefficients simplify to a constant term meaning that the power in each mode does not fluctuate as the initial polarization of the SMF input is rotated. The next step in the model used concepts from traditional coupled mode theory to explain how the modes exchange power as polarization is rotated. Under certain conditions, and simplifying the problem such that only total mode groups (not individual modes) were used, analytical solutions were found that showed the characteristic sinusoidal dependence (with a period of π) on polarization angle as seen in the experiment. These mode power solutions imply that the polarization effect might be due to an interference of modal fields at the interface where the polarization effect is seen. The model shows very good agreement with experiment for the low offset launch regime (≤ 7 μm) but breaks down for the higher offset regimes. This is because of the increasing complexity of the mode groups that are excited in higher offset regimes and because it is not understood how modes within a mode group interact with one another as polarization is rotated.

Future Work

To help resolve some of the issues of the current model, more experiments can be performed. To determine how an offset without the presence of fiber parameter mismatch might cause polarization dependence, a fiber spool can be used with a short length of fiber at the beginning cut off. This length of fiber, instead of being fused back on to the longer spool can then be aligned with the remaining fiber using a nanopositioner. The polarization can then be rotated as pulses are sent down this fiber system and monitored as the offset between the short length of MMF and the rest of the spool is increased. This simple experiment would help determine if the polarization dependence on performance is due to the offset between connectors in a MMF system or if the parameter mismatch between the patchcord and final length of fiber is the main cause. Another experiment would investigate the relationship between modes in a mode groups as polarization is rotated. This would require using either pulses much shorter than the 100 ps pulses used in this experiment or a fiber length considerably larger than 1 km. Either of these methods would allow for the possible resolution of individual modes clumped together in their mode groups and it would then be possible to monitor how their power fluctuates as a function of input polarization angle.

Though there remains much to accomplish, this thesis has presented the first steps to fully understanding the polarization dependence on performance in multimode fiber systems. It has presented quantitative measurements of this effect and created a mathematical model to help explain its causes and effects, creating a firm foundation upon which future investigations may be undertaken.

Chapter 7 References

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